MA3K4: Introduction to Group Theory

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Chapter 1

Element orders, subgroups and homomorphisms

The purpose of this chapter is to revise some of the theorems and definitions that you have seen in earlier modules. We recall the definition of a group, see some examples and non-examples of groups, and begin the study of three important aspects of group theory: elements (Section 1.2), subgroups (Sections 1.3 and 1.4) and homomorphisms (Section 1.5).

These notes are intended to be self-contained, so they include some revision. Much of this chapter and parts of Chapter 2 will be familiar from Algebra 1 and Algebra 3, or from Sets and Numbers and Groups and Rings, including some of the proofs. This is a good thing: repetition is key for learning, and it means that we start the new material on a stronger footing.

Throughout these notes, $\mathbb{N} = \{1, 2, 3, \dots\}$ denotes the set of natural numbers. In particular, $0 \notin \mathbb{N}$.

1.1 Basic definitions and examples

Definition 1.1. A *group* is a pair $(G, \circ)$ where G is a set, and $\circ : G \times G \to G$ is a binary operation on G that satisfies the following three properties:

(i) **Associativity**. For all $g, h, k \in G$, $(g \circ h) \circ k = g \circ (h \circ k)$.

(ii) **Identity**. There is an element of G, which we call 1_G, satisfying $g \circ 1_G = 1_G \circ g = g$ for all $g \in G$. We call 1_G an *identity element* in $(G, \circ)$.

(iii) **Inverse**. For all $g \in G$, there exists an element in G, which we denote by g^{-1}, satisfying $g \circ g^{-1} = g^{-1} \circ g = 1_G$, where 1_G is as in (ii). We call g^{-1} an *inverse* of g in $(G, \circ)$.

Implicit in part (ii) is that an identity element for an associative binary operation is unique. This is why we write it as 1_G, and from now on, call it *the* identity element in $(G, \circ)$.

Similarly, the inverse of an element g in an associative binary operation is unique. This is why we write it as g^{-1}, and from now on call it *the* inverse of g in $(G, \circ)$.

When checking whether a given pair $(G, \circ)$ is a group, note that you first need to check that $\circ$ really is a binary operation on G, that is, that $g \circ h \in G$ for all $g, h \in G$.

If $(G, \circ)$ is a group, then we call $\circ$ the *group operation*. When the context is clear, we omit $\circ$: we simply say that "G is a group", and we write gh instead of $g \circ h$, calling it the *product* of g and h.

Definition 1.2. The *order* of a group G, denoted $|G|$, is the number of elements of G. This may be finite or infinite.

Example 1.3. Let $G = \mathbb{Z}$. If we define $\circ : G \times G \to G$ by $g \circ h := g + h$, then we know that G is a group (the binary operation is associative, the identity element is 0, and the inverse of $g \in G$ is $-g$).

If we define $\circ : G \times G \to G$ by $g \circ h := gh$, then G is not a group. The binary operation is associative and there is an identity element (namely 1), but if $g \in G \setminus \{1, -1\}$, then g does not have an inverse.

Exercise. Can you think of other "natural" binary operations that do or do not yield a group?

Example 1.4. Let x be an indeterminate, let n be a positive integer, and let $G = \{x^i \mid 0 \leq i \leq n-1\}$. For $0 \leq i, j \leq n-1$, define a binary operation $\circ$ on G by $x^i \circ x^j = x^k$, where $0 \leq k \leq n-1$ is the unique integer satisfying $i + j \equiv k \pmod{n}$. Then $(G, \circ)$ is a group, written C_n, and called the *cyclic group of order n*. The group C_1 is called the *trivial group*.

The groups C_n, for $n \in \mathbb{N}$, satisfy $gh = hg$ for all $g, h \in C_n$. Groups with this property have a special name.

Definition 1.5. A group G is called *abelian* if $gh = hg$ for all $g, h \in G$. More generally, two elements g and h of G are said to *commute* if $gh = hg$.

Most groups we encounter in this course will *not* be abelian. Our first two examples of such groups are as follows.

Symmetric groups

Let X be a set. We write $\mathrm{Sym}(X)$ for the set of all bijections $f : X \to X$. If we define $f \circ g$ to be the usual composition of functions, then it is clear that $\circ : \mathrm{Sym}(X) \times \mathrm{Sym}(X) \to \mathrm{Sym}(X)$ is a binary operation. We now check the group axioms:

(i) Let $f, g, h \in \mathrm{Sym}(X)$. We need to prove that $(f \circ g) \circ h = f \circ (g \circ h)$. Since two maps are equal if and only if they agree on every element, it suffices to fix $x \in X$ and prove that $((f \circ g) \circ h)(x) = (f \circ (g \circ h))(x)$. Now, the left hand side of this equation is $(f \circ g)(h(x)) = f(g(h(x)))$ (by definition of composition). The right hand side is $f((g \circ h)(x)) = f(g(h(x)))$. Associativity follows.

(ii) The identity bijection $\mathrm{Id}_X : X \to X$ defined by $\mathrm{Id}_X(x) := x$ for all $x \in X$ is an identity element for $\mathrm{Sym}(X)$.

(iii) If $f \in \mathrm{Sym}(X)$, then define $f^{-1} : X \to X$ by $f^{-1}(x) = y$, where y is the unique element of X satisfying $f(y) = x$. One then checks that $f \circ f^{-1} = f^{-1} \circ f = \mathrm{Id}_X$.

Thus, for any set X we have shown that $\mathrm{Sym}(X)$ is a group (with group operation the usual composition of functions).

Example 1.6. Suppose that $X = \{1, 2, 3\}$, and let $\mathrm{Sym}(X)$ be as above. Then by listing all of the

possibilities for $f : X \to X$, we can see that $\mathrm{Sym}(X) = \{f_1, f_2, f_3, f_4, f_5, f_6\}$ where

$$\begin{aligned} f_1 &: 1,2,3 \longrightarrow 1,2,3 \\ f_2 &: 1,2,3 \longrightarrow 2,1,3 \\ f_3 &: 1,2,3 \longrightarrow 3,2,1 \\ f_4 &: 1,2,3 \longrightarrow 1,3,2 \\ f_5 &: 1,2,3 \longrightarrow 3,1,2 \\ f_6 &: 1,2,3 \longrightarrow 2,3,1 \end{aligned}$$

respectively. One checks that $f_2 \circ f_3 \neq f_3 \circ f_2$, so $\mathrm{Sym}(X)$ is a nonabelian group.

We can now define symmetric groups.

Definition 1.7. Let X be any set, and let $\mathrm{Sym}(X)$ denote the set of bijections from X to itself. Then $\mathrm{Sym}(X)$ is a group under composition of maps: it is called the *symmetric group* on X. Elements of $\mathrm{Sym}(X)$ are called *permutations*.

It is tedious to write down permutations of a given finite set, so we usually use shorthand notation, called *cycle notation*. You have seen this before, but we recall it here. We begin by defining *cycles* in $\mathrm{Sym}(X)$.

Definition 1.8. Let X be a set.

(i) If $a_1, \ldots, a_r$ are distinct elements of X, then $(a_1, a_2, \ldots, a_r)$ denotes the permutation $f \in \mathrm{Sym}(X)$ with

(1) $f(a_i) = a_{i+1}$ for $1 \leq i < r$;

(2) $f(a_r) = a_1$;

(3) $f(b) = b$ for $b \in X \setminus \{a_1, a_2, \ldots, a_r\}$.

If $f \in \mathrm{Sym}(X)$ is equal to $(a_1, \ldots, a_r)$ for some distinct elements $a_1, \ldots, a_r \in X$, then f is called a *cycle* in $\mathrm{Sym}(X)$.

(ii) Two cycles $(a_1, \ldots, a_r)$ and $(b_1, \ldots, b_s)$ in $\mathrm{Sym}(X)$ are called *disjoint* if the sets $\{a_1, \ldots, a_r\}$ and $\{b_1, \ldots, b_s\}$ are disjoint.

(iii) The *empty cycle*, written $()$, is defined as $() := \mathrm{Id}_X$.

As part (iii) of the definition suggests, we think of Id_X as a cycle. It is the identity element 1_G of $G = \mathrm{Sym}(X)$. Recall that the group operation in $\mathrm{Sym}(X)$ is composition of functions, so if f and g are cycles in $\mathrm{Sym}(X)$, then $fg = f \circ g$ means "apply g first, then apply f". Note also that cycle notation is not unique: different ordered tuples of distinct elements can define the same cycle. For example, $(1,2,3)$ and $(3,1,2)$ are different ordered 3-tuples of elements of $\{1,2,3\}$, but they define the same permutation in $\mathrm{Sym}(\{1,2,3\})$.

Example 1.9. Let $X = \{1,2,3\}$, and consider the cycles $f := (1,2,3)$ and $g := (1,2)$ in $\mathrm{Sym}(X)$. Since fg is a bijection, to determine fg we need to determine the image of every element of X under the map fg. To this end, we have

(1) $(fg)(1) = f(g(1)) = f(2) = 3$;

(2) $(fg)(2) = f(g(2)) = f(1) = 2$;

(3) $(fg)(3) = f(g(3)) = f(3) = 1$.

Thus, $h := fg$ is the permutation defined by $h(1) = 3$, $h(2) = 2$ and $h(3) = 1$. Hence, $h = fg$ is the cycle $(1, 3)$.

Theorem 1.10. Let X be a finite set. Then

(i) $|\mathrm{Sym}(X)| = |X|!$.

(ii) Every element of $\mathrm{Sym}(X)$ can be written as a product of disjoint cycles of length greater than 1, allowing the empty product, which represents the identity permutation. Moreover, this product is unique in the following sense: If $f \in \mathrm{Sym}(X)$ has the form $f = f_1 \dots f_m = g_1 \dots g_t$ where the f_i are disjoint cycles of length greater than 1, and the g_j are disjoint cycles of length greater than 1, then $m = t$ and $\{f_1, \dots, f_m\} = \{g_1, \dots, g_t\}$.

Proof. (i) We simply need to count the number of bijections from X to itself. Write $X = \{x_1, \dots, x_n\}$, where $|X| = n$, and let $f \in \mathrm{Sym}(X)$. Then the number of choices for $f(x_1)$ is n, the number of choices remaining for $f(x_2)$ is $n - 1$, and so on. Thus, $|\mathrm{Sym}(X)| = n!$.

(ii) Let $f \in \mathrm{Sym}(X)$. If $f = () = \mathrm{Id}_X$, then f is an empty product of disjoint cycles. So assume that $f \neq ()$, and let $Y := \{x \in X \mid f(x) \neq x\}$. Fix $a_1 \in Y$, and define $a_2 := f(a_1)$, $a_3 := f(a_2)$, and so on. Since X is finite, the a_i cannot all be distinct, and since f is injective, the first element to be repeated is a_1. Let m_1 be the least integer such that $f(a_{m_1}) = a_1$. Then $a_1, \dots, a_{m_1}$ are distinct.

If $Y = \{a_1, \dots, a_{m_1}\}$ (i.e. if $a_1, \dots, a_{m_1}$ are the only elements in Y), then we have $f = (a_1, \dots, a_{m_1})$ by definition. Thus, f is a product of disjoint cycles. So assume that $Y \setminus \{a_1, \dots, a_{m_1}\}$ is non-empty, and let a_{m_1+1} be an element of $Y \setminus \{a_1, \dots, a_{m_1}\}$. Define $a_{m_1+2} := f(a_{m_1+1})$, and so on, and as before let m_2 be the least integer such that $f(a_{m_2}) = a_{m_1+1}$. Notice that since f is injective, the sets $\{a_1, \dots, a_{m_1}\}$ and $\{a_{m_1+1}, \dots, a_{m_2}\}$ have empty intersection.

If $Y = \{a_1, \dots, a_{m_2}\}$, then we have $f = (a_1, \dots, a_{m_1})(a_{m_1+1}, \dots, a_{m_2})$. Therefore, f is a product of disjoint cycles, as required. Assume that $Y \setminus \{a_1, \dots, a_{m_2}\}$ is non-empty, and apply the same procedure as in the previous two paragraphs.

Since X is finite, the above process must terminate. When it does, we will have constructed f as a product of disjoint cycles, as required.

For uniqueness, suppose that $f = f_1 \dots f_m$ is a product of disjoint cycles, each of length greater than 1, and let a be a point moved by f_i. Since the cycles are disjoint, f_i agrees with f on the points it moves, and these points are exactly $a, f(a), f(f(a)), \dots$. Hence each f_i is determined by f, up to the order of the factors. □

We study the symmetric groups further in Chapter 2.

General linear groups

Definition 1.11. Let $\mathbb{F}$ be a field, and let n be a positive integer. We define

$$\mathrm{GL}_n(\mathbb{F}) := \{A \mid A \text{ an invertible } (n \times n)\text{-matrix with entries in } \mathbb{F}\}.$$

With the usual multiplication of matrices as the group operation, $\mathrm{GL}_n(\mathbb{F})$ is a group, called the *general linear group of dimension n over* $\mathbb{F}$.

We recall some facts that we need when working with general linear groups. A field $\mathbb{F}$ has an additive identity and a multiplicative identity, which we write as $0_{\mathbb{F}}$ and $1_{\mathbb{F}}$ respectively

(or 0 and 1, if the context is clear). An $(n \times n)$-matrix A over $\mathbb{F}$ is invertible if and only if $\det(A) \neq 0$, if and only if its rows are linearly independent, if and only if its columns are linearly independent.

Not every symmetric or general linear group is nonabelian, however.

Exercise. Prove that $\mathrm{Sym}(X)$ is abelian if and only if $|X| \leq 2$, and that $\mathrm{GL}_n(\mathbb{F})$ is abelian if and only if $n = 1$.

Theorem 1.12. Let $\mathbb{F}$ be a finite field, write $|\mathbb{F}| = q$, and let $n \in \mathbb{N}$. Then

$$|\mathrm{GL}_n(\mathbb{F})| = q^{\binom{n}{2}} \prod_{i=1}^{n} (q^i - 1).$$

Proof. **Non-examinable**, but see the Problem Sheets. The case where q is prime is Lemma 3.15. □

1.2 Orders of elements in groups

Now that we have two infinite families of examples, we turn to some basic properties of groups.

We begin with the order of an element in a group. For a group G, an element $g \in G$ and $n \in \mathbb{N}$, we write g^n for the product of n copies of g, that is,

$$g^n := \underbrace{g \circ g \circ \cdots \circ g}_{n \text{ copies}} = \underbrace{gg \cdots g}_{n \text{ copies}}.$$

We also define $g^0 := 1_G$, and $g^n := (g^{-1})^{-n}$ if n is a negative integer.

Definition 1.13. Let G be a group, and let $g \in G$. The *order* of g, denoted by $|g|$, is defined as follows:

$$|g| := \begin{cases} \infty \text{ if there is no positive integer } n \text{ satisfying } g^n = 1_G, \\ \min\{n \in \mathbb{N} \mid g^n = 1_G\} \text{ otherwise.} \end{cases}$$

We record some basic facts about orders of elements in a group G.

(1) The order of the identity element 1_G is 1. Moreover, 1_G is the unique element of order 1 in G.

(2) For all $g \in G$, we have $|g^{-1}| = |g|$.

(3) If $g \in G$ has finite order n and $n \mid \ell$, then $\ell = rn$ for some $r \in \mathbb{Z}$, so $g^\ell = (g^n)^r = (1_G)^r = 1_G$.

Example 1.14. We will determine the orders of the elements of some of the groups we have already seen:

(1) Consider the group $G = (\mathbb{Z}, +)$ from above. The identity element $1_G = 0$ has order 1, but what about the other elements? The group operation is addition, so for $g \in \mathbb{Z}$ we have $g^n = \underbrace{g + \cdots + g}_{n} = ng$. Since $ng = 0$ if and only if either n or g is 0, we deduce that $|g| = \infty$ for all $g \in \mathbb{Z} \setminus \{0\}$.

(2) Let X be a set, and let $f := (a_1, \ldots, a_r)$ be a cycle of length r in $\mathrm{Sym}(X)$. Then, by definition, $f^e(a_i) = a_j$, where $j \equiv i + e \pmod{r}$. It follows that $|f| = r$.

(3) Let $\mathbb{F}$ be a finite field, of order q say, and let α, β be elements of the multiplicative group $\mathbb{F}^\times$. Then

$$A := \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \in \mathrm{GL}_2(\mathbb{F}), \quad \text{and} \quad A^n = \begin{bmatrix} \alpha^n & 0 \\ 0 & \beta^n \end{bmatrix} \quad \text{for all } n \in \mathbb{N}.$$

Thus, $A^n = I_2$ if and only if $\alpha^n = \beta^n = 1$. We will see in Lemma 1.15(i) that this means that $|A|$ is the least common multiple of the orders of α and β in $\mathbb{F}^\times$.

We will see more examples like this later when we build up our theory a little more.

In the following lemma, we use some notation that you will have seen before: For an element g in a group G, define $\langle g \rangle := \{g^i \mid i \in \mathbb{Z}\}$.

Lemma 1.15. Let G be a group, and let a and b be two elements of G of finite order.

(i) If $\ell \in \mathbb{N}$, then $a^\ell = 1_G$ if and only if $|a|$ divides ℓ.

(ii) If $m \in \mathbb{N}$, then $|a^m| = |a|/\gcd(|a|, m)$.

(iii) Suppose that a and b commute (i.e. $ab = ba$). Then $|ab|$ divides $\mathrm{lcm}(|a|, |b|)$.

(iv) Suppose that a and b commute (i.e. $ab = ba$). If $\langle a \rangle \cap \langle b \rangle = \{1_G\}$, then $|ab| = \mathrm{lcm}(|a|, |b|)$.

Proof. We first prove (i). We have already noted that if $|a|$ divides ℓ, then $a^\ell = 1_G$. For the opposite direction, suppose that $a^\ell = 1_G$. By division with remainder, we can write $\ell = q|a| + r$, where $q \in \mathbb{Z}$ and $0 \leq r < |a|$. Then $1_G = a^\ell = a^{q|a|}a^r = a^r$. Since $r < |a|$ and $a^r = 1_G$, we must have $r = 0$ by the definition of the order of an element.

We now prove (ii). Assume first that m divides $|a|$. Then by part (i),

$$\begin{aligned} (a^m)^\ell = 1_G &\iff |a| \text{ divides } m\ell \\ &\iff |a|/m \text{ divides } \ell. \end{aligned}$$

Thus, the order of a^m is the least positive integer ℓ such that $|a|/m$ divides ℓ. Thus, $|a^m| = |a|/m$. We have therefore proved that

$$|a^m| = |a|/m \text{ if } m \text{ divides } |a|. \tag{1.1}$$

Let us now consider the general case. Let $k = \gcd(|a|, m)$, and write $m = ks$, for some $s \in \mathbb{Z}$. Then $m|a|/k = s|a|$, so $(a^m)^{|a|/k} = (a^{|a|})^s = 1_G$. Thus, $|a^m|$ divides $|a|/k$ by part (i).

On the other hand, by Bézout's lemma there exist $u, t \in \mathbb{Z}$ such that $k = u|a| + tm$. Hence, $a^{tm} = a^k$, since $a^{u|a|} = 1_G$. In particular, $|a^k| = |a^{tm}|$. But $(a^{tm})^{|a^m|} = (a^{m|a^m|})^t = 1_G$, so $|a^k| = |a^{tm}|$ divides $|a^m|$ by (i). Also, $|a^k| = |a|/k$ by (1.1). Hence $|a|/k$ divides $|a^m|$, and so $|a^m| = |a|/k$ by the previous paragraph.

Next, we prove (iii). Let $\ell := \mathrm{lcm}(|a|, |b|)$. Since $ab = ba$, we have $(ab)^\ell = a^\ell b^\ell = 1_G$. Thus, ab has finite order, and $|ab|$ divides ℓ, by part (i).

Finally, we prove (iv). So assume that $\langle a \rangle \cap \langle b \rangle = \{1_G\}$, and let $n = |ab|$. Then $1_G = (ab)^n = a^n b^n$, so $a^n = b^{-n}$. Thus, $a^n \in \langle b \rangle$. But then $a^n \in \langle a \rangle \cap \langle b \rangle = \{1_G\}$, so $a^n = 1_G$. In the same way, $b^n = 1_G$. Thus, $|a|, |b| \mid n$ by part (i), so $\mathrm{lcm}(|a|, |b|) \leq n$. It follows from part (iii) that $\mathrm{lcm}(|a|, |b|) = |ab|$. □

Exercise. Can you find an example of a group G and elements a, b of G with $ab = ba$ such that $|ab| \neq \mathrm{lcm}(|a|, |b|)$?

1.3 Subgroups and cosets

Informally speaking, a subset H of a group G is called a subgroup if it forms a group under the same operation as that of G. More precisely:

Definition 1.16. A subset H of a group G is called a *subgroup* of G if

(i) $h_1, h_2 \in H \Longrightarrow h_1 h_2 \in H$;

(ii) $1_G \in H$;

(iii) $h \in H \Longrightarrow h^{-1} \in H$.

In this case, we write $H \leq G$.

Our first example of a subgroup is a special kind of abelian subgroup.

Definition 1.17. For an element g in a group G, recall that $\langle g \rangle := \{g^i \mid i \in \mathbb{Z}\}$. We call $\langle g \rangle$ the *cyclic subgroup of G generated by g*. A group G is called *cyclic* if $G = \langle g \rangle$ for some $g \in G$. In this case, the element g is called a *generator* of G.

If $g \in G$ has finite order n, then $\langle g \rangle = \{1_G, g, \ldots, g^{n-1}\}$, and these n elements are distinct. Indeed, if $i \in \mathbb{Z}$, then writing $i = qn + r$ with $0 \leq r < n$ gives $g^i = g^r$, and if $g^i = g^j$ with $0 \leq i < j < n$, then $g^{j-i} = 1_G$ with $0 < j - i < n$, a contradiction. Similarly, if $|g| = \infty$, then the powers g^i are all distinct. Hence $|\langle g \rangle| = |g|$ in all cases. In particular, if G is cyclic with generator g, then $|G| = |g|$.

Example 1.18. Some examples and non-examples of cyclic groups:

(1) The group $(\mathbb{Z}, +)$ is a cyclic group, with a generator $g := 1$.

(2) If $X = \{1, 2\}$, then $\mathrm{Sym}(X)$ is a cyclic group, with a generator $g := (1, 2)$.

(3) If $X = \{1, 2, 3\}$, then the group $\mathrm{Sym}(X)$ is **not** cyclic. Indeed, suppose that $\mathrm{Sym}(X) = \langle g \rangle$. Then $|g| = |\mathrm{Sym}(X)| = 6$. But from the list of the six elements of $\mathrm{Sym}(X)$ in Section 1.1, every element of $\mathrm{Sym}(X)$ has order 1, 2 or 3. Alternatively, cyclic groups are abelian, and we saw in Section 1.1 that $\mathrm{Sym}(X)$ is not abelian.

Exercise. Let G be a group and $g \in G$. Using either the definition or the criterion in Lemma 1.19 below, you should check that the cyclic subgroup generated by g is indeed a subgroup.

The following lemma gives a quick way of checking whether a non-empty subset of a group is a subgroup.

Lemma 1.19. Let H be a non-empty subset of a group G. Then H is a subgroup of G if and only if $h_1 h_2^{-1} \in H$ whenever $h_1, h_2 \in H$.

Proof. ($\Longrightarrow$) Suppose first that H is a subgroup of G, and let $h_1, h_2 \in H$. Then since $h_2 \in H$, condition (iii) in Definition 1.16 guarantees that $h_2^{-1} \in H$. Condition (i) then guarantees that $h_1 h_2^{-1} \in H$, as required.

($\Longleftarrow$) For the opposite direction, suppose that $h_1 h_2^{-1} \in H$ whenever $h_1, h_2 \in H$. We need to prove that H is a subgroup of G, that is, that conditions (i), (ii), and (iii) in Definition 1.16 hold. For condition (ii), we may choose an element h of H, since H is non-empty. Then our condition guarantees that $hh^{-1} = 1_G \in H$. Thus, condition (ii) in Definition 1.16 holds. Condition (iii) follows, since if $h \in H$, then $1_G, h \in H$, so $h^{-1} = 1_G h^{-1} \in H$. Condition (i) then also follows, since if $h_1, h_2 \in H$, then $h_2^{-1} \in H$ by condition (iii), so $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$. □

Let us look at subgroups which are not cyclic.

Example 1.20. Let $\mathbb{F}$ be a field, and let $G := \mathrm{GL}_2(\mathbb{F})$. Let

$$H := \left\{ \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \;\middle|\; \alpha, \beta \in \mathbb{F}^\times \right\}.$$

Then H is a subgroup of G, often called the *diagonal subgroup* of G. Indeed,

$$h_1 := \begin{bmatrix} \alpha_1 & 0 \\ 0 & \beta_1 \end{bmatrix}, h_2 := \begin{bmatrix} \alpha_2 & 0 \\ 0 & \beta_2 \end{bmatrix} \in H \Rightarrow h_1 h_2^{-1} = \begin{bmatrix} \alpha_1 \alpha_2^{-1} & 0 \\ 0 & \beta_1 \beta_2^{-1} \end{bmatrix} \in H.$$

Thus, H is a subgroup by Lemma 1.19.

Exercise. Show that if $\mathbb{F}$ has more than two elements, then the diagonal subgroup H of $\mathrm{GL}_2(\mathbb{F})$ is not cyclic.

Our next example of a subgroup generalises the cyclic subgroups.

Definition 1.21. Let G be a group, and let A be a non-empty subset of G. The *subgroup of G generated by A*, written $\langle A \rangle$, is defined to be

$$\langle A \rangle := \{ a_1^{\epsilon_1} \dots a_m^{\epsilon_m} \mid m \in \mathbb{N}, a_i \in A, \epsilon_i \in \{\pm 1\} \}.$$

Note that $\langle A \rangle$ is indeed a subgroup of G. It is non-empty, since A is non-empty. Now, let $g, h \in \langle A \rangle$. Then by definition of $\langle A \rangle$, we have $g = a_1^{\epsilon_1} \dots a_m^{\epsilon_m}$ and $h = b_1^{\delta_1} \dots b_n^{\delta_n}$, for some $a_i, b_j \in A$, and $\epsilon_i, \delta_j \in \{\pm 1\}$. Thus, $h^{-1} = b_n^{-\delta_n} \dots b_1^{-\delta_1}$. It follows that $gh^{-1} = c_1^{\gamma_1} \dots c_{m+n}^{\gamma_{m+n}}$, where $c_i := a_i$ and $\gamma_i := \epsilon_i$ for $1 \leq i \leq m$, and $c_{m+i} := b_{n-i+1}$ and $\gamma_{m+i} := -\delta_{n-i+1} \in \{\pm 1\}$ for $1 \leq i \leq n$. Thus, $\langle A \rangle$ is a subgroup of G by Lemma 1.19.

We also define $\langle \varnothing \rangle := \{1_G\}$.

If $A = \{a_1, \dots, a_d\}$, then we sometimes write $\langle A \rangle = \langle a_1, \dots, a_d \rangle$ (i.e. we do not bother with the set brackets).

We now discuss cosets of a subgroup in a group.

Definition 1.22. Let G be a group, $H \leq G$, and $g \in G$. Then the *left coset* gH is the subset $\{gh \mid h \in H\}$ of G. (Similarly, the *right coset* Hg is the subset $\{hg \mid h \in H\}$, but we will deal mainly with left cosets in this course).

Lemma 1.23. Let G be a group and H be a subgroup of G. The following are equivalent for $g, k \in G$:

(i) $k \in gH$;

(ii) $gH = kH$;

(iii) $g^{-1}k \in H$.

Proof. Notice first that $hH = H$ whenever $h \in H$, since H is a subgroup.

If $k \in gH$, then $k = gh$ for some $h \in H$. Thus, $kH = ghH = gH$. So (i) implies (ii).

If $kH = gH$, then $k = k1_G \in kH$, so $k = gh$, for some $h \in H$. Thus, $g^{-1}k = h \in H$. So (ii) implies (iii).

Finally, if $g^{-1}k = h \in H$, then $k = gh \in gH$. So (iii) implies (i). □

Corollary 1.24. Let G be a group, and let H be a subgroup of G. For $g_1, g_2 \in G$, say that $g_1 \sim_H g_2$ if $g_1H = g_2H$. Then $\sim_H$ is an equivalence relation on G.

Proof. We need to prove that $\sim_H$ is reflexive, symmetric and transitive. Since $gH = gH$ for all $g \in G$, the relation $\sim_H$ is reflexive. If $g_1H = g_2H$, then $g_2H = g_1H$, so $\sim_H$ is symmetric. It remains to prove that $\sim_H$ is transitive.

Let $g_1, g_2, g_3 \in G$ with $g_1 \sim_H g_2$ and $g_2 \sim_H g_3$. We need to prove that $g_1 \sim_H g_3$. We have $g_1H = g_2H$ and $g_2H = g_3H$. But then $g_1H = g_3H$, so $g_1 \sim_H g_3$, as required. □

Corollary 1.25. Let G be a group and let H be a subgroup of G. Then

(i) If $g_1, g_2 \in G$, then either $g_1H = g_2H$, or $g_1H \cap g_2H = \varnothing$.

(ii) Let $\{g_iH \mid i \in I\}$ be the set of left cosets of H in G, where I is an index set and $g_iH \neq g_jH$ for $i \neq j$. Then $G = \bigsqcup_{i \in I} g_iH$ is the disjoint union of the sets g_iH, for $i \in I$.

Proof. By Lemma 1.23, the $\sim_H$-equivalence class of $g \in G$ is $\{k \in G \mid kH = gH\} = gH$. So the $\sim_H$-equivalence classes are precisely the left cosets of H in G. Since the distinct equivalence classes of an equivalence relation on G are pairwise disjoint and form a partition of G, both parts follow. □

Theorem 1.26 (Lagrange's theorem)**.** Let G be a finite group and H be a subgroup of G. Then the order of H divides the order of G.

Proof. Let $\{g_iH \mid i \in I\}$ be as in Corollary 1.25(ii). Then I is finite, since G is finite. For each $i \in I$, the map $H \to g_iH$ given by $h \mapsto g_ih$ is surjective by definition of g_iH, and injective since $g_ih_1 = g_ih_2$ implies $h_1 = h_2$. Hence $|g_iH| = |H|$ for all $i \in I$. Since $G = \bigsqcup_{i \in I} g_iH$, we have

$$|G| = \sum_{i \in I} |g_iH| = \sum_{i \in I} |H| = |I||H|.$$

Thus, $|H|$ divides $|G|$. □

Definition 1.27. Let G be a group, and let H be a subgroup of G.

(i) The set of left cosets of H in G is denoted by G/H. That is, $G/H := \{gH \mid g \in G\}$.

(ii) The number of distinct left cosets of H in G (i.e. the cardinality of the set G/H) is called the *index* of H in G and is written as $|G : H|$. If G is finite, then $|G : H| = |G|/|H|$.

We close this section with a corollary of Lagrange's theorem on orders of elements in finite groups.

Corollary 1.28. Let G be a finite group, and let $g \in G$. Then $|g|$ divides $|G|$.

Proof. The subgroup $H := \langle g \rangle$ has order $|g|$. The result now follows immediately from Lagrange's theorem. □

1.4 Normal subgroups and quotient groups

In this section, we study normal subgroups and quotient groups. We first introduce conjugation.

Definition 1.29. Let G be a group, let $g, x \in G$, and let $H \leq G$. The *conjugate* of x by g is ${}^g x := gxg^{-1}$, and the *conjugate* of H by g is

$$ {}^g H := gHg^{-1} := \{ghg^{-1} \mid h \in H\}.$$

Lemma 1.30. Let G be a group, and let $g, x \in G$. Then $|{}^g x| = |x|$.

Proof. For $n \in \mathbb{N}$, we have $({}^g x)^n = gx^n g^{-1}$, so $({}^g x)^n = 1_G$ if and only if $x^n = 1_G$. □

Lemma 1.31. Let G be a group, let $H \leq G$, and let $g \in G$. Then ${}^g H$ is a subgroup of G.

Proof. Let $g_1, g_2 \in gHg^{-1}$. By Lemma 1.19, it suffices to prove that $g_1 g_2^{-1} \in gHg^{-1}$, since gHg^{-1} is certainly non-empty. By definition, we can write $g_1 = gh_1g^{-1}$ and $g_2 = gh_2g^{-1}$, for some $h_1, h_2 \in H$. Then $g_2^{-1} = gh_2^{-1}g^{-1}$, so $g_1 g_2^{-1} = gh_1h_2^{-1}g^{-1} \in gHg^{-1}$. □

Definition 1.32. Let G be a group, and let H be a subgroup of G.

(i) We say that H is a *normal subgroup* of G if $gHg^{-1} = H$ for all $g \in G$. In this case, we write $H \trianglelefteq G$.

(ii) The *normaliser of H in G*, denoted $N_G(H)$, is defined as

$$N_G(H) := \{g \in G \mid gHg^{-1} = H\}.$$

If $H \leq G$, then $N_G(H)$ is a subgroup of G containing H, and $H \trianglelefteq N_G(H)$ (see the Problem Sheets).

Lemma 1.33. Let G be a group, and let $H \leq G$. Then $H \trianglelefteq G$ if and only if $xHx^{-1} \subseteq H$ for all $x \in G$.

Proof. If $H \trianglelefteq G$, then $xHx^{-1} = H$ for all $x \in G$, so certainly $xHx^{-1} \subseteq H$ for all $x \in G$.

Conversely, suppose that $xHx^{-1} \subseteq H$ for all $x \in G$, and let $g \in G$. Then $gHg^{-1} \subseteq H$. Also, $g^{-1}Hg = (g^{-1})H(g^{-1})^{-1} \subseteq H$, and multiplying on the left by g and on the right by g^{-1} gives $H \subseteq gHg^{-1}$. Thus, $H = gHg^{-1}$ for all $g \in G$, so $H \trianglelefteq G$. □

Every group G has at least two normal subgroups, namely $\{1_G\}$ and G itself.

Definition 1.34. A group G is called *simple* if $G \neq \{1_G\}$ and the only normal subgroups of G are $\{1_G\}$ and G.

For a group G and subsets A and B of G, we will write $AB := \{ab \mid a \in A, b \in B\}$ for the *product* of A and B. Note that AB is a subset of G, but in general it is not a subgroup, even if both A and B are subgroups.

Lemma 1.35. Let N be a normal subgroup of a group G, and let $g, h \in G$. Then $(gN)(hN) = ghN$.

Proof. We first prove that $(gN)(hN) \subseteq ghN$. So let $x = gn_1 \in gN$ and $y = hn_2 \in hN$ $(n_1, n_2 \in N)$. Then

$$xy = gn_1hn_2 = gh(h^{-1}n_1h)n_2 \in ghN$$

since $h^{-1}n_1h \in h^{-1}Nh = N$.

For the opposite containment, let $x = ghn \in ghN$, with $n \in N$. Then $x = (g1_G)(hn) \in (gN)(hN)$. Hence $(gN)(hN) = ghN$. □

The point of the previous lemma is that if N is a normal subgroup of a group G, then we have a natural binary operation on the set $G/N = \{gN \mid g \in G\}$ of left cosets of N in G.

Definition 1.36. Let G be a group, and let $N \trianglelefteq G$. The binary operation $\circ : G/N \times G/N \to G/N$ defined by $(gN) \circ (hN) = ghN$ is called the *natural binary operation* of G/N.

Proposition 1.37. Let N be a normal subgroup of a group G. Then G/N is a group under the natural binary operation, with identity $1_GN = N$, and with $(gN)^{-1} = g^{-1}N$ for all $g \in G$.

Proof. By Lemma 1.35, the natural binary operation is a binary operation on G/N, and $(gN)(hN) = ghN$ for all $g, h \in G$. We check the group axioms.

(i) Associativity. Let $g, h, k \in G$. By associativity in G,

$$((gN)(hN))(kN) = (ghN)(kN) = (gh)kN = g(hk)N = (gN)(hkN) = (gN)((hN)(kN)).$$

(ii) Identity. Let $g \in G$. Then $(gN)(1_GN) = (g1_G)N = gN = (1_Gg)N = (1_GN)(gN)$.

(iii) Inverses. Let $g \in G$. Then $(gN)(g^{-1}N) = gg^{-1}N = 1_GN = g^{-1}gN = (g^{-1}N)(gN)$.

□

Definition 1.38. Let G be a group, and let $N \trianglelefteq G$. The group G/N of Proposition 1.37 is called the *quotient* (or *factor*) *group* of G by N.

If G is finite, then $|G/N| = |G : N| = |G|/|N|$, by the proof of Theorem 1.26.

1.5 Homomorphisms

Definition 1.39. Let G and H be groups.

(i) A map $\phi : G \to H$ is called a *homomorphism* provided $\phi(g_1g_2) = \phi(g_1)\phi(g_2)$ for all $g_1, g_2 \in G$.

(ii) If $\phi : G \to H$ is a homomorphism and is a bijection, ϕ is called an *isomorphism*, and we say that the groups G and H are isomorphic and write $G \cong H$.

(iii) If $\phi : G \to H$ is a homomorphism, then the *kernel* and the *image* of ϕ are defined respectively as

$$\ker(\phi) := \{g \in G \mid \phi(g) = 1_H\},$$
$$\mathrm{Im}(\phi) := \{\phi(g) \mid g \in G\}.$$

Definition 1.40. Let G be a group and $N \trianglelefteq G$. Then the map $\pi : G \to G/N$ defined by $\pi(g) = gN$ is a surjective homomorphism (this follows immediately from Lemma 1.35). It is called the *natural homomorphism* from G to G/N.

Theorem 1.41 (First Isomorphism Theorem)**.** Let G and H be groups, and let $\phi : G \to H$ be a homomorphism. Then the following hold:

(i) $\ker(\phi)$ is a normal subgroup of G;

(ii) $\mathrm{Im}(\phi)$ is a subgroup of H;

(iii) the map $g \ker(\phi) \mapsto \phi(g)$ is a well-defined isomorphism $G/\ker(\phi) \to \mathrm{Im}(\phi)$. In particular, $G/\ker(\phi) \cong \mathrm{Im}(\phi)$.

Proof. You saw the proof of this in Algebra 3 or Groups and Rings, so we will not repeat it here. □

Example 1.42. Let $\mathbb{F}$ be a field, and let n be a positive integer. Then $\det : \mathrm{GL}_n(\mathbb{F}) \to \mathbb{F}^\times$ is a homomorphism, with kernel

$$\mathrm{SL}_n(\mathbb{F}) := \{A \mid A \in \mathrm{GL}_n(\mathbb{F}) \text{ with } \det(A) = 1_\mathbb{F}\}.$$

The group $\mathrm{SL}_n(\mathbb{F})$ is called the *special linear group of dimension n over* $\mathbb{F}$.

Theorem 1.43 (Second Isomorphism Theorem)**.** Let G be a group, $H \leq G$ and $N \trianglelefteq G$. Then the following hold:

(i) $NH = HN \leq G$;

(ii) $H \cap N$ is a normal subgroup of H;

(iii) $H/(H \cap N) \cong NH/N$.

Proof. This is an exercise on the Problem Sheets. □

Theorem 1.44 (Third Isomorphism Theorem)**.** Let G be a group, and let N and K be normal subgroups of G with $N \leq K$. Then $K/N \trianglelefteq G/N$, and $(G/N)/(K/N) \cong G/K$.

Proof. This is an exercise on the Problem Sheets. □

Theorem 1.45 (Correspondence Theorem)**.** Let G be a group and $N \trianglelefteq G$. Then there is a one-to-one correspondence between the subgroups of G/N and the subgroups of G containing N. More precisely:

(i) the map $f : \{J \leq G \mid N \leq J\} \to \{X \mid X \leq G/N\}$, $J \mapsto J/N$, is a bijection;

(ii) if $N \leq J \leq G$, then $J \trianglelefteq G$ if and only if $J/N \trianglelefteq G/N$.

Proof. (i) Let $A := \{J \leq G \mid N \leq J\}$ and $B := \{X \mid X \leq G/N\}$, and let $f : A \to B$ be as in (i). Let $J \in A$. Then $J \leq G$ with $N \leq J$, and so, as $N \trianglelefteq G$, we have $N \trianglelefteq J$. Thus, we may consider J/N, and $J/N \in B$. So the map f is well-defined. We now show that f is a bijection.

If $J_1, J_2 \in A$ with $J_1 \neq J_2$, then without loss of generality there exists $x \in J_1$ with $x \notin J_2$. We claim that $xN \notin J_2/N$. Indeed, otherwise $xN = jN$ for some $j \in J_2$, so $j^{-1}x \in N \leq J_2$ by Lemma 1.23, and hence $x \in J_2$, a contradiction. Thus, $xN \in (J_1/N) \setminus (J_2/N)$, so $f(J_1) \neq f(J_2)$, and f is injective.

Next, we prove that f is surjective. Take any $X \in B$, and let $Y := \{g \in G \mid gN \in X\}$. We show that $Y \in A$ and $f(Y) = X$. First, $N \subseteq Y$, since $nN = N = 1_{G/N} \in X$ for all $n \in N$. In particular, $Y \neq \varnothing$. Next, let $y_1, y_2 \in Y$. Then $y_1N, y_2N \in X$, so $y_1y_2^{-1}N = (y_1N)(y_2N)^{-1} \in X$ by Proposition 1.37, since X is a subgroup of G/N. Thus, $y_1y_2^{-1} \in Y$, so $Y \leq G$ by Lemma 1.19,

and hence $Y \in A$. Finally, $f(Y) = Y/N = \{yN \mid y \in Y\} = X$ by definition of Y. So f is surjective.

(ii) Suppose that $J \trianglelefteq G$, and let $g \in G$. Then $(gN)(J/N)(gN)^{-1} = \{gjg^{-1}N \mid j \in J\} = J/N$, so $J/N \trianglelefteq G/N$. Conversely, suppose that $J/N \trianglelefteq G/N$, and let $g \in G$ and $j \in J$. Then $gjg^{-1}N = (gN)(jN)(gN)^{-1} \in J/N$, so $gjg^{-1}N = j'N$ for some $j' \in J$. Hence $gjg^{-1} \in j'N \subseteq J$, so $J \trianglelefteq G$ by Lemma 1.33. □

Chapter 2

Permutation groups and group actions

2.1 Permutation groups

In this section, we study an important class of groups called *permutation groups*.

Definition 2.1. Let X be any set. A subgroup of $\mathrm{Sym}(X)$ is called a *permutation group* on X.

So for example, $\mathrm{Sym}(X)$ itself is a permutation group on X.

To define two important classes of examples of permutation groups (the dihedral and alternating groups) we need some preparatory results about permutations.

Definition 2.2. Let X be a set.

(i) For $g \in \mathrm{Sym}(X)$, the *support of* g is the set

$$\mathrm{supp}(g) := \{x \in X \mid g(x) \neq x\} \subseteq X.$$

(ii) Let G be a permutation group on X. The *support of* G is the set

$$\mathrm{supp}(G) := \{x \in X \mid g(x) \neq x \text{ for some } g \in G\} = \bigcup_{g \in G} \mathrm{supp}(g) \subseteq X.$$

If $G = \langle g \rangle \leq \mathrm{Sym}(X)$, then $\mathrm{supp}(\langle g \rangle) = \mathrm{supp}(g)$. Note also that if

$$g = (a_1, \ldots, a_{m_1}) \ldots (a_{m_{t-1}+1}, \ldots, a_{m_t})$$

is a product of disjoint cycles of length greater than 1, then we can read off $\mathrm{supp}(g)$: we have

$$\mathrm{supp}(g) = \{a_1, \ldots, a_{m_1}, a_{m_1+1}, \ldots, a_{m_{t-1}+1}, \ldots, a_{m_t}\}.$$

Exercise. Prove that if G is a permutation group on X, and $H \leq G$, then $\mathrm{supp}(H) \subseteq \mathrm{supp}(G)$. Deduce that if G_1 and G_2 are permutation groups on X and $\mathrm{supp}(G_1) \cap \mathrm{supp}(G_2) = \varnothing$, then $G_1 \cap G_2 = \{\mathrm{Id}_X\}$.

Example 2.3. In this example, we will prove that if G is a permutation group on X and $G = \langle A \rangle$ for some $\varnothing \neq A \subseteq G$, then

$$\mathrm{supp}(G) = \bigcup_{a \in A} \mathrm{supp}(a).$$

So if we know a generating set for G, then we can read off $\mathrm{supp}(G)$ from the generators, using the observation above about products of disjoint cycles.

We now prove this. First, $\bigcup_{a\in A}\mathrm{supp}(a) \subseteq \mathrm{supp}(G)$ by definition, so it remains to prove the opposite containment. So let $x \in \mathrm{supp}(G)$. In pursuit of a contradiction, assume that $x \notin \bigcup_{a\in A}\mathrm{supp}(a)$. Then $a(x) = x$ for all $a \in A$. Hence, it is also true that $a^{-1}(x) = x$ for all $a \in A$. That is, $a^{\epsilon}(x) = x$ for all $a \in A, \epsilon \in \{\pm 1\}$ $(*)$.

Now, since $x \in \mathrm{supp}(G)$, we have $g(x) \neq x$ for some $g \in G$. Since $g \in G = \langle A\rangle$, we also have $g = a_1^{\epsilon_1} \dots a_m^{\epsilon_m}$ for some $a_i \in A, \epsilon_i \in \{\pm 1\}, m \in \mathbb{N}$. But then

$$x \neq g(x) = a_1^{\epsilon_1} \circ \dots \circ a_m^{\epsilon_m}(x) = x,$$

with the latter equality following from $(*)$. This contradiction completes the proof.

Theorem 2.4. Let X be a finite set.

(i) Disjoint cycles in $\mathrm{Sym}(X)$ commute.

(ii) If $f = (a_1, \dots, a_r) \in \mathrm{Sym}(X)$ is a cycle of length r, then $|f| = r$. More generally, if $f \in \mathrm{Sym}(X)$ and $f = f_1 \dots f_m$ is a decomposition of f into a product of disjoint cycles, then $|f| = \mathrm{lcm}(|f_1|, \dots, |f_m|)$.

(iii) Let $f = (a_1, \dots, a_r)$ be a cycle in $\mathrm{Sym}(X)$, and let $g \in \mathrm{Sym}(X)$. Then

$$^g f = (g(a_1), \dots, g(a_r)).$$

Proof. (i) Suppose that $f = (a_1, \dots, a_r)$ and $g = (b_1, \dots, b_s)$ are disjoint cycles in $\mathrm{Sym}(X)$. It suffices to prove that $(f \circ g)(x) = (g \circ f)(x)$ for all $x \in X$. So let $x \in X$. Suppose first that $x \in \{b_1, \dots, b_s\}$. Then $g(x) \in \{b_1, \dots, b_s\}$ by definition, so $g(x) \notin \{a_1, \dots, a_r\}$ since the cycles are disjoint. Thus, $(f \circ g)(x) = f(g(x)) = g(x)$. Also, $x \notin \{a_1, \dots, a_r\}$, so $f(x) = x$. Hence, $(g \circ f)(x) = g(f(x)) = g(x)$.

Similarly, if $x \in \{a_1, \dots, a_r\}$, then $(f \circ g)(x) = f(x) = (g \circ f)(x)$. Finally, if x lies in neither set, then $f(x) = g(x) = x$, so $(f \circ g)(x) = x = (g \circ f)(x)$.

(ii) We noted in Chapter 1 that if $f = (a_1, \dots, a_r)$ is a cycle of length $r \geq 1$, then $|f| = r$.

Next, let us consider the general case. Write $f = f_1 \dots f_m$, where the f_i are disjoint cycles. We will prove that $|f| = \mathrm{lcm}(|f_1|, \dots, |f_m|)$ by induction on m. If $m = 1$, then $|f| = |f_1|$ by the case above.

So assume that $m > 1$, and that the result holds for smaller values of m. Let $g := f_1 \dots f_{m-1}$. We have $|g| = \mathrm{lcm}(|f_1|, \dots, |f_{m-1}|)$ by the inductive hypothesis. Also, if we write $f_i = (a_{i,1}, \dots, a_{i,t_i})$, then $\mathrm{supp}(g) = \{a_{i,j} \mid 1 \leq i \leq m-1, 1 \leq j \leq t_i\}$, while $\mathrm{supp}(f_m) = \{a_{m,1}, \dots, a_{m,t_m}\}$. Thus, $\langle g\rangle \cap \langle f_m\rangle = \{\mathrm{Id}_X\}$ by the exercise above. Also, f_m commutes with each of $f_1, \dots, f_{m-1}$ by part (i), so it commutes with g. Hence Lemma 1.15(iv) gives

$$|f| = \mathrm{lcm}(|g|, |f_m|) = \mathrm{lcm}(\mathrm{lcm}(|f_1|, \dots, |f_{m-1}|), |f_m|) = \mathrm{lcm}(|f_1|, \dots, |f_m|)$$

as required.

(iii) Let $f = (a_1, \dots, a_r)$ be a cycle in $\mathrm{Sym}(X)$, where $a_1, \dots, a_r$ are distinct elements in X. Let g be another element of $\mathrm{Sym}(X)$. We need to prove that

$$^g f = (g(a_1), \dots, g(a_r)).$$

For $1 \leq i \leq r$, define $b_i := g(a_i)$. Then, if $1 \leq i \leq r-1$, we have

$$\begin{aligned}(^g f)(b_i) &= (gfg^{-1})(b_i)\\ &= g(f(g^{-1}(g(a_i))))\\ &= g(f(a_i))\\ &= g(a_{i+1}) = b_{i+1}.\end{aligned}$$

In the same way, $({}^g f)(b_r) = b_1$. Finally, suppose that $b \in X \setminus \{b_1, \ldots, b_r\}$, and let $c \in X$ such that $g(c) = b$. If $f(c) \neq c$, then c lies in $\{a_1, \ldots, a_r\}$, by definition of a cycle. But then $b = g(c) = g(a_i) = b_i$ for some i, contradicting our choice of b. Thus, we must have $f(c) = c$. It follows that:

$$\begin{aligned}({}^g f)(b) &= (gfg^{-1})(b)\\ &= g(f(g^{-1}(g(c))))\\ &= g(f(c))\\ &= g(c) = b.\end{aligned}$$

Thus, ${}^g f$ is the cycle $(b_1, \ldots, b_r)$, as claimed. □

We can now recall the dihedral groups.

Definition 2.5. Let $n \in \mathbb{N}$ with $n \geq 3$, and set $X := \{1, \ldots, n\}$. Define the permutations $\sigma, \tau \in \mathrm{Sym}(X)$ by

$$\begin{aligned}\sigma &:= (1, \ldots, n) \text{ and}\\ \tau &:= \prod_{i=1}^{\lfloor n/2 \rfloor} (i, n-i+1).\end{aligned}$$

The *dihedral group* of order $2n$, written D_{2n}, is the subgroup of $\mathrm{Sym}(X)$ defined by

$$D_{2n} := \langle \{\sigma, \tau\} \rangle.$$

Some authors, including those of Groups and Rings, write D_n for the dihedral group of order $2n$. We always write D_{2n}, so that the subscript is the order of the group.

For example, if $n = 7$, then

$$D_{14} = \langle \{(1,2,3,4,5,6,7), (1,7)(2,6)(3,5)\} \rangle,$$

while if $n = 8$, then

$$D_{16} = \langle \{(1,2,3,4,5,6,7,8), (1,8)(2,7)(3,6)(4,5)\} \rangle.$$

Lemma 2.6. Let G be a group, and let $H = \langle A \rangle$ and $K = \langle B \rangle$ be subgroups of G, where $A, B \subseteq G$ and H is finite. Then $K \leq N_G(H)$ if and only if ${}^b a \in H$ for all $a \in A$ and $b \in B$.

Proof. If $K \leq N_G(H)$, then ${}^b a \in {}^b H = H$ for all $a \in A$ and $b \in B$. Conversely, suppose that ${}^b a \in H$ for all $a \in A$ and $b \in B$, and let $b \in B$. Every $h \in H$ can be written as $h = a_1^{\epsilon_1} \cdots a_m^{\epsilon_m}$ with $a_i \in A$, $\epsilon_i \in \{\pm 1\}$ and $m \in \mathbb{N}$. Since conjugation by b is a homomorphism (see the Problem Sheets),

$${}^b h = ({}^b a_1)^{\epsilon_1} \cdots ({}^b a_m)^{\epsilon_m} \in H.$$

Thus ${}^b H \subseteq H$. Since $|{}^b H| = |H|$ and H is finite, ${}^b H = H$, so $b \in N_G(H)$. Hence $B \subseteq N_G(H)$, and since $N_G(H)$ is a subgroup of G (see the Problem Sheets), $K = \langle B \rangle \leq N_G(H)$. □

Theorem 2.7. Let $n \geq 3$, and let $D_{2n} = \langle \sigma, \tau \rangle$ be as defined in Definition 2.5. Then

(i) $|D_{2n}| = 2n$;

(ii) $\langle \sigma \rangle$ is a normal subgroup of D_{2n} of order n. In particular, D_{2n} is not simple.

Proof. Part (i) is an exercise on the Problem Sheets, so we prove only (ii). By Theorem 2.4(iii), ${}^{\tau}\sigma = (n, n-1, \ldots, 1) = \sigma^{-1} \in \langle\sigma\rangle$, and ${}^{\sigma}\sigma = \sigma \in \langle\sigma\rangle$. Hence $D_{2n} = \langle\sigma, \tau\rangle \leq N_{D_{2n}}(\langle\sigma\rangle)$ by Lemma 2.6, that is, $\langle\sigma\rangle \trianglelefteq D_{2n}$. Finally, $|\langle\sigma\rangle| = |\sigma| = n$ by Theorem 2.4(ii), and $1 < n < 2n = |D_{2n}|$, so D_{2n} is not simple. □

Next, we have the alternating groups.

Definition 2.8. Let X be a finite set.

(i) Let $f \in \mathrm{Sym}(X)$, and write $f = f_1 \ldots f_r$ where the f_i are disjoint cycles. We say that f is *even* if the number of cycles of even length in $\{f_1, \ldots, f_r\}$ is even (this includes the trivial element $()$, since it has 0 cycles of even length). Otherwise, f is said to be *odd*.

(ii) The set $\mathrm{Alt}(X) := \{f \in \mathrm{Sym}(X) \mid f \text{ is even}\}$ is called the *alternating group* on X.

As the name suggests, the alternating group is in fact a group. The proof is an exercise on the Problem Sheets, but we state the result here.

Proposition 2.9. Let X be a finite set with $|X| \geq 2$. Then $\mathrm{Alt}(X)$ is a subgroup of $\mathrm{Sym}(X)$, and it has order $|X|!/2$. In other words, $|\mathrm{Sym}(X) : \mathrm{Alt}(X)| = 2$.

Proof. This is an exercise on the Problem Sheets. □

Proposition 2.10. Let X and Y be finite sets. If $|X| = |Y|$, then the groups $\mathrm{Sym}(X)$ and $\mathrm{Sym}(Y)$ are isomorphic.

Proof. Let $\phi : Y \to X$ be a bijection. One checks that the map $\theta : \mathrm{Sym}(X) \to \mathrm{Sym}(Y)$ defined by $\theta(f) = \phi^{-1} \circ f \circ \phi$ is an isomorphism. □

We finish this section with some notation. We let S_n denote the group $\mathrm{Sym}(\{1, \ldots, n\})$, and A_n denote the group $\mathrm{Alt}(\{1, \ldots, n\})$. (So by Proposition 2.10, S_n is isomorphic to $\mathrm{Sym}(X)$ whenever $|X| = n$.)

2.2 Group actions

When groups were first defined by Galois in the nineteenth century, the motivation was to have algebraic objects which measured the symmetry of various different mathematical structures. In this section, we will make this idea rigorous, by defining what it means for a group to "act (as symmetries)" on a set X.

Definition 2.11. Let G be a group and X a set. An *action* of G on X is a map $\cdot : G \times X \to X$, which satisfies the properties:

(i) $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G,\ x \in X$;

(ii) $1_G \cdot x = x$ for all $x \in X$.

In this case, we also say that G *acts on* X, and that X is a G*-set*. We often write $g.x$ instead of $g \cdot x$.

Example 2.12. We will frequently use the following three group actions.

(1) **Action of a group on itself by left multiplication**. Let G be a group, let $X = G$ and define the action $. : G \times X \to X$ by $g.x = gx$. Then

(i) $(gh).x = ghx$ and $g.(h.x) = g.(hx) = ghx$, so $(gh).x = g.(h.x)$ for all $g, h \in G$, $x \in X$;

(ii) $1_G.x = 1_G x = x$ for all $x \in X$.

Thus, . is a group action.

(2) **Action of a group on itself by conjugation**. Let G be a group, let $X = G$ and define the action $. : G \times X \to X$ by $g.x = {}^g x = gxg^{-1}$. Then

(i) $(gh).x = ghx(gh)^{-1} = ghxh^{-1}g^{-1}$ and $g.(h.x) = g.(hxh^{-1}) = ghxh^{-1}g^{-1}$, so $(gh).x = g.(h.x)$ for all $g, h \in G$, $x \in X$;

(ii) $1_G.x = 1_G x 1_G^{-1} = 1_G x 1_G = x$ for all $x \in X$.

So again, . is a group action.

(3) **Action of a group on the set of left cosets of a subgroup**. Let G be a group, let H be a subgroup of G, and let $X = G/H = \{gH \mid g \in G\}$ be the set of left cosets of H in G. Define $. : G \times X \to X$ by $g.xH := gxH$. This is well-defined. Indeed, if $xH = yH$, then $y = xh$ for some $h \in H$, so $gyH = gxhH = gxH$. Arguing as in the previous examples, we see that . is a group action.

What we have defined is a *left* action of G on X. A *right* action can be defined analogously as a map $X \times G \to X$.

There is a strong link between group actions and the symmetric groups $\mathrm{Sym}(X)$, which is encoded in the following proposition.

Proposition 2.13. Let $\cdot$ be an action of the group G on the set X. For $g \in G$, define the map $\phi(g) : X \to X$ by $\phi(g)(x) = g \cdot x$. Then $\phi(g) \in \mathrm{Sym}(X)$, and $\phi : G \to \mathrm{Sym}(X)$ is a homomorphism.

Proof. Let $g, h \in G$, and $x \in X$. Then

$$\phi(gh)(x) = (gh).x = g.(h.x) = \phi(g)(h.x) = (\phi(g) \circ \phi(h))(x),$$

so $\phi(gh) = \phi(g) \circ \phi(h)$. Also, $\phi(1_G) = \mathrm{Id}_X$, since $1_G.x = x$ for all $x \in X$. Hence $\phi(g) \circ \phi(g^{-1}) = \phi(1_G) = \mathrm{Id}_X = \phi(g^{-1}) \circ \phi(g)$, so $\phi(g)$ is a bijection, that is, $\phi(g) \in \mathrm{Sym}(X)$. □

This leads naturally to the definition of the *kernel* and *image* of an action.

Definition 2.14. Let $\cdot$ be an action of a group G on a set X.

(i) The *kernel* of the action $\cdot$, denoted $\ker(G, X, \cdot)$, is defined to be the kernel $\ker(\phi)$ of the homomorphism $\phi : G \to \mathrm{Sym}(X)$ defined in Proposition 2.13. So

$$\ker(G, X, \cdot) = \{\, g \in G \mid g \cdot x = x \text{ for all } x \in X \,\}.$$

(ii) The *image* of the action $\cdot$, denoted $\mathrm{Im}(G, X, \cdot)$, is defined to be the image of the homomorphism $\phi : G \to \mathrm{Sym}(X)$ defined in Proposition 2.13. So

$$\mathrm{Im}(G, X, \cdot) = \{\phi(g) \mid g \in G\}.$$

(iii) The action $\cdot$ is said to be *faithful* if $\ker(G, X, \cdot) = \{1_G\}$, and *trivial* if $\ker(G, X, \cdot) = G$.

Example 2.15.

(1) If G acts on itself by left multiplication, then $\ker(G, G, \cdot) = \{1_G\}$, since $g.1_G = g$ for all $g \in G$.

(2) If G acts on itself by conjugation, then the action is trivial if and only if $gxg^{-1} = x$ for all $g, x \in G$, i.e. if and only if G is abelian.

(3) If G acts on G/H for a subgroup $H \leq G$, then the action is trivial if and only if $gH = H$ for all $g \in G$, i.e. if and only if $G = H$. So if H is a proper subgroup of G, then $\ker(G, G/H, \cdot)$ is a proper normal subgroup of G. We will use this often.

If $\cdot : G \times X \to X$ is an action of a group G on a set X, then the First Isomorphism Theorem implies that $G/\ker(G, X, \cdot) \cong \mathrm{Im}(G, X, \cdot) \leq \mathrm{Sym}(X)$. We state this as a proposition. We write $H \lneq G$ to mean that H is a proper subgroup of G, that is, $H \leq G$ and $H \neq G$.

Proposition 2.16. Let $\cdot$ be an action of G on X. Then

(i) $G/\ker(G, X, \cdot)$ is isomorphic to a subgroup of $\mathrm{Sym}(X)$. In particular, if X is finite, then $|G : \ker(G, X, \cdot)|$ divides $|X|!$.

(ii) Let H be a *proper* subgroup of G, i.e. $H \lneq G$. If $X = G/H$ and $\cdot$ are as in Example 2.12(3), then $\ker(G, X, \cdot)$ is a *proper* normal subgroup of G, and if $|G : H|$ is finite then $|G : \ker(G, X, \cdot)|$ divides $|G : H|!$.

Proof. Part (i) is immediate from the First Isomorphism Theorem. Part (ii) follows from (i) and Example 2.15(3). □

We now define the orbit and the stabiliser of a point in X.

Definition 2.17. Let $\cdot$ be an action of a group G on a set X. Let $x \in X$.

(i) The *orbit* of x, denoted $\mathrm{Orb}_G(x)$, is

$$\mathrm{Orb}_G(x) := \{\, g \cdot x \mid g \in G \,\};$$

(ii) The *stabiliser* of x in G, denoted $\mathrm{Stab}_G(x)$, is

$$\mathrm{Stab}_G(x) := \{\, g \in G \mid g \cdot x = x \,\}.$$

Proposition 2.18. Let G act on X. Then

(i) $\mathrm{Stab}_G(x)$ is a subgroup of G for all $x \in X$;

(ii) $\bigcap_{x \in X} \mathrm{Stab}_G(x)$ is the kernel of the action of G on X.

Proof. This is an exercise on the Problem Sheets. □

The next theorem is fundamental in group theory.

Theorem 2.19 (The Orbit–Stabiliser Theorem)**.** Let the finite group G act on X, and let $x \in X$. Then $|G| = |\mathrm{Orb}_G(x)|\,|\mathrm{Stab}_G(x)|$.

Proof. Let $Y := G/\mathrm{Stab}_G(x)$ be the set of left cosets of $\mathrm{Stab}_G(x)$ in G. Define $\phi : Y \to \mathrm{Orb}_G(x)$ by $g\,\mathrm{Stab}_G(x) \mapsto g.x$. We claim that ϕ is a bijection.

To see this, we first need to prove that ϕ is well-defined. So assume that $g_1\,\mathrm{Stab}_G(x) = g_2\,\mathrm{Stab}_G(x)$. Then $g_2 = g_1 s$, for some $s \in \mathrm{Stab}_G(x)$ by Lemma 1.23. Thus,

$$\phi(g_2\,\mathrm{Stab}_G(x)) = g_2.x = (g_1 s).x = g_1.(s.x) = g_1.x = \phi(g_1\,\mathrm{Stab}_G(x)),$$

so ϕ is well-defined.

Next, we show that ϕ is injective. Suppose that $\phi(g_1 \operatorname{Stab}_G(x)) = \phi(g_2 \operatorname{Stab}_G(x))$. Then $g_1.x = g_2.x$, so $(g_2^{-1}g_1).x = x$. Hence, $g_2^{-1}g_1 \in \operatorname{Stab}_G(x)$, so $\operatorname{Stab}_G(x) = g_2^{-1}g_1 \operatorname{Stab}_G(x)$. Hence, $g_2 \operatorname{Stab}_G(x) = g_1 \operatorname{Stab}_G(x)$. Thus, ϕ is injective.

To see that ϕ is surjective, let $y \in \operatorname{Orb}_G(x)$. Then by definition, $y = g.x$ for some $g \in G$. Then $\phi(g \operatorname{Stab}_G(x)) = g.x$. Thus, ϕ is surjective, so ϕ is bijective.

Since $|Y| = |G|/|\operatorname{Stab}_G(x)|$, the proof is complete. □

Definition 2.20. Let G be a group, and let $x \in G$.

(i) The *centraliser* of x in G is $C_G(x) := \{g \in G \mid gxg^{-1} = x\}$;

(ii) The *conjugacy class* of x in G is ${}^G x := \{{}^g x \mid g \in G\}$;

(iii) The *centre* of G is $Z(G) := \{g \in G \mid gyg^{-1} = y \text{ for all } y \in G\}$.

Let us compute the orbits and stabilisers of the group actions from Example 2.12.

Example 2.21.

(1) Let G act on $X := G$ by left multiplication, and let $x \in X$. Then, if $y \in X$, we have $y = (yx^{-1}).x$, so $y \in \operatorname{Orb}_G(x)$. Thus, $\operatorname{Orb}_G(x) = X$ for all $x \in X$. Also, for $g \in G$, $g.x = x$ if and only if $gx = x$, which is true if and only if $g = 1_G$. Thus, $\operatorname{Stab}_G(x) = \{1_G\}$ for all $x \in G$.

(2) Let G act on $X := G$ by conjugation, and let $x \in X$. Then $\operatorname{Orb}_G(x) = {}^G x$, and $g.x = x$ if and only if $gxg^{-1} = x$, so $\operatorname{Stab}_G(x) = C_G(x)$. Also, $\ker(G, X, \cdot) = Z(G)$.

(3) Let H be a subgroup of G, and let G act on $X := G/H$ by left multiplication. Then $\operatorname{Stab}_G(xH)$ is the set

$$\{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gxH = H\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1} = {}^x H.$$

Also, if $x_1H, x_2H \in X$, then $(x_1x_2^{-1}).x_2H = x_1H$, so $\operatorname{Orb}_G(xH) = X$ for all $xH \in X$.

The following is arguably the most useful consequence of the Orbit–Stabiliser Theorem.

Corollary 2.22. Let G be a finite group acting on a set X, and let $x, y \in X$. Then

(i) Either $\operatorname{Orb}_G(x) = \operatorname{Orb}_G(y)$ or $\operatorname{Orb}_G(x) \cap \operatorname{Orb}_G(y) = \varnothing$;

(ii) The set $\{\operatorname{Orb}_G(x) \mid x \in X\}$ forms a partition of X;

(iii) $|\operatorname{Orb}_G(x)|$ divides $|G|$.

Proof. Define a relation $\sim$ on X by $x \sim y$ if and only if $y = g.x$ for some $g \in G$. One checks that $\sim$ is an equivalence relation on X, and that the equivalence classes are $\{\operatorname{Orb}_G(x) \mid x \in X\}$. Parts (i) and (ii) follow.

Part (iii) follows immediately from the Orbit–Stabiliser Theorem and Lagrange's theorem since $|\operatorname{Orb}_G(x)| = |G|/|\operatorname{Stab}_G(x)|$. □

Part (iii) of Corollary 2.22 is far from obvious. In particular, it shows that the size $|{}^G x|$ of a conjugacy class in a finite group G divides $|G|$. We now discuss some applications.

Applications of the Orbit–Stabiliser Theorem from Example 2.12(1)

Here, we will consider the action of G on itself by left multiplication.

Theorem 2.23 (Cayley's theorem)**.** Every group G is isomorphic to a subgroup of $\mathrm{Sym}(G)$. In particular, if G is finite of order n, then G is isomorphic to a subgroup of S_n.

Proof. Let G act on $X := G$ by left multiplication. The kernel of this action is $\{g \in G \mid gx = x \text{ for all } x \in G\}$. If g lies in the kernel, then $g = g1_G = 1_G$. Thus, the kernel is trivial, so $G \cong G/\{1_G\}$ is isomorphic to a subgroup of $\mathrm{Sym}(G)$ by Proposition 2.16(i). The final statement follows since $\mathrm{Sym}(G) \cong S_n$ by Proposition 2.10. □

Applications of the Orbit–Stabiliser Theorem from Example 2.12(2)

In this part of Section 2.2, we will consider the action of a group G on itself by conjugation.

Theorem 2.24. Let p be a prime, let $n \in \mathbb{N}$, and let G be a finite group with $|G| = p^n$. Then $|Z(G)| > 1$.

Proof. Corollary 2.22 shows that $|^G x|$ is a power of p for all $x \in G$. Also, by definition, $Z(G) = \{x \in G \mid |^G x| = 1\}$. Suppose that $|Z(G)| = 1$. Then Corollary 2.22(ii) implies that

$$p^n = |G| = 1 + p^{a_1} + \cdots + p^{a_k}$$

for some $a_i \in \mathbb{N}$. This is a contradiction however, since the left hand side of the equation above is divisible by p (i.e. congruent to 0 modulo p), while the right hand side is congruent to 1 modulo p. □

Corollary 2.25. Let p be a prime, let $n \in \mathbb{N}$, and let G be a finite group with $|G| = p^n$.

(i) If $n = 2$, i.e. $|G| = p^2$, then G is abelian;

(ii) If $n = 3$, i.e. $|G| = p^3$, then either G is abelian, or $|Z(G)| = p$.

Proof. We first prove (i). So assume that $|G| = p^2$. Recall from the Problem Sheets that if $G/Z(G)$ is cyclic, then G is abelian. Now, by Lagrange's theorem and Theorem 2.24, the centre of G has order either p or p^2. But since every group of prime order is cyclic, $|Z(G)| = p$ would imply that $G/Z(G)$ is cyclic, so G is abelian by the result mentioned above. If $|Z(G)| = p^2$ then $G = Z(G)$, so again, G is abelian.

Part (ii) is similar: By Lagrange's theorem and Theorem 2.24, the centre of G has order either p, p^2, or p^3. Suppose that G is nonabelian. Then by arguing as in part (i), we see that $Z(G)$ cannot have order p^2 or p^3. Thus, $|Z(G)| = p$, and this completes the proof. □

We finish with the following:

Theorem 2.26 (Cauchy's theorem)**.** Let G be a finite group, and let p be a prime divisor of $|G|$. Then G contains an element of order p. In fact, the number of elements of G of order p is congruent to -1 modulo p.

Proof. Define

$$X := \{(x_1, \ldots, x_p) \mid x_i \in G \text{ and } x_1 \ldots x_p = 1_G\}.$$

Note that

$$\begin{aligned}(x_1, \ldots, x_p) \in X &\Rightarrow x_1 \ldots x_p = 1_G \\ &\Rightarrow x_p x_1 \ldots x_{p-1} = x_p(x_1 \ldots x_p)x_p^{-1} = x_p 1_G x_p^{-1} = 1_G \\ &\Rightarrow (x_p, x_1, \ldots, x_{p-1}) \in X.\end{aligned}$$

Repeatedly applying the above shows that X is closed under cyclically permuting the entries of its elements.

Now, let $C := \langle (1, \ldots, p) \rangle \leq S_p$ be the subgroup of the symmetric group S_p generated by the p-cycle $(1, \ldots, p)$. Define $. : C \times X \to X$ by

$$f.(x_1, \ldots, x_p) := (x_{f^{-1}(1)}, \ldots, x_{f^{-1}(p)})$$

for $f \in C$ and $(x_1, \ldots, x_p) \in X$. By the paragraph above, $f.(x_1, \ldots, x_p) \in X$, since the elements of C permute the entries cyclically. The identity Id fixes every entry, so $\mathrm{Id}.(x_1, \ldots, x_p) = (x_1, \ldots, x_p)$. Also, if $f, g \in C$ and $x = (x_1, \ldots, x_p) \in X$, then the i-th entry of $f.(g.x)$ is the $f^{-1}(i)$-th entry of $g.x$, namely $x_{g^{-1}(f^{-1}(i))} = x_{(fg)^{-1}(i)}$, which is the i-th entry of $(fg).x$. So $.$ is an action of C on X.

The idea of the proof is now to use Corollary 2.22(ii) to partition X into a disjoint union of C-orbits. To do this, we split the orbits into three types.

(1) The p-tuple $(1_G, \ldots, 1_G)$ lies in X, and is fixed by every element of C. Hence, if we let $R := \mathrm{Orb}_C((1_G, \ldots, 1_G))$, we have $|R| = 1$.

(2) Let x be an element of G of order p. Then $(x, \ldots, x)$ lies in X, and as in (1), $S_x := \mathrm{Orb}_C((x, \ldots, x))$ has size 1. We claim that if S is any C-orbit in X of size 1, then either $S = R$, or $S = S_x$ for some $x \in G$ with $|x| = p$. To see this, let $(x_1, \ldots, x_p) \in S$. Then the Orbit–Stabiliser Theorem implies that $\mathrm{Stab}_C((x_1, \ldots, x_p)) = C$. Thus, taking $f = (1, 2, \ldots, p) \in C$, we have

$$(x_1, \ldots, x_p) = f.(x_1, \ldots, x_p) = (x_p, x_1, \ldots, x_{p-1}).$$

That is, $x_1 = x_2 = \cdots = x_p$. Thus, $x_1^p = 1_G$, so $|x_1| \in \{1, p\}$ by Lemma 1.15(i), and our claim follows.

(3) Let $T_1, \ldots, T_n$ be the distinct orbits in X of size greater than 1. Since $|C| = p$, Corollary 2.22(iii) implies that $|T_i| = p$ for all i.

We have now listed all of the C-orbits in X. To complete the proof, we need to know $|X|$. We claim that

$$|X| = |G|^{p-1}. \tag{2.1}$$

To see this, define $\phi : X \to G^{p-1}$ by $\phi((x_1, \ldots, x_p)) := (x_1, \ldots, x_{p-1})$. Then ϕ is injective, since

$$(x_1, \ldots, x_p) \in X \Rightarrow x_p = (x_1 \ldots x_{p-1})^{-1}.$$

Furthermore, ϕ is surjective, since

$$(x_1, \ldots, x_{p-1}) \in G^{p-1} \Rightarrow (x_1, \ldots, x_{p-1}, y) \in X,$$

where $y := (x_1 \ldots x_{p-1})^{-1}$. Thus, $|X| = |G^{p-1}| = |G|^{p-1}$, as claimed.

We can now complete the proof. Let $\mathcal{P} := \{x \in G \mid |x| = p\}$. Then Corollary 2.22(ii) and (2.1) imply that

$$\begin{aligned} |G|^{p-1} = |X| &= |R| + \sum_{x \in \mathcal{P}} |S_x| + \sum_{i=1}^{n} |T_i| \\ &= 1 + |\mathcal{P}| + np. \end{aligned}$$

Since p divides $|G|$ and $p - 1 \geq 1$, we deduce that $|\mathcal{P}| \equiv -1 \pmod{p}$. In particular, $|\mathcal{P}| > 0$, that is, G contains an element of order p. □

Applications of the Orbit–Stabiliser Theorem from Example 2.12(3)

We now consider the action of G by left multiplication on the set of left cosets of a subgroup. Applying the Orbit–Stabiliser Theorem to this action gives two useful results. The first is as follows. Recall that the product $HK := \{hk \mid h \in H, k \in K\}$ of two subgroups in a group is not necessarily a subgroup. However:

Proposition 2.27. Let G be a finite group, and let H and K be subgroups of G. Then

$$|HK| = |KH| = \frac{|K||H|}{|H \cap K|}.$$

Proof. Let $X := G/H$ be the set of left cosets of H in G, and consider the action of K on X given by $k.(gH) := kgH$. This is the restriction of the action of G on X by left multiplication, so it is indeed an action. We claim first that

$$\frac{|KH|}{|H|} \text{ is equal to the size of the } K\text{-orbit } \mathrm{Orb}_K(H) \text{ of } H \in X. \tag{2.2}$$

To see this, note that by Corollary 1.25, if $k_1, k_2 \in K$, then $k_1H = k_2H$ or $k_1H \cap k_2H = \varnothing$. Thus, if we let $k_1H, \ldots, k_sH$ be the distinct elements of the set $\{kH \mid k \in K\}$, then we have

$$KH = \bigsqcup_{i=1}^{s} k_iH.$$

Thus, $|KH| = \sum_{i=1}^{s} |k_iH| = \sum_{i=1}^{s} |H| = s|H|$, so $|KH|/|H| = s$.

On the other hand, if $gH \in X$, then gH lies in the K-orbit of H if and only if $gH = kH$ for some $k \in K$. Thus, $\mathrm{Orb}_K(H) = \{kH \mid k \in K\} = \{k_1H, \ldots, k_sH\}$. Hence, $|\mathrm{Orb}_K(H)| = s = |KH|/|H|$, as claimed in (2.2).

Finally, we will prove that $\mathrm{Stab}_K(H) = H \cap K$. To see this, note that if $k \in K$ then

$$k \in \mathrm{Stab}_K(H) \Leftrightarrow kH = H \Leftrightarrow k \in H.$$

Thus, $\mathrm{Stab}_K(H) = H \cap K$, and the Orbit–Stabiliser Theorem gives $|KH|/|H| = |\mathrm{Orb}_K(H)| = |K|/|H \cap K|$. Finally, since $(hk)^{-1} = k^{-1}h^{-1}$, the map $x \mapsto x^{-1}$ is a bijection $HK \to KH$, so $|HK| = |KH|$. □

Proposition 2.28. Let G be a finite group, and let H and K be subgroups of G. Then $|G : H \cap K| \leq |G : H||G : K|$.

Proof. Let $X_1 := G/H$ and $X_2 := G/K$ be the sets of left cosets of H and K in G, respectively, and let $X := X_1 \times X_2$. Then G acts on X via $g.(y_1H, y_2K) := (gy_1H, gy_2K)$. Let $x := (H, K) \in X$. For $g \in G$, we have

$$\begin{aligned} g \in \mathrm{Stab}_G(x) &\Leftrightarrow gH = H \text{ and } gK = K \\ &\Leftrightarrow g \in H \cap K \end{aligned}$$

by Lemma 1.23. Thus, $\mathrm{Stab}_G(x) = H \cap K$. It follows from the Orbit–Stabiliser Theorem that

$$|G : H \cap K| = |\mathrm{Orb}_G(x)| \leq |X| = |G : H||G : K|.$$

□

2.3 Fixed points

We close this chapter with an interesting application of group actions. First, we need a definition.

Definition 2.29. Let G be a group acting on a set X, and let $g \in G$.

(i) We say that $x \in X$ is a *fixed point* of g, or that g *fixes* x, if $g.x = x$. The set of all fixed points of g is written $\mathrm{fix}_X(g) := \{x \in X \mid g.x = x\}$.

(ii) We say that $g \in G$ is *fixed point free* if $\mathrm{fix}_X(g) = \varnothing$.

We also need a lemma. This is sometimes referred to in the literature as "not Burnside's lemma".

Lemma 2.30 (the "not Burnside" lemma)**.** Let G be a finite group acting on a finite set X, and let $r := |\{\mathrm{Orb}_G(x) \mid x \in X\}|$ be the number of orbits of G. Then

$$r = \frac{1}{|G|} \sum_{g \in G} |\mathrm{fix}_X(g)|.$$

Proof. The proof uses a "double counting" argument. Indeed, we will determine the cardinality of the set

$$\Lambda := \{(g, x) \in G \times X \mid g.x = x\}$$

in two different ways.

Counting the pairs $(g, x) \in \Lambda$ first by g and then by x, we obtain

$$\sum_{g \in G} |\mathrm{fix}_X(g)| = |\Lambda| = \sum_{x \in X} |\mathrm{Stab}_G(x)|.$$

Now, write $\mathrm{Orb}_G(x_1), \ldots, \mathrm{Orb}_G(x_r)$ for the distinct orbits in X. Then by Corollary 2.22, we have $X = \mathrm{Orb}_G(x_1) \sqcup \cdots \sqcup \mathrm{Orb}_G(x_r)$. Also, $|\mathrm{Stab}_G(x_i)| = |\mathrm{Stab}_G(x)|$ whenever $x \in \mathrm{Orb}_G(x_i)$ by the Orbit–Stabiliser Theorem. Thus

$$\begin{aligned} \sum_{g \in G} |\mathrm{fix}_X(g)| &= \sum_{x \in X} |\mathrm{Stab}_G(x)| \\ &= \sum_{i=1}^{r} \sum_{x \in \mathrm{Orb}_G(x_i)} |\mathrm{Stab}_G(x)| \\ &= \sum_{i=1}^{r} |\mathrm{Stab}_G(x_i)||\mathrm{Orb}_G(x_i)| \\ &= \sum_{i=1}^{r} |G| = r|G|. \end{aligned}$$

The result follows. □

Lemma 2.30 has the following consequence.

Corollary 2.31. Let G be a finite group acting on a finite set X with $|X| > 1$, and suppose that G has precisely 1 orbit (G is said to be *transitive* in this case). Then G contains a fixed point free element.

Proof. Suppose that G contains no fixed point free elements. Then $|\operatorname{fix}_X(g)| \geq 1$ for all $g \in G$. Thus, since $|\operatorname{fix}_X(1_G)| = |X|$, Lemma 2.30 implies that

$$|G| = \sum_{g \in G} |\operatorname{fix}_X(g)| = |X| + \sum_{g \in G \setminus \{1_G\}} |\operatorname{fix}_X(g)| \geq |X| + |G| - 1.$$

This is a contradiction, since $|X| > 1$. Hence there exists some $g \in G$ with $|\operatorname{fix}_X(g)| = 0$. □

If we remove both finiteness assumptions from Corollary 2.31, then the result no longer holds. See the Problem Sheets for more details.

Chapter 3

The Sylow theorems

In Chapter 1 we saw Lagrange's theorem, which states that if H is a subgroup of a finite group G, then $|H|$ divides $|G|$. A natural question is whether the converse holds. That is, if G is a finite group and r divides $|G|$, then does G contain a subgroup of order r? This is true when G is cyclic (see the Problem Sheets), but it is not true in general.

For example, if G is a nonabelian finite simple group of even order, then G has no subgroup of order $|G|/2$. (Such a subgroup H would have index 2, and hence would be a proper normal subgroup of G. But $|G| > 2$, since G is nonabelian, so $1 < |H| < |G|$, contradicting the simplicity of G.)

However, Sylow's theorems show that if r is a certain prime power dividing $|G|$, then G *does* have a subgroup of order r. These theorems are the focus of this chapter.

Throughout this chapter, let G denote a finite group and p a prime.

Notation. We will use $|G|_p$ to denote the highest power of p that divides $|G|$. That is, if $|G| = p^n \cdot m$ where $\gcd(m, p) = 1$, then $|G|_p = p^n$.

Definition 3.1. Let $H \leq G$.

(i) We say that H is a *p-subgroup* of G if $|H|$ is a power of p.

(ii) Let $P \leq G$, and suppose that $|P| = |G|_p$. Then P is called a *Sylow p-subgroup* of G.

(iii) We write $\mathrm{Syl}_p(G) := \{P \leq G \mid P \text{ a Sylow } p\text{-subgroup of } G\}$ for the set of Sylow p-subgroups of G.

By definition, if $p \nmid |G|$, then $\{1_G\}$ is a Sylow p-subgroup of G. All of our applications will concern the case $p \mid |G|$, but for some proofs, it is useful to bear this fact in mind.

Example 3.2. The following are some examples of p-subgroups and Sylow p-subgroups.

(1) Take $G = S_4$. Then $|G| = 4! = 2^3 \cdot 3$, so $|G|_2 = 2^3$ and $|G|_3 = 3$. We therefore have:
   (i) The subgroup $\{1, (1,2,3), (3,2,1)\}$ is a Sylow 3-subgroup of G;
   (ii) The subgroup $V_4 := \{1, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\}$ is a 2-subgroup of G, but *not* a Sylow 2-subgroup of G;
   (iii) The subgroup $D_8 = \langle \sigma, \tau \rangle$, where $\sigma := (1,2,3,4)$ and $\tau := (1,4)(2,3)$ (see Theorem 2.7), is a Sylow 2-subgroup of G;
   (iv) The subgroup A_4 is not a p-subgroup of G for any prime p, while $\{1_G\}$ is a p-subgroup of G for all primes p.

(2) Let $G = C_{p^n \cdot m}$ where p is a prime and $\gcd(p, m) = 1$. Since p^n divides $|G|$, there exists a unique subgroup $H \leq G$ of order p^n (see the Problem Sheets). By definition, H is a Sylow p-subgroup of G. So in this example, G has a unique Sylow p-subgroup.

The following result is due to Ludwig Sylow and dates back to 1872. It is fundamental to finite group theory.

Theorem 3.3 (The Sylow theorems)**.** Let G be a finite group, and let p be prime. Then the following conditions hold:

(1) G has at least one Sylow p-subgroup;

(2) All the Sylow p-subgroups of G are conjugate;

(3) Any p-subgroup of G is contained in a Sylow p-subgroup of G;

(4) The number of Sylow p-subgroups of G is congruent to 1 modulo p.

There are various proofs of this result. In this course we will give two proofs.

3.1 Wielandt's proof of Sylow's theorems 1 and 4

Our first proof is due to Wielandt. Before we begin, we require the following number theoretic lemma.

Lemma 3.4. Let p be prime, and let n and m be integers with $\gcd(m, p) = 1$. Then

(i) p divides $\binom{p}{i}$ for all $1 \leq i \leq p-1$;

(ii) $\binom{p^n m}{p^n} \equiv m \pmod{p}$.

Proof. (i) By definition,

$$\binom{p}{i} = \frac{p(p-1)\cdots(p-i+1)}{i(i-1)\cdots 1}.$$

Let $a := (p-1)\cdots(p-i+1)$ and $b := i(i-1)\cdots 1 = i!$, so that $\binom{p}{i} = pa/b$. Since b is a product of integers less than p, all prime divisors of b are less than p. Thus, $p \nmid b$. Since $b\binom{p}{i} = pa$, the prime p divides $b\binom{p}{i}$, and since $p \nmid b$, it follows that p divides $\binom{p}{i}$.

(ii) Let $\mathbb{F}$ be the field $\mathbb{Z}/p\mathbb{Z}$. Then we may think of $\mathbb{F}$ as $\mathbb{F} = \{0, 1, \ldots, p-1\}$, where addition in $\mathbb{F}$ is defined as $x + y := z$, where z is the unique element of $\{0, \ldots, p-1\}$ satisfying $x + y \equiv z \pmod{p}$, and multiplication is defined in the same way: $xy := w$, where w is the unique element of $\{0, \ldots, p-1\}$ satisfying $xy \equiv w \pmod{p}$.

Consider the associated polynomial ring $\mathbb{F}[X]$ (where X is an indeterminate), and the polynomial $(1+X)^p \in \mathbb{F}[X]$. By the binomial theorem, we have

$$\begin{aligned}(1+X)^p &= \sum_{i=0}^{p} \binom{p}{i} X^i \\ &= \binom{p}{0} X^0 + \binom{p}{p} X^p \\ &= 1 + X^p, \end{aligned} \tag{3.1}$$

where the penultimate equality above follows from (i), since multiples of p are 0 in the field $\mathbb{F}$.

Applying (3.1) again with X replaced by X^p, we get

$$(1+X)^{p^2} = ((1+X)^p)^p = (1+X^p)^p = 1+(X^p)^p = 1+X^{p^2}.$$

Repeatedly applying the above then yields

$$(1+X)^{p^n} = 1+X^{p^n}.$$

Hence,

$$(1+X)^{p^n m} = (1+X^{p^n})^m. \tag{3.2}$$

Using the binomial theorem to expand the left and right hand sides of (3.2), we get

$$\sum_{i=0}^{p^n m} \binom{p^n m}{i} X^i = \sum_{j=0}^{m} \binom{m}{j} X^{p^n j}. \tag{3.3}$$

Comparing the coefficients of $X^{p^n j}$ on the left and right hand sides above, we get

$$\binom{p^n m}{p^n j} = \binom{m}{j} \text{ in the field } \mathbb{F} \text{ whenever } 0 \leq j \leq m.$$

In other words,

$$\binom{p^n m}{p^n j} \equiv \binom{m}{j} \pmod{p} \text{ whenever } 0 \leq j \leq m.$$

Applying the above with $j = 1$ then gives the result. □

We now prove Sylow's theorems 1 and 4.

Proposition 3.5. Let G be a finite group and p a prime. Write $|G| = p^n m$, where $\gcd(p, m) = 1$. Recall that $\mathrm{Syl}_p(G)$ is the set of Sylow p-subgroups of G (i.e. the set of subgroups of G of order p^n). Then $|\mathrm{Syl}_p(G)| \equiv 1 \pmod{p}$. In particular, Sylow's theorems 1 and 4 hold.

Proof. It suffices to prove that $|\mathrm{Syl}_p(G)| \equiv 1 \pmod{p}$: this is Sylow's theorem 4, and it implies that $\mathrm{Syl}_p(G) \neq \varnothing$, which is Sylow's theorem 1.

Let

$$X = \{S \subseteq G \mid |S| = p^n\}.$$

Then

$$|X| = \binom{p^n \cdot m}{p^n},$$

so $|X| \equiv m \pmod{p}$ by Lemma 3.4(ii). In particular,

$$p \nmid |X|. \tag{3.4}$$

Consider the action of G on X by left multiplication, defined by

$$g \cdot S := gS = \{gs \mid s \in S\}$$

for $g \in G$ and $S \in X$. (Checking that this defines a group action is an exercise on the Problem Sheets.)

Now, for each $S \in X$, there exists $T \in \mathrm{Orb}_G(S)$ such that $1_G \in T$ (if $x \in S$, then $T := x^{-1} \cdot S$ is in $\mathrm{Orb}_G(S)$ and contains 1_G, for example). In this case, $\mathrm{Orb}_G(S) = \mathrm{Orb}_G(T)$ by Corollary 2.22(i). Thus, letting $\mathrm{Orb}_G(S_1), \ldots, \mathrm{Orb}_G(S_t)$ be the distinct orbits in X, we may assume that $1_G \in S_i$ for each i.

Applying Corollary 2.22(ii), we have

$$X = \mathrm{Orb}_G(S_1) \sqcup \cdots \sqcup \mathrm{Orb}_G(S_t).$$

In particular,

$$|X| = \sum_{i=1}^{t} |\mathrm{Orb}_G(S_i)|. \tag{3.5}$$

We want to study $|X|$ modulo p, so we will now analyse the orbits $\mathrm{Orb}_G(S_i)$ with the property that p does not divide $|\mathrm{Orb}_G(S_i)|$. Note first that by the equation above and (3.4), at least one of the orbits $\mathrm{Orb}_G(S_i)$ has the property that $p \nmid |\mathrm{Orb}_G(S_i)|$. Let r be the number of orbits $\mathrm{Orb}_G(S_i)$ with the property that $p \nmid |\mathrm{Orb}_G(S_i)|$, so that $1 \leq r \leq t$. To make our notation a little easier, we will assume, without loss of generality, that $p \nmid |\mathrm{Orb}_G(S_i)|$ for $1 \leq i \leq r$, and $p \mid |\mathrm{Orb}_G(S_i)|$ for $r < i \leq t$.

We now prove two claims:

Claim 1: Fix $1 \leq i \leq r$. Then $\mathrm{Stab}_G(S_i) = S_i$. In particular, S_i is a subgroup, hence $S_i = \mathrm{Stab}_G(S_i) \in \mathrm{Syl}_p(G)$ for each $1 \leq i \leq r$.

Proof: Fix $1 \leq i \leq r$. If $g \in \mathrm{Stab}_G(S_i)$, then $gS_i = S_i$ by definition. Since $1_G \in S_i$, we have $g = g1_G \in gS_i = S_i$. Thus, $\mathrm{Stab}_G(S_i) \subseteq S_i$.

Next, by the Orbit–Stabiliser Theorem, we have $p \nmid |\mathrm{Orb}_G(S_i)| = |G : \mathrm{Stab}_G(S_i)|$. Thus, since $|G| = p^n m = |G : \mathrm{Stab}_G(S_i)||\mathrm{Stab}_G(S_i)|$, we must have that p^n divides $|\mathrm{Stab}_G(S_i)|$. It follows in particular that $p^n \leq |\mathrm{Stab}_G(S_i)|$. Thus, $|S_i| = p^n \leq |\mathrm{Stab}_G(S_i)|$. We deduce from the above that $\mathrm{Stab}_G(S_i) = S_i$, as required.

Claim 2: We have $r = |\mathrm{Syl}_p(G)|$.

Proof: Define a map $f : \{\mathrm{Orb}_G(S_i) \mid 1 \leq i \leq r\} \to \mathrm{Syl}_p(G)$ by $f(\mathrm{Orb}_G(S_i)) = S_i$. By Claim 1, $\mathrm{Stab}_G(S_i) = S_i$ is a subgroup of G of order p^n, so this map is well-defined. We will now prove that f is a bijection.

First, we prove that f is one-to-one. Let $1 \leq i, j \leq r$, and suppose that $S_i = S_j$. Then $\mathrm{Orb}_G(S_i) = \mathrm{Orb}_G(S_j)$. Thus, f is one-to-one.

Finally, let $P \in \mathrm{Syl}_p(G)$. Then $P \in X$, and $\mathrm{Stab}_G(P) = P$ by Lemma 1.23. It follows that p does not divide $m = |G : P| = |G : \mathrm{Stab}_G(P)| = |\mathrm{Orb}_G(P)|$, so $\mathrm{Orb}_G(P) = \mathrm{Orb}_G(S_i)$ for some $1 \leq i \leq r$. Hence, $P \in \mathrm{Orb}_G(S_i)$, so $P = gS_i$ for some $g \in G$. Hence $1_G \in gS_i$, so $g^{-1} \in S_i$. Since S_i is a subgroup by Claim 1, we deduce that $g \in S_i$, so $gS_i = S_i$. Thus, $P = S_i = f(\mathrm{Orb}_G(S_i))$, so f is onto.

This proves that f is a bijection, so the proof of our claim is complete.

Finally, we have $|X| \equiv m$ modulo p by Lemma 3.4(ii). Also, we have

$$|X| \equiv \sum_{i=1}^{r} |\mathrm{Orb}_G(S_i)| \pmod{p}.$$

Since $|\mathrm{Orb}_G(S_i)| = |G : \mathrm{Stab}_G(S_i)| = |G|/|S_i| = m$ by Claim 1, and $r = |\mathrm{Syl}_p(G)|$ by Claim 2, we deduce that

$$m \equiv m|\mathrm{Syl}_p(G)| \pmod{p}.$$

Since p does not divide m, m has an inverse, say u, modulo p. Multiplying both sides of the above equation by u then completes the proof. □

3.2 Proof of Sylow's theorems 2 and 3

We now prove Sylow's theorems 2 and 3. To do this, we first prove the following statement.

Proposition 3.6. Let G be a finite group and p a prime. Let $H \leq G$ and let P be a Sylow p-subgroup of G. Then there exists $g \in G$ such that $H \cap gPg^{-1}$ is a Sylow p-subgroup of H.

Proof. Consider the set $X := G/P = \{xP \mid x \in G\}$ of left cosets of P in G. Then G acts on X by left multiplication (Example 2.12(3)) and so H acts on X. Let us study the action of H on X. Take any $xP \in X$. Then

$$\begin{aligned}\mathrm{Stab}_H(xP) &= \{h \in H \mid h(xP) = xP\} = \{h \in H \mid x^{-1}hxP = P\} \\ &= \{h \in H \mid x^{-1}hx \in P\} = \{h \in H \mid h \in xPx^{-1}\} = H \cap xPx^{-1}.\end{aligned}$$

Since P is a Sylow p-subgroup of G, $|X| = |G : P|$ is coprime to p. Let $\mathrm{Orb}_H(x_1P), \ldots, \mathrm{Orb}_H(x_sP)$ denote the distinct orbits of H in its action on X. Then by Corollary 2.22(ii), we have

$$|X| = \sum_{i=1}^{s} |\mathrm{Orb}_H(x_iP)|.$$

If every such orbit has length divisible by p, then $|X|$ would be divisible by p, a contradiction. Hence, there exists $i \in \{1, \ldots, s\}$ such that $|\mathrm{Orb}_H(x_iP)|$ is not divisible by p.

Let $Q := \mathrm{Stab}_H(x_iP) = H \cap x_iPx_i^{-1}$. Then Q is a p-subgroup, since it is a subgroup of $x_iPx_i^{-1}$. Also, $|H : Q| = |\mathrm{Orb}_H(x_iP)|$ is not divisible by p. Thus, Q is a Sylow p-subgroup of H. This completes the proof. □

Corollary 3.7. Sylow's theorems 2 and 3 hold.

Proof. **Part 2:** Let H and P be Sylow p-subgroups of G. By Proposition 3.6, there exists $g \in G$ such that $H \cap gPg^{-1}$ is a Sylow p-subgroup of H. But $|H|$ is a power of p, so $\mathrm{Syl}_p(H) = \{H\}$. Thus, $H = H \cap gPg^{-1}$. Since $|H| = |gPg^{-1}|$, it follows that $H = gPg^{-1}$. This completes the proof of Sylow's theorem 2.

Part 3: Let H be a p-subgroup of G, and let P be a Sylow p-subgroup of G. By Proposition 3.6, there exists $g \in G$ such that $H \cap gPg^{-1}$ is a Sylow p-subgroup of H. But $|H|$ is a power of p, so $\mathrm{Syl}_p(H) = \{H\}$. Thus, $H = H \cap gPg^{-1} \leq gPg^{-1}$, which proves Sylow's theorem 3. □

3.3 Consequences of Sylow's theorems

The group G acts on $\mathrm{Syl}_p(G)$ by conjugation, since each conjugate of a Sylow p-subgroup is again a Sylow p-subgroup. By Sylow's theorem 2, $\mathrm{Orb}_G(P) = \mathrm{Syl}_p(G)$ for any $P \in \mathrm{Syl}_p(G)$. The stabiliser of P under this action is its normaliser:

$$\mathrm{Stab}_G(P) = N_G(P) = \{g \in G \mid gPg^{-1} = P\}.$$

Corollary 3.8. Let G be a finite group, p a prime, and $P \in \mathrm{Syl}_p(G)$. Then the following conditions hold:

(i) $|\operatorname{Syl}_p(G)| = |G : N_G(P)|$;

(ii) $|\operatorname{Syl}_p(G)|$ divides $|G|/|G|_p$;

(iii) $P \trianglelefteq G$ if and only if $|\operatorname{Syl}_p(G)| = 1$.

Proof.

(i) By the Orbit–Stabiliser Theorem, $|\operatorname{Syl}_p(G)| = |\operatorname{Orb}_G(P)| = |G : N_G(P)|$.

(ii) Since $P \leq N_G(P)$, by Lagrange's theorem, $|N_G(P)| = |P| \cdot |N_G(P) : P|$. Hence, $|\operatorname{Syl}_p(G)| = |G : N_G(P)| = \frac{|G|}{|N_G(P)|} = \frac{|G|}{|P| \cdot |N_G(P):P|}$ which divides $\frac{|G|}{|P|} = \frac{|G|}{|G|_p}$.

(iii) Finally, $P \trianglelefteq G$ if and only if $G = N_G(P)$ if and only if $|\operatorname{Syl}_p(G)| = 1$.

□

Corollary 3.9. Let G be a finite group, and let p be a prime. Define

$$F_p(G) := \{x \in G \mid x \neq 1_G, \text{ and } |x| \text{ is a power of } p\}$$

Then

(i) $F_p(G) = \bigcup_{P \in \operatorname{Syl}_p(G)} (P \setminus \{1_G\})$;

(ii) $|F_p(G)| \geq |G|_p - 1$, with equality if and only if $|\operatorname{Syl}_p(G)| = 1$ (i.e. if and only if G has a normal Sylow p-subgroup);

(iii) If $|G|_p = p$, then $|F_p(G)| = |\operatorname{Syl}_p(G)|(p-1)$.

Proof. This is an exercise on the Problem Sheets. □

3.4 Applications of Sylow's theorems

We now consider two applications of the Sylow theorems. We will sometimes call these applications *the Sylow game*.

Application 1: proving that groups of a particular order are not simple (Version 1 of the Sylow game)

(1) Let G be a group of order 20. Can G be simple?

In pursuit of a contradiction, assume that G is simple. Thus, $|\operatorname{Syl}_p(G)| > 1$ for all primes $p \mid |G|$ by Corollary 3.8(iii). Note that $|G| = 2^2 \cdot 5$. By Sylow's theorem 4, $|\operatorname{Syl}_5(G)| \equiv 1 \pmod 5$. By Corollary 3.8(ii), $|\operatorname{Syl}_5(G)| \mid \frac{20}{5} = 4$ and so $|\operatorname{Syl}_5(G)| = 1$. This contradicts our assumption above.

(2) Let G be a group of order 56. Can G be simple?

In pursuit of a contradiction, assume that G is simple. Thus, $|\operatorname{Syl}_p(G)| > 1$ for all primes $p \mid |G|$ by Corollary 3.8(iii). Note that $|G| = 2^3 \cdot 7$. By Sylow's theorem 4, $|\operatorname{Syl}_7(G)| \equiv 1 \pmod 7$. By Corollary 3.8(ii), $|\operatorname{Syl}_7(G)| \mid \frac{56}{7} = 8$. Therefore, since $|\operatorname{Syl}_7(G)| > 1$, we have $|\operatorname{Syl}_7(G)| = 8$.

It follows from Corollary 3.9(iii) that

$$|F_7(G)| = 8(7-1) = 48.$$

But then $|F_2(G)| \leq |G| - 1 - |F_7(G)| = 7 = |G|_2 - 1$. We deduce from Corollary 3.9(ii) that $|\operatorname{Syl}_2(G)| = 1$, which is a contradiction.

(3) Let G be a group of order 36. Can G be simple?

In pursuit of a contradiction, assume that G is simple. Thus, $|\operatorname{Syl}_p(G)| > 1$ for all primes $p \mid |G|$ by Corollary 3.8(iii). Note that $|G| = 2^2 \cdot 3^2$. By Sylow's theorem 4, $|\operatorname{Syl}_3(G)| \equiv 1 \pmod 3$. By Corollary 3.8(ii), $|\operatorname{Syl}_3(G)| \mid \frac{36}{9} = 4$. Therefore, since $|\operatorname{Syl}_3(G)| > 1$, we must have $|\operatorname{Syl}_3(G)| = 4$.

Let G act on $X := \operatorname{Syl}_3(G)$ by conjugation (see the discussion at the beginning of Section 3.3 above). Let $K := \ker(G, X, \cdot)$, so that $K \leq \operatorname{Stab}_G(P) = N_G(P)$ for all $P \in X$. Then $4 = |\operatorname{Syl}_3(G)| = |G : N_G(P)|$, so $K \leq N_G(P)$ implies $K \neq G$. Since G is simple, we deduce that $K = \{1_G\}$.

Finally, Proposition 2.16 implies that $36 = |G| = |G/K|$ divides $|\operatorname{Sym}(X)| = |S_4| = 24$. This contradiction shows that G cannot be simple.

To summarise: suppose that we want to prove that no group of some particular order n is simple, where n is not a prime power (the methods here do not work for groups of prime power order). We start by assuming that G is a simple group of order n. Since n is not a prime power, a Sylow p-subgroup P of G satisfies $1 \neq P \neq G$ for every prime p dividing n, so simplicity forces $|\operatorname{Syl}_p(G)| > 1$ by Corollary 3.8(iii). We then saw the following three strategies.

Strategy 1 (the basic method). Fix a prime p dividing $|G|$, and use the two facts

(1) $|\operatorname{Syl}_p(G)|$ divides $|G|/|G|_p$ (Corollary 3.8(ii));

(2) $|\operatorname{Syl}_p(G)| \equiv 1 \pmod p$ (Sylow's theorem 4)

to deduce that $|\operatorname{Syl}_p(G)| = 1$, that is, that a Sylow p-subgroup P of G is normal in G. Since n is not a prime power, $1 \neq P \neq G$, so G is not simple.

Strategy 2 (element counting). Suppose that p is a prime with $|G|_p = p$. Assume also that Strategy 1 did not work: that is, we have failed to prove that $|\operatorname{Syl}_p(G)| = 1$ using Corollary 3.8(ii) and Sylow's theorem 4. In any case, Corollary 3.8(ii) and Sylow's theorem 4 give us a list of possibilities for $|\operatorname{Syl}_p(G)|$. Let $F_p(G)$ be as in Corollary 3.9. Then by Corollary 3.9(iii), we have

$$|F_p(G)| = |\operatorname{Syl}_p(G)|(p-1).$$

Let $q \neq p$ be another prime dividing $|G|$. Then $F_q(G) \subseteq G \setminus (F_p(G) \sqcup \{1_G\})$, so $|F_q(G)| \leq |G| - |\operatorname{Syl}_p(G)|(p-1) - 1$. If this latter number is *at most* $|G|_q - 1$, then Corollary 3.9(ii) tells us that G has a normal Sylow q-subgroup, say Q. Since both p and q divide $|G|$, we have $1 \neq Q \neq G$, so G is not simple.

Strategy 3 (the homomorphism method). Assume that Strategy 1 did not work: that is, we have failed to prove that $|\operatorname{Syl}_p(G)| = 1$ for some prime p using Corollary 3.8(ii) and Sylow's theorem 4. In any case, Corollary 3.8(ii) and Sylow's theorem 4 give us a list of possibilities for $|\operatorname{Syl}_p(G)|$. Since G is simple, $|\operatorname{Syl}_p(G)| > 1$. Let $P \in \operatorname{Syl}_p(G)$, and let K be the kernel of the conjugation action of G on $\operatorname{Syl}_p(G)$. Then, as in the example of order 36 above, we have

$$K \neq G, \text{ since } |\operatorname{Syl}_p(G)| = |G : N_G(P)| > 1 \text{ and } K \leq N_G(P).$$

In other words, G acts non-trivially on $\mathrm{Syl}_p(G)$. Thus, $K = \{1_G\}$, since we are assuming that G is simple.

Now let $n = |\mathrm{Syl}_p(G)|$. Since $K = \{1_G\}$, Proposition 2.16(i) tells us that G is isomorphic to a subgroup of S_n. Thus, Lagrange's theorem implies that $|G|$ divides $n!$. So if $|G|$ does not divide $n!$, then we have a contradiction.

Application 2: proving that a particular group is simple (Version 2 of the Sylow game)

We can also use Sylow's theorems to prove that various groups *are* simple. Our example is the simplicity of A_5, which some of you saw in Groups and Rings. First, we require the following.

Corollary 3.10. Let G be a finite group, let p be a prime divisor of $|G|$, let $F_p(G)$ be as in Corollary 3.9, and let N be a normal subgroup of G.

(i) If $x \in N$, then ${}^G x \subseteq N$.

(ii) Suppose that p does not divide $|G : N|$. Then

(1) $\mathrm{Syl}_p(N) = \mathrm{Syl}_p(G)$;

(2) $F_p(G) = F_p(N)$.

Proof. Part (i) follows since $x \in N \Rightarrow {}^g x \in {}^g N = N$ for all $g \in G$.

For (ii), suppose that p does not divide $|G : N|$. Then all Sylow p-subgroups of G are contained in N (see the Problem Sheets). Also, each Sylow p-subgroup of N is contained in a Sylow p-subgroup of G, by Sylow's theorem 3. Since $p \nmid |G : N|$, we have $|N|_p = |G|_p$, so a Sylow p-subgroup of G contained in N is a Sylow p-subgroup of N, and conversely. Hence, $\mathrm{Syl}_p(G) = \mathrm{Syl}_p(N)$, so (1) holds.

Finally, part (2) follows immediately from part (1) and Corollary 3.9(i). □

Proposition 3.11. Let $G = A_5$. Then G is a simple group.

Proof. Assume the contrary. Then there exists $N \trianglelefteq G$ with $\{1_G\} \neq N \neq G$. By Lagrange's theorem, $|N|$ divides $|G| = 60$. Hence, the prime divisors of $|N|$ are among 2, 3 and 5.

Note first three facts about G:

(1) There are 24 elements of order 5, since the elements of order 5 in G are precisely the 5-cycles, and there are $4! = 24$ of these.

(2) There are 20 elements of order 3, since the elements of order 3 in G are precisely the 3-cycles, and there are $2\binom{5}{3} = 20$ of these.

(3) The only elements of order 2 in G are those of the form $(a, b)(c, d)$, for a, b, c, d distinct elements of $\{1, 2, 3, 4, 5\}$. There are 15 of them, and they are all conjugate in G. Indeed, by Theorem 2.4(iii) any two are conjugate by some $g \in S_5$. If g is odd, replace it by $g(a, b)$, which is even and conjugates $(a, b)(c, d)$ to the same element, since (a, b) commutes with $(a, b)(c, d)$.

We now distinguish our arguments into two cases.

Case 1: 3 divides $|N|$ or 5 divides $|N|$. Let $p \in \{3, 5\}$, and suppose that $p \mid |N|$. Then $p \nmid |G : N|$, so we can use Corollary 3.10(ii): We have $F_p(G) = F_p(N)$.

Suppose that $p = 5$. Then $|F_p(N)| = 24$, so $|N| \geq 25$. Since $|N|$ divides 60 and is less than 60, it follows that $|N| = 30$.

Suppose that $p = 3$. Then $|F_p(N)| = 20$, so $|N| \geq 21$. Since $|N|$ divides 60 and is less than 60, it follows again that $|N| = 30$.

We have therefore shown that if either 3 or 5 divides $|N|$, then both 3 and 5 divide $|N|$, $|N| = 30$, $|F_3(N)| = 20$ and $|F_5(N)| = 24$. But then $30 = |N| \geq 1 + |F_3(N)| + |F_5(N)| = 45$, a contradiction.

Case 2: Neither 3 **nor** 5 **divides** $|N|$. Then $|N|$ divides 4 by Lagrange's theorem. By Cauchy's theorem, we can choose $x \in N$ of order 2. We then use Corollary 3.10(i) and Fact (3) above: $4 \geq |N| \geq |^G x| = 15$, another contradiction. This completes the proof. □

For more information about the thinking and strategies behind the *Sylow game*, see Appendix B.

3.5 Simplicity of A_n, $n \geq 5$

In this section we are going to prove the following result that was first proved by C. Jordan in the 1870s.

Theorem 3.12. For $n \geq 5$, the group A_n is simple.

We proved the result for $n = 5$ in Proposition 3.11, and we prove Theorem 3.12 by induction on n. First, we need some facts about A_n.

The group S_n acts on $X_n := \{1, 2, \ldots, n\}$ by $f \cdot i := f(i)$, and hence so does its subgroup A_n. We call this the *natural action*.

Lemma 3.13. The following conditions hold:

(i) Let $n \geq 3$, and let X be the set of 3-cycles in S_n. Note that $X \subseteq A_n$ by definition of A_n. We have $A_n = \langle X \rangle$.

(ii) Suppose that $n \geq 5$. Then any two 3-cycles are conjugate in A_n.

Proof. (i) Recall from the Problem Sheets that every element of A_n is a product of an even number of transpositions (i.e. cycles of length 2). Thus, to prove part (i) of the lemma, it is enough to show that a product of two transpositions can be written as a product of 3-cycles.

Consider a product $(a, b)(c, d)$ of two transpositions, with $a, b, c, d \in \{1, \ldots, n\}$ and $a \neq b$, $c \neq d$. We distinguish three possible cases according to the size of $\{a, b\} \cap \{c, d\}$:

(1) Suppose that $\{a, b\} \cap \{c, d\}$ is empty. Then by definition, (a, b) and (c, d) are disjoint cycles. By direct calculation, we have

$$(a, b)(c, d) = (a, b, c)(b, c, d),$$

which is a product of 3-cycles.

(2) Suppose that $|\{a, b\} \cap \{c, d\}| = 1$. Without loss of generality, we may assume that $a = c$. Then $(a, b)(c, d) = (a, b)(a, d) = (a, d, b)$, which is a 3-cycle.

(3) Suppose finally that $|\{a, b\} \cap \{c, d\}| = 2$, that is, $\{a, b\} = \{c, d\}$. Then $(a, b)(c, d) = (a, b)^2 = 1_{A_n} = (a, b, e)^3$ for any $e \in \{1, \ldots, n\} \setminus \{a, b\}$. This completes the proof of (i).

(ii) By Theorem 2.4(iii), ${}^g x$ is a 3-cycle for all $g \in A_n$ and all 3-cycles x. Thus, A_n acts on the set X by conjugation. That is, the map $\cdot : A_n \times X \to X$ given by $g \cdot x := {}^g x$ for $g \in A_n$, $x \in X$, is a group action (see also Example 2.12(2)).

Thus, the statement of (ii) is equivalent to the statement that A_n has precisely one orbit in its action on X. Equivalently,

$$X = \mathrm{Orb}_{A_n}(x) \tag{3.6}$$

where x is the 3-cycle $x := (1,2,3)$.

We now prove (3.6). Let $\sigma = (a,b,c) \in A_n$ be a 3-cycle. Again, we distinguish four cases:

(1) Suppose that $\{a,b,c\} \cap \{1,2,3\}$ is empty. Then $n \geq 6$, and $(1,a)$, $(2,b)$, $(3,c)$ are transpositions. Let $g := (1,2)(1,a)(2,b)(3,c)$. Then $g \in A_n$, and by Theorem 2.4(iii) we have

$${}^g(1,2,3) = {}^{(1,2)}(a,b,c) = (a,b,c)$$

with the last equality above following since $\{a,b,c\}$ and $\{1,2\}$ are disjoint (and disjoint cycles commute).

(2) Suppose that $|\{a,b,c\} \cap \{1,2,3\}| = 1$. Without loss of generality, we may assume that $a = 1$. Then by direct calculation using Theorem 2.4(iii), we have

$${}^g(1,2,3) = (a,b,c)$$

where $g := (2,b)(3,c) \in A_n$.

(3) Suppose that $|\{a,b,c\} \cap \{1,2,3\}| = 2$. Assume first that $a = 1$ and $b = 2$. Let $d \in \{1,\ldots,n\} \setminus \{1,2,3,c\}$. Then as in case 1 above we have

$${}^g(1,2,3) = (a,b,c)$$

where $g := (3,d)(3,c) \in A_n$. The remaining arrangements are handled similarly. For example, if $\sigma = (1,3,c)$, then ${}^g(1,2,3) = (1,3,c)$ where $g := (2,3,c) \in A_n$.

(4) Suppose finally that $\{a,b,c\} = \{1,2,3\}$. If $(a,b,c) = (1,2,3)$, then ${}^{1_{A_n}}(1,2,3) = (a,b,c)$, as required. If $(a,b,c) = (1,3,2)$, then ${}^g(1,2,3) = (a,b,c)$, where $g := (2,3)(4,5) \in A_n$.

The result follows. □

Our proof of Theorem 3.12 below follows the proof given by D. Dummit and R. Foote in their book *Abstract Algebra*.

Lemma 3.14. For $n \geq 5$, any $1 \neq \sigma \in A_n$ has a conjugate $\sigma' = {}^\tau\sigma$, with $\tau \in A_n$, such that $\sigma \neq \sigma'$ and $\sigma(i) = \sigma'(i)$ for some $i \in \{1,2,\ldots,n\}$.

Proof. Take any $1 \neq \sigma \in A_n$. Let r be the longest length of a disjoint cycle in σ. Relabelling if necessary, we may write

$$\sigma = (1,2,\ldots,r)\pi$$

where $(1,2,\ldots,r)$ and π are disjoint permutations.

If $r \geq 3$, let $\tau = (3,4,5)$ and $\sigma' = \tau\sigma\tau^{-1}$. Then $\sigma(1) = 2$ and $\sigma'(1) = 2$, while $\sigma(2) = 3$ and $\sigma'(2) = 4$. Hence, $\sigma \neq \sigma'$ while $\sigma(1) = \sigma'(1)$.

If $r = 2$, then σ is a product of disjoint transpositions. If there are at least 3 disjoint transpositions involved, then $n \geq 6$ and after relabelling we can write $\sigma = (1,2)(3,4)(5,6)\cdots$. Let $\tau =$

$(1,2)(3,5)$ and $\sigma' = \tau\sigma\tau^{-1}$. Then $\sigma(1) = 2 = \sigma'(1)$ while $\sigma(3) = 4$ and $\sigma'(3) = 6$. Hence, again $\sigma \neq \sigma'$ while $\sigma(1) = \sigma'(1)$.

Finally, suppose that $r = 2$ and σ is a product of two disjoint transpositions. Relabelling if necessary, we may write $\sigma = (1,2)(3,4)$. This time, let $\tau = (1,3,2)$ and $\sigma' = \tau\sigma\tau^{-1}$. Then $\sigma' = (1,3)(2,4)$, so $\sigma \neq \sigma'$ and $\sigma(5) = 5 = \sigma'(5)$. □

Proof of Theorem 3.12. By Proposition 3.11, A_5 is simple. Thus we may suppose that $n \geq 6$. Recall that A_n acts on X_n in the natural way. For each $i \in X_n$, let $H_i := \mathrm{Stab}_{A_n}(i) \cong A_{n-1}$. By induction, H_i is a simple group. Observe that H_i contains a 3-cycle (based on three points of X_n other than i).

Assume there exists a non-trivial proper normal subgroup N of A_n. Take any $1 \neq \sigma \in N$. By Lemma 3.14, there exists $\sigma' \in A_n$ such that $\sigma \neq \sigma'$, σ' is conjugate to σ in A_n, and $\sigma(i) = \sigma'(i)$ for some $i \in X_n$. Since $N \trianglelefteq A_n$, $\sigma' \in N$. Hence, $\sigma^{-1}\sigma' \in N$, $\sigma^{-1}\sigma' \neq 1_{A_n}$ and $\sigma^{-1}\sigma'(i) = i$. Thus $\sigma^{-1}\sigma' \in H_i$ and so $N \cap H_i \neq \{1_{A_n}\}$. Now, $N \trianglelefteq A_n$, and so $N \cap H_i \trianglelefteq H_i$ by Theorem 1.43(ii). But H_i is a simple group, and so $N \cap H_i = H_i$. Therefore $H_i \leq N$. But H_i contains a 3-cycle, and so N contains a 3-cycle. As $N \trianglelefteq A_n$, by Lemma 3.13(ii), N contains all 3-cycles of A_n. The result now follows by Lemma 3.13(i). □

3.6 A Sylow p-subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$, and an application

In this section, we find a Sylow p-subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$, and use it to give Serre's proof of Sylow's theorem 1.

We begin with some preparation.

The group $\mathrm{GL}_n(\mathbb{F}_p)$

Let p be a prime, $n \in \mathbb{N}$ and $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$, so that $\mathbb{F}_p$ is a finite field of order p.

Recall from Chapter 1 that $\mathrm{GL}_n(\mathbb{F}_p)$ is the group of invertible $n \times n$ matrices over $\mathbb{F}_p$, under matrix multiplication.

Let $V = \mathbb{F}_p^n$ be the n-dimensional column vector space over $\mathbb{F}_p$, and let

$$G := \{T : V \to V \mid T \text{ an invertible linear map}\}.$$

Then G is a group under composition of functions, and we saw in MA106 that G is isomorphic to $\mathrm{GL}_n(\mathbb{F}_p)$. Indeed, for $A \in \mathrm{GL}_n(\mathbb{F}_p)$, define $\theta(A) : V \to V$ by $\theta(A)(v) := Av$. Then it follows from the definition of matrix multiplication that θ is a homomorphism. If $\{e_1, \ldots, e_n\}$ is the standard basis for V, and $T \in G$, then define A_T to be the matrix of T with respect to $\{e_1, \ldots, e_n\}$. Then the map $\varphi : G \to \mathrm{GL}_n(\mathbb{F}_p)$ defined by $\varphi(T) := A_T$ is both a left and right inverse to θ.

Since $\mathbb{F}_p$ is finite, so is $\mathrm{GL}_n(\mathbb{F}_p)$. The next lemma uses a group action to find its order.

Lemma 3.15. The order of $\mathrm{GL}_n(\mathbb{F}_p)$ is $p^{\binom{n}{2}} \prod_{i=1}^{n}(p^i - 1)$.

Proof. Let V and G be as above. Since $\mathrm{GL}_n(\mathbb{F}_p) \cong G$, it suffices to prove that $|G| = p^{\binom{n}{2}} \prod_{i=1}^{n}(p^i - 1)$.

Let $E = (e_1, \ldots, e_n)$ be an ordered basis of V. Then for any $T \in G$, we have that $(T(e_1), \ldots, T(e_n))$ is again an ordered basis of V, since T is invertible. Let

$$\mathcal{B} = \{E \mid E \text{ is an ordered basis of } V\}.$$

We may therefore define a map $\cdot : G \times \mathcal{B} \to \mathcal{B}$ by $T \cdot E = (T(e_1), \ldots, T(e_n))$ for $T \in G$ and $E = (e_1, \ldots, e_n) \in \mathcal{B}$. One checks that this is an action of G on $\mathcal{B}$.

Take any two ordered bases $E = (e_1, \ldots, e_n)$ and $F = (f_1, \ldots, f_n)$ of V. From MA106, there exists a unique linear map $T : V \to V$ with $T(e_i) = f_i$ for $1 \leq i \leq n$. Since $\{f_1, \ldots, f_n\}$ is a basis of V, this map is invertible, so $T \in G$ and $T \cdot E = F$. Hence,

$$\mathrm{Orb}_G(E) = \mathcal{B}.$$

Moreover, by uniqueness, the identity map Id_V is the only linear map T with

$$T \cdot (e_1, \ldots, e_n) = (e_1, \ldots, e_n).$$

That is,

$$\mathrm{Stab}_G(E) = \{\mathrm{Id}_V\}.$$

By the Orbit–Stabiliser Theorem, it follows that

$$|G| = |\mathrm{Orb}_G(E)| \cdot |\mathrm{Stab}_G(E)| = |\mathcal{B}| \cdot 1 = |\mathcal{B}|.$$

To count the ordered bases $(e_1, \ldots, e_n)$ of V, note that we can choose any of the $|V| - 1 = p^n - 1$ non-zero elements of V as e_1, then any of the $|V| - p = p^n - p$ elements of $V \setminus \{\lambda e_1 \mid \lambda \in \mathbb{F}_p\}$ as e_2, and so on. Continuing in this way, we obtain

$$\begin{aligned} |\mathcal{B}| &= (p^n - 1)(p^n - p) \ldots (p^n - p^{n-1}) \\ &= p^{\frac{n(n-1)}{2}}(p^n - 1)(p^{n-1} - 1) \ldots (p - 1) = p^{\frac{n(n-1)}{2}} \prod_{i=1}^{n} (p^i - 1), \end{aligned}$$

and so

$$|\mathrm{GL}_n(\mathbb{F}_p)| = |G| = p^{\frac{n(n-1)}{2}} \prod_{i=1}^{n} (p^i - 1) = p^{\binom{n}{2}} \prod_{i=1}^{n} (p^i - 1)$$

as required. □

Next, consider the set

$$U_n(\mathbb{F}_p) := \{A = (a_{ij})_{n \times n} \mid a_{ij} \in \mathbb{F}_p \text{ for all } i, j,\ a_{ij} = 0 \text{ for } j < i, \text{ and } a_{ii} = 1 \text{ for all } i\}$$

of upper unitriangular matrices over $\mathbb{F}_p$. It is an exercise on the Problem Sheets to show that the upper triangular matrices with non-zero diagonal entries form a subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$. The diagonal entries of a product of two such matrices are the products of the corresponding diagonal entries, and the diagonal entries of the inverse of such a matrix are the inverses of its diagonal entries. Hence $U_n(\mathbb{F}_p)$ is also a subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$. Counting the possible entries above the diagonal, we see that $|U_n(\mathbb{F}_p)| = p^{\binom{n}{2}}$. Since $p^i - 1 \equiv -1 \pmod{p}$ for all $i \geq 1$, the prime p does not divide $\prod_{i=1}^{n}(p^i - 1)$, so $|\mathrm{GL}_n(\mathbb{F}_p)|_p = p^{\binom{n}{2}}$. Therefore, $U_n(\mathbb{F}_p) \in \mathrm{Syl}_p(\mathrm{GL}_n(\mathbb{F}_p))$. We have proved the following statement.

Lemma 3.16. The group $U_n(\mathbb{F}_p)$ is a Sylow p-subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$.

An application: Serre's proof of Sylow's theorem 1

We now use our work on $\mathrm{GL}_n(\mathbb{F}_p)$ to give Serre's proof of Sylow's theorem 1. We first need the following lemma.

Lemma 3.17. The group $\mathrm{GL}_n(\mathbb{F}_p)$ contains a subgroup isomorphic to S_n.

Proof. For $\sigma \in S_n$ let I_σ be the $n \times n$ matrix over $\mathbb{F}_p$ with

$$a_{ij} = \begin{cases} 1, & \text{if } i = \sigma(j), \\ 0, & \text{otherwise.} \end{cases}$$

Since I_σ is a permutation matrix, $\det(I_\sigma) = \pm 1 \neq 0$, and so $I_\sigma \in \mathrm{GL}_n(\mathbb{F}_p)$. In particular, we have a map

$$f : S_n \to \mathrm{GL}_n(\mathbb{F}_p)$$

defined by $f(\sigma) = I_\sigma$. Since $I_{\sigma_1\sigma_2} = I_{\sigma_1} I_{\sigma_2}$ for $\sigma_1, \sigma_2 \in S_n$, we have $f(\sigma_1\sigma_2) = f(\sigma_1)f(\sigma_2)$, and so f is a group homomorphism. Now $\ker(f) = \{\sigma \in S_n \mid I_\sigma = I_n\} = \{1_{S_n}\}$. Therefore, by the First Isomorphism Theorem, $S_n \cong S_n/\{1_{S_n}\} \cong \mathrm{Im}(f) \leq \mathrm{GL}_n(\mathbb{F}_p)$. □

Serre's proof of Sylow's theorem 1. Let G be a finite group of order $p^n \cdot m$ where p is a prime, $n \in \mathbb{N}$ and $\gcd(p, m) = 1$. Let $N = p^n \cdot m$. Then by Theorem 2.23, G is isomorphic to a subgroup of S_N. Now by Lemma 3.17, there exists a subgroup of $\mathrm{GL}_N(\mathbb{F}_p)$ isomorphic to S_N. Therefore, there exists a subgroup G_0 of $\mathrm{GL}_N(\mathbb{F}_p)$ isomorphic to G. By Lemma 3.16, $\mathrm{GL}_N(\mathbb{F}_p)$ has a Sylow p-subgroup $U_N(\mathbb{F}_p)$. Therefore, by Proposition 3.6, G_0 has a Sylow p-subgroup, and hence so does G. □

Chapter 4

Classifying groups of small order

In this chapter, we are going to classify the groups of order up to 19 (apart from those of order 16). In order to do so, we will need some results on special types of groups. We will begin with groups called *semidirect products*. As the name suggests, they are a generalisation of direct products.

4.1 Semidirect products

We start by defining the direct product of groups.

Definition 4.1. Let H and K be groups, and define a binary operation $\cdot : (H \times K) \times (H \times K) \to H \times K$ by $(h_1, k_1) \cdot (h_2, k_2) := (h_1h_2, k_1k_2)$. Then $(H \times K, \cdot)$ is a group, called the *direct product* of H and K. We denote it by $H \times K$.

One can extend this definition to the direct product of finitely many groups $G_1, \ldots, G_n$: we define multiplication in $G_1 \times \cdots \times G_n$ by $(g_1, \ldots, g_n) \cdot (g_1', \ldots, g_n') = (g_1g_1', \ldots, g_ng_n')$.

One can check that $G_1 \times \cdots \times G_n$ with the operation defined above is indeed a group, with identity $(1_{G_1}, \ldots, 1_{G_n})$ and with $(g_1, \ldots, g_n)^{-1} = (g_1^{-1}, \ldots, g_n^{-1})$.

The following lemma gives a useful way to tell whether a group is isomorphic to the direct product of two of its subgroups.

Lemma 4.2. Let G have normal subgroups $H, K \trianglelefteq G$ such that $G = HK$ and $H \cap K = \{1_G\}$. Then

(i) x commutes with y for all $x \in H$, $y \in K$ (so if H and K are abelian, then G is abelian);

(ii) $G \cong H \times K$.

Proof. Take $h \in H$, $k \in K$. Consider $g = hkh^{-1}k^{-1}$. Since both H and K are normal in G we have

$$g = hkh^{-1}k^{-1} = (hkh^{-1})k^{-1} = h(kh^{-1}k^{-1}) \in H \cap K = \{1_G\},$$

so $g = 1_G$. Thus $hk = kh$, and all elements of H commute with all elements of K. This proves (i).

Now define $\varphi : H \times K \to G$ by $\varphi((h, k)) = hk$. Then

$$\varphi((h_1, k_1)(h_2, k_2)) = \varphi((h_1h_2, k_1k_2)) = h_1h_2k_1k_2 = h_1k_1h_2k_2 = \varphi((h_1, k_1))\varphi((h_2, k_2)),$$

so φ is a homomorphism.

Notice that φ is surjective because $G = HK$. Suppose that $\varphi((h,k)) = 1_G$. Then $hk = 1_G$, so $h = k^{-1} \in H \cap K = \{1_G\}$. Thus $h = k = 1_G$. So φ is injective (see the Problem Sheets). Hence φ is an isomorphism, and $H \times K \cong G$. □

In the situation of Lemma 4.2, we write $G = H \times K$ rather than $G \cong H \times K$.

We would now like to generalise the direct product of two finite groups. First, we need a definition.

Definition 4.3. Let G be a group. An *automorphism* of G is an isomorphism $\phi : G \to G$. The set $\{\phi \mid \phi \text{ is an automorphism of } G\}$ is a group under composition of functions. It is called the *automorphism group* of G, and is denoted by $\mathrm{Aut}(G)$.

We can now define semidirect products, which generalise direct products.

Definition 4.4. Let H and K be groups, and let $\phi : H \to \mathrm{Aut}(K)$ be a homomorphism. Write the image of $h \in H$ under ϕ as ϕ_h, and define a binary operation $\cdot : (H \times K) \times (H \times K) \to H \times K$ by

$$(h_1, k_1) \cdot (h_2, k_2) := (h_1 h_2, \phi_{h_2^{-1}}(k_1) k_2).$$

Then $(H \times K, \cdot)$ is a group called the *semidirect product of H and K with respect to ϕ*. We denote it by $H \ltimes_\phi K$.

It can be tricky to imagine maps $H \to \mathrm{Aut}(K)$ for given groups H and K, so we now look at some examples.

Example 4.5 (Three examples of homomorphisms $H \to \mathrm{Aut}(K)$)**.**

(1) **The trivial homomorphism.** Let H and K be any groups. Then the map $\phi : H \to \mathrm{Aut}(K)$ given by $\phi(h) := \phi_h := \mathrm{Id}_K$ for all $h \in H$ (i.e. $\phi_h(k) := k$ for all $h \in H, k \in K$) is a homomorphism. The resulting semidirect product is isomorphic to the direct product:

$$H \ltimes_\phi K \cong H \times K.$$

(2) **The inversion homomorphism.** Let $H = \langle x \rangle \cong C_2$, and let K be any *abelian* group. Then the map $\phi : H \to \mathrm{Aut}(K)$ given by $\phi(1_H) := \mathrm{Id}_K$ and $\phi(x) := \phi_x : K \to K$ with $\phi_x(k) := k^{-1}$ for all $k \in K$, is a homomorphism. If $K \cong C_n$, then the resulting semidirect product is isomorphic to the dihedral group:

$$H \ltimes_\phi K \cong D_{2n}.$$

That is, $C_2 \ltimes_\phi C_n \cong D_{2n}$, where $\phi : C_2 \to \mathrm{Aut}(C_n)$ is the inversion homomorphism (see Example 4.7 for a proof).

(3) **The conjugation homomorphism.** Let H and K be subgroups of a group G, with K *normal in* G. Then the map $\phi : H \to \mathrm{Aut}(K)$ given by $\phi(h) := \phi_h : K \to K$, with $\phi_h(k) := hkh^{-1}$ for all $h \in H, k \in K$, is a homomorphism.

The following result gives us a useful way to tell if a group is isomorphic to the semidirect product of two subgroups, and relies on the conjugation homomorphism above.

Lemma 4.6. Let G be a group, and let K and H be subgroups of G, with $K \trianglelefteq G$. Define

$$\phi : H \to \mathrm{Aut}(K)$$

by $\phi_h(k) := hkh^{-1}$ for $h \in H, k \in K$. Assume that $G = HK$ and that $K \cap H = \{1_G\}$. Then

$$G \cong H \ltimes_\phi K.$$

Proof. Define $f : H \ltimes_\phi K \to G$ by $f((h,k)) := hk$. Note that

$$(h_1k_1)(h_2k_2) = h_1h_2{}^{h_2^{-1}}(k_1)k_2,$$

for $h_1, h_2 \in H$, $k_1, k_2 \in K$. We therefore have

$$\begin{aligned} f((h_1,k_1)(h_2,k_2)) &= f((h_1h_2, \phi_{h_2^{-1}}(k_1)k_2)) \\ &= h_1h_2\phi_{h_2^{-1}}(k_1)k_2 = h_1h_2(h_2^{-1}k_1h_2)k_2 = h_1k_1h_2k_2 = f((h_1,k_1))f((h_2,k_2)). \end{aligned}$$

by definition of multiplication in $H \ltimes_\phi K$. Since $G = HK$, the map f is surjective, so it remains to prove that f is injective. But if $f((h,k)) = 1_G$ for $h \in H$, $k \in K$, then $1_G = hk$, so $h = k^{-1} \in H \cap K = \{1_G\}$. That is, $h = k = 1_G$. Thus, $\ker(f) = \{1_G\}$, i.e. f is injective (see the Problem Sheets). □

Example 4.7 (Dihedral groups as semidirect products)**.** Let $n \in \mathbb{N}$ with $n \geq 3$, and consider the dihedral group $G := D_{2n} = \langle \sigma, \tau \rangle$ from Definition 2.5. Here σ and τ are the elements

$$\begin{aligned} \sigma &:= (1, \ldots, n) \text{ and} \\ \tau &:= \prod_{i=1}^{\lfloor n/2 \rfloor} (i, n-i+1) \end{aligned}$$

of S_n. Let $K := \langle \sigma \rangle$ and $H := \langle \tau \rangle$, and recall that ${}^\tau\sigma = \sigma^{-1}$. It follows that ${}^\tau k = k^{-1}$ for all $k \in K$.

Since $|\tau| = 2$, $|\sigma| = n$, and $D_{2n} = K \sqcup \tau K$, we have $G = HK$ and $H \cap K = \{1_G\}$. Hence, $G \cong H \ltimes_\phi K$ by Lemma 4.6, where $\phi : H = \{1_G, \tau\} \to \mathrm{Aut}(K)$ is given by $\phi_{1_G}(k) := k$ and $\phi_\tau(k) := k^{-1}$, for $k \in K$.

Lemma 4.8. Let K be a group, and let A be a non-empty subset of K with $K = \langle A \rangle$.

(i) The group K is abelian if and only if $ab = ba$ for all $a, b \in A$.

(ii) Suppose that K is abelian, and that $f : K \to K$ is an automorphism satisfying $f(a) = a^{-1}$ for all $a \in A$. Then $f(k) = k^{-1}$ for all $k \in K$.

Proof. (i) If K is abelian, then certainly $ab = ba$ for all $a, b \in A$. Conversely, if the elements of A commute with each other, then so do the elements of $A \cup A^{-1}$. Every element of K is a product of elements of $A \cup A^{-1}$, so any two elements of K commute.

(ii) Since K is abelian, the map $\iota : K \to K$ given by $\iota(k) = k^{-1}$ is a homomorphism. The set $\{k \in K \mid f(k) = \iota(k)\}$ contains 1_K and is closed under products and inverses, since f and ι are homomorphisms. So it is a subgroup of K containing A, and hence it is equal to K. □

Dihedral groups appear often in our classification of groups of small order, so we record the following lemma, which recognises them.

Lemma 4.9. Let G be a nonabelian finite group, and suppose that

(1) G has a cyclic subgroup K of order $n := |G|/2$;

(2) $G \setminus K$ contains an element of order 2;

(3) If $i \in \{0, \ldots, n-1\}$ satisfies $i^2 \equiv 1 \pmod{n}$, then $i \equiv \pm 1$ modulo n.

Then $G \cong D_{2n}$.

Proof. Let G be a group satisfying (1), (2), and (3) in the statement of the lemma, with $K \leq G$ cyclic of order $n := |G|/2$. Then $|G : K| = 2$, so $K \trianglelefteq G$.

Let x be an element of $G \setminus K$ of order 2, and let $H = \langle x \rangle$. Then $H \cap K = \{1_G\}$, so $|HK| = |H||K|$ by Proposition 2.27. It follows that $|HK| = |G|$, so $HK = G$. Let $\phi : H \to \mathrm{Aut}(K)$ be the homomorphism defined by $\phi_h(k) := hkh^{-1}$, for $h \in H$, $k \in K$. Thus, we have

$$G \cong H \ltimes_\phi K$$

by Lemma 4.6.

By Example 4.7, we are therefore done if we can prove that $\phi_x(k) = k^{-1}$ for all $k \in K$. Write $K = \langle y \rangle$, where $|y| = n$. Then $xyx^{-1} = y^i$, for some $0 \leq i \leq n-1$ so $y = x^2yx^{-2} = xy^ix^{-1} = (xyx^{-1})^i = y^{i^2}$. It follows that $y^{i^2-1} = 1_G$, so $n = |y|$ divides $i^2 - 1$ by Lemma 1.15(i). In other words, $i^2 - 1 \equiv 0 \pmod{n}$. It follows from hypothesis (3) that $i \equiv \pm 1 \pmod{n}$. Since $0 \leq i \leq n-1$, we deduce that $i \in \{1, n-1\}$.

If $i = 1$, then we would have $xyx^{-1} = y^1 = y$, so $xy = yx$. Then G is abelian, a contradiction. So we must have $xyx^{-1} = y^{n-1} = y^{-1}$. Thus, the homomorphism ϕ is given by $\phi_1(k) = k$, and $\phi_x(k) = k^{-1}$, for all $k \in K$. This proves that $G \cong H \ltimes_\phi K \cong D_{2n}$, as required. □

Example 4.10. The following are some examples of positive integers n which satisfy Hypothesis (3) in Lemma 4.9.

(1) Let $n = 6$. Then $2^2, 3^2, 4^2 \equiv 4, 3, 4 \pmod{n}$ respectively, so $i^2 \equiv 1 \pmod{n}$ if and only if $i \equiv \pm 1 \pmod{6}$.

(2) Let $n = p$, where p is prime. Then since $\mathbb{Z}/p\mathbb{Z} = \{0, \ldots, p-1\}$ is a field, it has no zero divisors. That is, $i^2 - 1 = 0$ in $\mathbb{Z}/p\mathbb{Z}$ if and only if $(i-1)(i+1) \equiv 0 \pmod{p}$, which is true if and only if $i \equiv \pm 1 \pmod{p}$.

(3) Let $n = p^2$, where p is prime. Then $\mathbb{Z}/p^2\mathbb{Z}$ is not a field, so we cannot use the argument above. If $p = 2$, then since $2^2 \not\equiv 1 \pmod{4}$, we have that $i^2 \equiv 1 \pmod{4}$ if and only if $i \equiv \pm 1 \pmod{4}$.

Assume now that p is odd, and let i be an element of $\{0, 1, \ldots, p^2 - 1\}$ such that $i^2 \equiv 1 \pmod{p^2}$. Then p^2 divides $(i-1)(i+1)$. Also, since p is odd, we see that either p does not divide $i-1$ or p does not divide $i+1$ (if it divided both, then it would divide $i+1-(i-1) = 2$). Thus, since p^2 divides $i^2 - 1$, we must have that p^2 divides $i-1$ or p^2 divides $i+1$. Hence, since $0 \leq i \leq p^2 - 1$, the only possibilities are $i = 1$ or $i = p^2 - 1$, i.e. $i \equiv \pm 1 \pmod{p^2}$.

4.2 Semidirect products of abelian and cyclic groups, and a lemma of Fitting

In this section, we consider the following specific case: Let G be a finite group with $|G|/2$ odd, and suppose that G has an abelian normal subgroup K of order $|G|/2$. Since $|G|/2$ is odd, we have $|G|_2 = 2$. Thus, if we fix $H \in \mathrm{Syl}_2(G)$, then $|H| = 2$, so we may write $H = \langle x \rangle$.

The following is a powerful result of Fitting.

Lemma 4.11 (Fitting's lemma)**.** Let G be a finite group with $|G|/2$ odd, let K be an abelian normal subgroup of G of order $|G|/2$, and let $H = \langle x \rangle \in \mathrm{Syl}_2(G)$. For $v \in K$, define $[v, x] := vxv^{-1}x^{-1}$ and $[K, x] := \langle [v, x] \mid v \in K \rangle$. Then

(i) ${}^x a = a^{-1}$ for all $a \in [K, x]$;

(ii) $K = C_K(x) \times [K, x]$;

(iii) $G \cong (H \ltimes_\theta [K, x]) \times C_K(x)$, where $\theta : H \to \mathrm{Aut}([K, x])$ is the inversion homomorphism.

Proof. (i) Note that since K is normal in G, the group $[K, x]$ is indeed contained in K. We first show that ${}^x[v, x] = [v, x]^{-1}$ for all $v \in K$. To see this, fix $v \in K$. Then

$$ {}^x[v, x] = x(vxv^{-1}x^{-1})x = xvxv^{-1} = [v, x]^{-1} $$

with the last equality above following since $x = x^{-1}$. Hence conjugation by x maps $[K, x]$ to itself, so it restricts to an automorphism of $[K, x]$ which inverts every generator $[v, x]$. Since $[K, x]$ is abelian, Lemma 4.8(ii) gives ${}^x a = a^{-1}$ for all $a \in [K, x]$.

(ii) Define $f : K \to [K, x]$ by $f(k) := [k, x]$, for $k \in K$. Then

$$ \begin{aligned} f(k_1k_2) &= k_1k_2xk_2^{-1}k_1^{-1}x^{-1} \\ &= k_1k_2xk_2^{-1}x^{-1}xk_1^{-1}x^{-1} \\ &= k_1k_2({}^xk_2^{-1})({}^xk_1^{-1}) \\ &= k_1({}^xk_1^{-1})k_2({}^xk_2^{-1}) && \text{since } K \text{ is abelian} \\ &= [k_1, x][k_2, x] = f(k_1)f(k_2). \end{aligned} $$

Thus, f is a homomorphism. Since $[k, x] = 1_G$ if and only if $kx = xk$, we have $\ker(f) = C_K(x)$. Also, since $\mathrm{Im}(f)$ is a subgroup of $[K, x]$ and contains a generating set for $[K, x]$, we have $\mathrm{Im}(f) = [K, x]$. Thus, by the First Isomorphism Theorem, we have

$$ |K| = |\ker(f)||\mathrm{Im}(f)| = |C_K(x)||[K, x]|. $$

Finally, since x inverts every element of $[K, x]$, and K has odd order (so the only element k of K with $k = k^{-1}$ is 1_G), we have $[K, x] \cap C_K(x) = \{1_G\}$. It follows from Proposition 2.27 and the above that

$$ |K| = |C_K(x)||[K, x]| = |C_K(x)[K, x]|. $$

Thus, $K = C_K(x)[K, x]$. Since K is abelian (so every subgroup is normal), we deduce from Lemma 4.2 that $K = C_K(x) \times [K, x]$, as required.

(iii) This is an exercise on the Problem Sheets. □

4.3 Abelian groups

In this short section, we state (without proof) the Fundamental Theorem of Finite Abelian Groups, which you have seen in an earlier module.

Theorem 4.12 (The Fundamental Theorem of Finite Abelian Groups)**.** Let G be a finite abelian group. Then there exist divisors $d_1, \ldots, d_t$ of $|G|$ such that $d_1 \mid d_2 \mid \cdots \mid d_t$, and

$$G \cong (\mathbb{Z}/d_1\mathbb{Z}) \times (\mathbb{Z}/d_2\mathbb{Z}) \times \cdots \times (\mathbb{Z}/d_t\mathbb{Z}).$$

4.4 Groups of order p, p^2, or $2p$, for p prime

Lemma 4.13. If $|G| = p$ is prime, then $G \cong C_p$.

Proof. Let $1 \neq g \in G$. By Lagrange's theorem, $|g|$ divides p. Since $g \neq 1$, $|g| = p$ and so $G = \langle g \rangle$. □

Lemma 4.14. If $|G| = p^2$ with p prime, then either $G \cong C_{p^2}$ or $G \cong C_p \times C_p$.

Proof. We already proved in Corollary 2.25 that all groups of order p^2 are abelian. The result then follows immediately from the Fundamental Theorem of Finite Abelian Groups. □

Lemma 4.15. If $|G| = 2p$ with p an odd prime, then G is either cyclic ($G \cong C_{2p}$) or dihedral ($G \cong D_{2p}$).

Proof. Let G be a group of order $2p$, with p prime. If G is abelian, then we have $G \cong C_{2p}$ by the Fundamental Theorem of Finite Abelian Groups.

So assume that G is nonabelian. Let $P \in \mathrm{Syl}_p(G)$. Since $|\mathrm{Syl}_p(G)| \mid |G|/|P| = 2$, and $|\mathrm{Syl}_p(G)| \equiv 1 \pmod{p}$, we must have $|\mathrm{Syl}_p(G)| = 1$, that is, $P \trianglelefteq G$.

By Theorem 2.26, G has an element of order 2, and since p is odd, every such element lies in $G \setminus P$. Also, $P \cong C_p$ by Lemma 4.13, and since $\mathbb{Z}/p\mathbb{Z}$ is a field, the only solutions to $i^2 = 1$ in $\mathbb{Z}/p\mathbb{Z}$ are $i = \pm 1$. It then follows from Lemma 4.9 that $G \cong D_{2p}$. □

4.5 Groups of order $2p^2$, where p is an odd prime

Definition 4.16. Let p be an odd prime, let $H := C_2 = \langle x \rangle$ and let $K := C_p \times C_p$. Let

$$\theta : H \to \mathrm{Aut}(K)$$

be the inversion homomorphism. That is, θ is defined by $\theta_{x^i}(k) := k^{(-1)^i}$ for $i \in \{0, 1\}$. The group $H \ltimes_\theta K$ is called the *generalised dihedral group of order* $2p^2$, and denoted by GD_{2p^2}.

Lemma 4.17. Let p be an odd prime, and let G be a group of order $2p^2$. Then G is isomorphic to precisely one of the following groups:

$$C_{2p^2}, C_{2p} \times C_p, D_{2p} \times C_p, D_{2p^2} \text{ or } \mathrm{GD}_{2p^2}.$$

Proof. The groups listed in the statement are pairwise non-isomorphic, as the reader can check from their definitions (for example, by comparing element orders). So it suffices to show that G is isomorphic to one of them.

Suppose first that G is abelian. Then by the Fundamental Theorem of Finite Abelian Groups, we see that G is isomorphic to precisely one of:

$$C_{2p^2} \text{ or } C_{2p} \times C_p.$$

So we may assume that G is nonabelian. Let $K \in \mathrm{Syl}_p(G)$ and $H = \langle x \rangle \in \mathrm{Syl}_2(G)$, with $|x| = 2$. Then $|K| = p^2$, so $|G : K| = 2$. It follows that $K \trianglelefteq G$. Also, K is isomorphic to precisely one of C_{p^2} or $C_p \times C_p$ by Lemma 4.14.

Case 1: Suppose that $K \cong C_{p^2}$. Then G satisfies all of the conditions of Lemma 4.9, so $G \cong D_{2p^2}$ (condition (3) in Lemma 4.9 holds here by Example 4.10(3)).

Case 2: Suppose that $K \cong C_p \times C_p$. Then G satisfies all of the conditions of Lemma 4.11 (Fitting's lemma), so we have

$$G \cong (H \ltimes_\theta [K, x]) \times C_K(x),$$

where $\theta : H \to \mathrm{Aut}([K, x])$ is the inversion homomorphism. By Lagrange's theorem, the possibilities for $|C_K(x)|$ are $1, p$, or p^2. However, $|C_K(x)|$ cannot be p^2. Indeed, otherwise x would commute with every element of K, and since K is abelian and $G = \langle K \cup \{x\} \rangle$, Lemma 4.8(i) would imply that G is abelian, a contradiction. Thus, we have

$$|C_K(x)| \in \{1, p\}.$$

Note that $|[K, x]| = |K|/|C_K(x)|$ by part (ii) of Fitting's lemma (so $[K, x] = K$ if $|C_K(x)| = 1$).

Case 2(a): $|C_K(x)| = 1$**.** In this case, we have $G \cong H \ltimes_\theta K$, since $[K, x] = K$ and $C_K(x) = \{1_G\}$. Hence, since $K \cong C_p \times C_p$, we have $G \cong \mathrm{GD}_{2p^2}$ by definition of GD_{2p^2}.

Case 2(b): $|C_K(x)| = p$**.** In this case, we have $|C_K(x)| = p = |[K, x]|$. Thus, $H \ltimes_\theta [K, x]$ is a nonabelian group of order $2p$ (it is nonabelian since G is nonabelian, and a direct product of abelian groups is itself abelian). Hence, $H \ltimes_\theta [K, x] \cong D_{2p}$ by Lemma 4.15. Since $C_K(x) \cong C_p$ by Lemma 4.13, we deduce that $G \cong D_{2p} \times C_p$. This completes the proof. □

4.6 Groups of order pq, where $p < q$ are primes with $p \nmid q - 1$

Lemma 4.18. Let p and q be primes with $p < q$, and $p \nmid (q - 1)$. Let G be a group of order pq. Then $G \cong C_{pq}$.

Proof. By Corollary 3.8(ii) and Sylow's theorem 4, $|\mathrm{Syl}_p(G)|$ divides q, and is congruent to 1 modulo p. If $|\mathrm{Syl}_p(G)| = q$, then it would follow that $q \equiv 1 \pmod{p}$, i.e. that p divides $q - 1$. This is a contradiction. Thus, $|\mathrm{Syl}_p(G)| = 1$.

Next, again using Corollary 3.8(ii) and Sylow's theorem 4, $|\mathrm{Syl}_q(G)|$ divides p, and is congruent to 1 modulo q. Since $q > p$, it follows that $|\mathrm{Syl}_q(G)| = 1$.

We have therefore shown that G has a normal Sylow p-subgroup, say H, and a normal Sylow q-subgroup, say K. We have $H \cap K = \{1_G\}$ by Lagrange's theorem, and hence $HK = G$ by Proposition 2.27 (see also the Problem Sheets). We therefore have $G \cong H \times K$ by Lemma 4.2. Let x and y be generators of H and K respectively. Then x and y commute, so $|xy| = |x||y| = pq$ by Lemma 1.15(iv). Thus, $G = \langle xy \rangle$. □

We have now classified (in particular) all groups of the following orders:

$$1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15, 17, 18, 19.$$

Thus, apart from order 16 (which we omit), all that remains is to classify groups of orders 8 and 12. We distinguish these cases now.

4.7 Groups of order 8

We begin by defining a special group of order 8.

Definition 4.19. Consider the set $Q_8 := \{\pm 1, \pm i, \pm j, \pm k\}$ of eight symbols. Define a binary operation $\cdot : Q_8 \times Q_8 \to Q_8$ on Q_8 by

$$1 \cdot g = g \cdot 1 = g \text{ and } (-1) \cdot g = g \cdot (-1) = -g \text{ for all } g \in Q_8 \quad (1)$$
$$i \cdot j = k,\ j \cdot k = i \text{ and } k \cdot i = j \quad (2)$$
$$j \cdot i = -k,\ k \cdot j = -i \text{ and } i \cdot k = -j \quad (3)$$
$$(\pm 1)^2 = 1 \text{ and } g^2 = -1 \text{ for all } g \in \{\pm i, \pm j, \pm k\} \quad (4)$$

Under this binary operation, Q_8 is a group, called the *quaternion group*. Relation (3) follows from relations (2) and (4).

Lemma 4.20. Let $G = Q_8$. Then

(1) $Z(G) = \{\pm 1\}$;

(2) G has precisely 1 element of order 2 (namely -1), and 6 elements of order 4 (namely $\pm i$, $\pm j$ and $\pm k$);

(3) $G = \langle i, j\rangle = \langle j, k\rangle = \langle k, i\rangle$;

(4) the Cayley table of G is *completely* determined by the relations (1)–(4) in Definition 4.19.

Proof. Each part follows by direct calculation from the relations (1)–(4). □

In particular, Q_8 is not isomorphic to D_8, since D_8 has 5 elements of order 2 (and 2 elements of order 4).

You saw this classification in Algebra 3 or Groups and Rings, so we will not cover it in lectures. It is included here for completeness.

Lemma 4.21. Let G be a group of order 8. Then G is isomorphic to precisely one of the following groups:

$$C_2 \times C_2 \times C_2, \quad C_4 \times C_2, \quad C_8, \quad D_8, \quad Q_8.$$

Proof. As before, the listed groups are easily seen to be pairwise non-isomorphic, so we need only show that G is one of them.

If G is abelian, then G is isomorphic to precisely one of

$$C_2 \times C_2 \times C_2, \quad C_4 \times C_2, \quad C_8$$

by the Fundamental Theorem of Finite Abelian Groups.

So assume that G is nonabelian. Then $G \ncong C_8$, so no element of G has order 8. Also, if $x^2 = 1_G$ for all $x \in G$, then $xy = (xy)^{-1} = y^{-1}x^{-1} = yx$ for all $x, y \in G$, so G would be abelian. Thus, we may choose some $u \in G$ with $|u| > 2$. We then have $|u| = 4$ by Lagrange's theorem.

Let $K = \langle u \rangle$, and let $v \in G \setminus K$, with $|v|$ minimal. Then $|v| \in \{2, 4\}$.

Before distinguishing these two possibilities for $|v|$, we note that $G = \langle u, v \rangle$, since $v \notin K = \langle u \rangle$, and $|K| = 4$. Also, $K \trianglelefteq G$, since $|G : K| = 2$. It follows that ${}^v u \in K = \{1_G, u, u^2, u^3\}$. If ${}^v u = u$, then v commutes with u, so G is abelian by Lemma 4.8(i), since $G = \langle u, v \rangle$. This is a contradiction. So ${}^v u \in \{1_G, u^2, u^3\}$. Since $u^3 = u^{-1}$ is the only element of the latter set of order 4, and $|{}^v u| = |u| = 4$, we must therefore have ${}^v u = u^{-1}$.

We now distinguish the two possibilities for the order of v.

Case 1: $|v| = 2$. Suppose that $|v| = 2$. Then G satisfies the three hypotheses of Lemma 4.9 (for condition (3), see Example 4.10(3)). Thus, $G \cong D_8$.

Case 2: $|v| = 4$. Suppose that $|v| = 4$. Then since $v \notin K$ and $|G : K| = 2$, we must have $G/K = \{K, vK\}$. Hence, $G = K \sqcup vK$. Since $4 = |v|$ is the minimal order of an element in $G \setminus K$, we deduce that every element of $vK = G \setminus K$ has order 4. Hence, G has precisely 6 elements of order 4 (namely $vK \sqcup \{u, u^{-1}\}$) and 1 element of order 2 (namely $z := u^2$). It follows that

$$x^2 = z \text{ for all } x \in vK \sqcup \{u, u^{-1}\}.$$

Hence, $z \in Z(G)$ and $zx = x^3 = x^{-1}$ for all $x \in vK \sqcup \{u, u^{-1}\}$. Let

$$w := uv = v^{v^{-1}} u = vu^{-1} \in vK.$$

Note that $u^{\pm 1}, v^{\pm 1}$ and $w^{\pm 1}$ are distinct elements of G of order 4. Indeed, $u^{\pm 1} \in K$ and $v^{\pm 1}, w^{\pm 1} \in vK$, so the only possibility is $v^{\pm 1} = w = uv$, which would imply that either $u = 1_G$ or $v^2 = u$. These are both contradictions, since $u \neq 1_G$, and $|v| = |u| = 4$ (so $|v^2| = 2$).

It follows that

$$G = \{1_G, z, u^{\pm 1}, v^{\pm 1}, w^{\pm 1}\} = \{1_G, z, u, zu, v, zv, w, zw\}.$$

Note that

$$uv = vv^{-1}uv = v^{v^{-1}} u = vu^{-1} = vu^3 = vuz = zvu.$$

The identical argument with u and v replaced by any distinct pair of elements of $\{u, v, w\}$ yields

$$gh = zhg \text{ for all } g, h \in \{u, v, w\}.$$

Since $u^2 = v^2 = w^2 = z \in Z(G)$, it follows that

$$zg = gz \text{ for all } g \in G \tag{1}$$
$$uv = w,\ vw = v(uv) = zv^2u = u \text{ and } wu = (uv)u = zvu^2 = v \tag{2}$$
$$vu = zw,\ wv = zu \text{ and } uw = zv \text{ in the same way} \tag{3}$$
$$1_G^2 = z^2 = 1_G \text{ and } g^2 = z \text{ for all } g \in \{u, zu, v, zv, w, zw\} \tag{4}$$

The Cayley table of G is therefore completely determined by the relations (1)–(4) above. Comparing these with the relations (1)–(4) in Definition 4.19, we see that the map $\phi : G \to Q_8$ given by

$$1_G \mapsto 1, \quad u \mapsto i, \quad v \mapsto j, \quad w \mapsto k, \quad z \mapsto -1, \quad zu \mapsto -i, \quad zv \mapsto -j, \quad zw \mapsto -k$$

is a homomorphism. Since ϕ is a bijection, we have $G \cong Q_8$, as required. □

We omit the classification of groups of order 16. There are 14 of them, and the number of groups of order 2^n, up to isomorphism, grows rapidly with n, as the following table shows.

n	1	2	3	4	5	6	7	8	9	10
groups of order 2^n	1	2	5	14	51	267	2328	56092	10494213	49487367289

4.8 Groups of order 12

The following exercise is typical of an exam question on this chapter.

Exercise. Let G be a finite group of order 12, and suppose that G has no element of order 6.

(1) Show that G has at least 4 elements of order 3. (Hint: use the tools in Appendix A.)

(2) Hence or otherwise, deduce that G has 4 Sylow 3-subgroups. (Hint: Corollary 3.9.)

We have already seen two nonabelian groups of order 12: namely, A_4 and D_{12}. We now define another.

Definition 4.22. Let $H := C_4$ and $K := C_3$, and write $H = \langle h \rangle$. Define $\phi : H \to \mathrm{Aut}(K)$ by $\phi_{h^i}(k) := k^{(-1)^i}$, for $0 \leq i \leq 3$ and $k \in K$. The resulting group $H \ltimes_\phi K$ is called the *dicyclic group of order* 12, and is denoted by Dic_{12}.

Lemma 4.23. Let G be a group of order 12. Then G is isomorphic to precisely one of the groups:

$$C_{12}, C_6 \times C_2, \mathrm{Dic}_{12}, D_{12}, A_4.$$

Proof. Again, the listed groups are pairwise non-isomorphic. If G is abelian, then we have

$$G \cong C_{12} \text{ or } G \cong C_6 \times C_2$$

by Theorem 4.12.

So assume that G is nonabelian. We will distinguish our cases according to whether or not G contains an element of order 6.

Case 1: G contains an element of order 6. Let a be such an element, and let $K := \langle a \rangle$. Suppose first that $G \setminus K$ has an element of order 2. Then G satisfies the three hypotheses of Lemma 4.9 (for condition (3), see Example 4.10(1)). Thus, $G \cong D_{12}$.

Assume now that no element of $G \setminus K$ has order 2, and let $H \in \mathrm{Syl}_2(G)$. Then $H \not\leq K$, since $|K|_2 = 2$. Also, $|H| = 4$. If every non-identity element of H has order 2, then $H \setminus K$ would consist of elements of order 2, contrary to our assumption that $G \setminus K$ has no elements of order 2. Thus, H must have an element of order 4. Hence, $H \cong C_4$.

Now, let $K_1 := \langle a^2 \rangle$. Then $|K_1| = |a^2| = 3$, by Lemma 1.15. Thus, $K_1 \in \mathrm{Syl}_3(G)$. Also, $K_1 \trianglelefteq G$ (see the Problem Sheets). We now have

(1) $H \leq G$, $K_1 \trianglelefteq G$;

(2) $H \cap K_1 = \{1_G\}$ by Lagrange's theorem;

(3) $|HK_1| = |H||K_1| = |G|$ by (2) and Proposition 2.27. Thus, $HK_1 = G$.

It then follows from Lemma 4.6 that $G \cong H \ltimes_\phi K_1$, where ϕ is the map $\phi : H \to \mathrm{Aut}(K_1)$ given by $\phi_h(k) := hkh^{-1}$. Finally, note that if $H = \langle h \rangle$ and $k \in K_1 \setminus \{1_G\}$, then $G = \langle h, k \rangle$, so $hkh^{-1} \neq k$ by Lemma 4.8(i), since G is nonabelian. Thus, since $hkh^{-1} \in K_1$ and $|hkh^{-1}| = |k|$, we must have $hkh^{-1} = k^{-1}$. Thus,

$$h^i k h^{-i} = k^{(-1)^i}.$$

Hence, $G \cong \mathrm{Dic}_{12}$ by definition of Dic_{12}.

Case 2: G **has no element of order** 6**.** By Cauchy's theorem, we may choose an element x of G of order 3. Since $x \in C_G(x)$, and G has no element of order 6, we see that 2 cannot divide $|C_G(x)|$. Indeed, if it did, then we could choose an element y of $C_G(x)$ of order 2, again by Cauchy's theorem. But then $|xy| = 6$ by Lemma 1.15, a contradiction.

Thus, we must have $|C_G(x)| = 3$, by Lagrange's theorem. Hence, $|{}^G x| = |G : C_G(x)| = 4$ by the Orbit–Stabiliser Theorem. Since ${}^G x$ consists of 4 elements of order 3, and G has precisely $2|\operatorname{Syl}_3(G)|$ elements of order 3 by Corollary 3.9(iii), we deduce that $|\operatorname{Syl}_3(G)| > 1$.

Finally, let $H \in \operatorname{Syl}_3(G)$, and consider the action of G on $G/H = \{gH \mid g \in G\}$ by left multiplication. Let K be the kernel of this action. Then $K \leq H$ (see the Problem Sheets). But H is not normal in G, since $H \in \operatorname{Syl}_3(G)$ and $|\operatorname{Syl}_3(G)| > 1$. So we must have $|K| = 1$, since $|H| = 3$. Thus,

$$G \cong G/K \text{ is isomorphic to a subgroup of } S_4.$$

But the only subgroup of S_4 of order 12 is A_4 (see the Problem Sheets). Thus, $G \cong A_4$. This completes the proof. □

4.9 The unique simple group of order 60

In this section, we prove the following result.

Theorem 4.24. Let G be a finite simple group of order 60. Then $G \cong A_5$.

Proof. Let G be a finite simple group with $|G| = 60$. We will subdivide the proof into a series of claims.

Claim 1: The group G does not contain a proper subgroup of index $n \leq 4$.

Proof: Assume the contrary, and let $H \lneq G$ with $n := |G : H| \leq 4$. Then $2 \leq n \leq 4$, and G acts non-trivially by left multiplication on the set G/H of left cosets of H in G. Since G is simple, the kernel of this action is trivial. Thus, by Proposition 2.16(i), G is isomorphic to a subgroup of S_n. But $|S_n| \leq 4! = 24 < 60 = |G|$, a contradiction.

Claim 2: The group G contains a subgroup of index 5.

Proof: Assume the contrary. Let $P \in \operatorname{Syl}_2(G)$. By Sylow's theorem 4 and Corollary 3.8(ii), $|\operatorname{Syl}_2(G)|$ is odd and divides 15. Moreover, $|\operatorname{Syl}_2(G)| \neq 1$ since G is simple. Also, $|\operatorname{Syl}_2(G)| = |G : N_G(P)|$ by Corollary 3.8(i), so $|\operatorname{Syl}_2(G)| \neq 3$ by Claim 1.

Thus,

$$|\operatorname{Syl}_2(G)| \in \{5, 15\}.$$

If $|\operatorname{Syl}_2(G)| = 5$, then we are done, since then $|G : N_G(P)| = 5$ by Corollary 3.8(i). So assume that $|\operatorname{Syl}_2(G)| = 15$.

Let $g \in G \setminus N_G(P)$, and let $H := \langle P \cup {}^g P \rangle$. Then H contains at least two Sylow 2-subgroups (namely, P and ${}^g P$). So $|\operatorname{Syl}_2(H)|$, being an odd number by Sylow's theorem 4, is at least 3.

Since $P \in \mathrm{Syl}_2(H)$, Corollary 3.8(ii) shows that $|\mathrm{Syl}_2(H)|$ divides $|H : P|$, so $|H|$ is divisible by $4|\mathrm{Syl}_2(H)|$. By Claim 1, we have either $H = G$, or $|H| < |G|/4 = 15$. Thus, we must have either $H \neq G$ and $|\mathrm{Syl}_2(H)| = 3$, or $H = G$. If $H \neq G$ and $|\mathrm{Syl}_2(H)| = 3$, then 12 divides $|H|$, as noted above, so we must have $|H| = 12$ by Claim 1. Thus, $|G : H| = 5$, as required.

So assume that for all $g \in G \setminus N_G(P)$, we have $G = \langle P \cup {}^gP \rangle$. If $x \in P \cap {}^gP$, then x is centralised by both P and ${}^gP$, since groups of order 4 are abelian. Thus, x is centralised by G, since $G = \langle P \cup {}^gP \rangle$. Since $Z(G) = \{1_G\}$ (as G is simple), we deduce that $P \cap {}^gP = \{1_G\}$ for all $g \in G \setminus N_G(P)$. Since every Sylow 2-subgroup of G is conjugate to P, it follows (by conjugating) that any two distinct Sylow 2-subgroups of G intersect trivially. Now,

$$F_2(G) = \bigcup_{Q \in \mathrm{Syl}_2(G)} (Q \setminus \{1_G\})$$

by Corollary 3.9(i). By our work above, this is a disjoint union. Thus, $|F_2(G)| = |\mathrm{Syl}_2(G)|(4 - 1) = 45$.

Finally, by Sylow's theorem 4 and Corollary 3.8(ii), we have $|\mathrm{Syl}_5(G)|$ is congruent to 1 modulo 5, and divides 12. Since $|\mathrm{Syl}_5(G)| > 1$ (as G is simple), we deduce that $|\mathrm{Syl}_5(G)| = 6$. But then

$$|F_5(G)| = 6(5 - 1) = 24$$

by Corollary 3.9(iii). So $60 = |G| \geq 1 + |F_2(G)| + |F_5(G)| = 70$, a contradiction. This completes the proof of Claim 2.

Let $H \leq G$ be a subgroup of index 5. Then G acts non-trivially by left multiplication on the set of left cosets G/H of H in G. Since G is simple, the kernel of this action is trivial, and so by Proposition 2.16(i), G is isomorphic to a subgroup of S_5 of order 60. The only such subgroup of S_5 is A_5 (see the Problem Sheets). This completes the proof. □

Chapter 5

Soluble groups and the Jordan–Hölder theorem

The purpose of this chapter is twofold. First, we state and prove the Jordan–Hölder theorem, which shows why finite simple groups play a central role in finite group theory.

Second, we study an important class of groups: the *soluble groups*. These arise in Galois theory: a polynomial with rational coefficients is *solvable by radicals* if and only if its Galois group is soluble. Since S_5 is not soluble, this leads to the Abel–Ruffini theorem, which states that the general polynomial of degree 5 is not solvable by radicals. We will not go into the details of Galois theory here.

5.1 Composition series and the Jordan–Hölder theorem

Recall that $H \lneq G$ means that H is a *proper* subgroup of G. Similarly, we write $H \trianglelefteq\!\!\!\!\!\!\cdot\;\; G$ to mean that H is a *proper* normal subgroup of G, while $H \trianglelefteq G$ continues to mean that H is a normal subgroup of G.

Definition 5.1. A *composition series* of a group G is a series

$$\{1_G\} = G_0 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_1 \trianglelefteq\!\!\!\!\!\!\cdot\;\; \cdots \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_r = G$$

such that for each $1 \leq i \leq r$, the group G_i/G_{i-1} is simple. Moreover, in this case, r is called the *length* of the composition series.

Example 5.2. The following are examples of composition series of some groups we have seen in the course.

(1) Let p be an odd prime, and let $G = D_{2p} = \langle \sigma, \tau \rangle$, where σ and τ are as in Definition 2.5. Let $G_0 := \{1_G\}$, $G_1 = \langle \sigma \rangle$ and $G_2 = G$. Then $G_0 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_1$ and $G_1 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_2$. Moreover, $G_1/G_0 \cong G_1 \cong C_p$ and $G_2/G_1 \cong C_2$ are simple. Hence, the series

$$\{1_G\} = G_0 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_1 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_2 = G$$

is a composition series of G of length 2.

(2) Let $n \geq 5$, and let $G = S_n$. Let $G_0 = \{1_G\}$, $G_1 := A_n$, and $G_2 := S_n$. Then $G_0 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_1$ and $G_1 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_2$. Moreover, $G_1/G_0 \cong G_1 = A_n$ and $G_2/G_1 \cong C_2$ are simple. Hence, the series

$$\{1_G\} = G_0 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_1 \trianglelefteq\!\!\!\!\!\!\cdot\;\; G_2 = G$$

is a composition series of G of length 2.

(3) Let $G = D_8 = \langle \sigma, \tau \rangle$ (again, where σ and τ are as in Definition 2.5). Let $G_0 := \{1_G\}$, $G_1 := \langle \sigma^2 \rangle \leq Z(G)$, $G_2 := \langle \sigma \rangle$ and $G_3 := G$. Then $G_0 \trianglelefteq G_1$, $G_1 \trianglelefteq G_2$, and $G_2 \trianglelefteq G_3$. Moreover, G_1/G_0, G_2/G_1 and G_3/G_2 all have order 2, so each of these groups is isomorphic to C_2. Hence, each is simple, so the series

$$\{1_G\} = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq G_3 = G$$

is a composition series of G of length 3.

For technical reasons, we assume if G is the trivial group (i.e. $G = \{1_G\}$), then the series

$$\{1_G\} = G_0 = G$$

is a composition series of G of length 0.

Theorem 5.3. Every finite group has a composition series.

Proof. We will prove the theorem by induction on $|G|$. As mentioned above, if $|G| = 1$, then G has a composition series.

So assume that $|G| > 1$, and that every group of order less than $|G|$ has a composition series. If G is simple, then $\{1_G\} =: G_0 \trianglelefteq G_1 := G$ is a composition series of G.

So assume that G is not simple. Then G has a normal subgroup N with $\{1_G\} \neq N \neq G$. Then $|N| < |G|$ and $|G/N| < |G|$, so our inductive hypothesis implies that:

(1) N has a composition series $\{1_G\} = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_r = N$;

(2) G/N has a composition series $\{1_{\overline{G}}\} = \overline{G}_0 \trianglelefteq \overline{G}_1 \trianglelefteq \cdots \trianglelefteq \overline{G}_s = G/N$.

Fix $0 \leq i \leq s$. By the Correspondence Theorem (Theorem 1.45), there exists a subgroup X_i of G containing N such that $\overline{G}_i = X_i/N$. If $i < s$, then $\overline{G}_i$ is a normal subgroup of $\overline{G}_{i+1}$, so X_i is a normal subgroup of X_{i+1} by Theorem 1.45(ii), applied to $N \trianglelefteq X_{i+1}$. Moreover, by the Third Isomorphism Theorem, we have

$$\frac{\overline{G}_{i+1}}{\overline{G}_i} = \frac{X_{i+1}/N}{X_i/N} \cong \frac{X_{i+1}}{X_i},$$

so X_{i+1}/X_i is simple.

Now, notice that $X_s = G$ and $N_r = N = X_0$. Let $G_i = N_i$ for $0 \leq i \leq r$, and let $G_i = X_{i-r}$ for $r+1 \leq i \leq r+s$. Then by our work above,

$$G_0 := \{1_G\} \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_{r+s} = G$$

is a composition series for G. This completes the induction, and the result follows. □

Corollary 5.4. Let G be a finite group, and let $N \trianglelefteq G$. Suppose that

$$\{1_G\} =: N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_r = N \text{ and}$$
$$\{1_{G/N}\} =: \frac{X_0}{N} \trianglelefteq \frac{X_1}{N} \trianglelefteq \cdots \trianglelefteq \frac{X_s}{N} = \frac{G}{N}$$

are composition series for N and G/N, respectively, where each X_i is a subgroup of G containing N (in particular, $X_0 = N$ and $X_s = G$ by the Correspondence Theorem). For $0 \leq i \leq r$, set $G_i := N_i$, and for $1 \leq i \leq s$, set $G_{r+i} := X_i$. Then

$$\{1_G\} := G_0 \trianglelefteq \cdots \trianglelefteq G_{r+s} = G$$

is a composition series for G.

A finite group may have more than one composition series. For example, consider $G = D_8 = \langle \sigma, \tau \rangle$ as above.

Example 5.5. We saw that with $G_0 = \{1_G\}$, $G_1 = \langle \sigma^2 \rangle$, $G_2 = \langle \sigma \rangle$ and $G_3 = G$, we get a composition series

$$\{1_G\} = G_0 \lneq G_1 \lneq G_2 \lneq G_3 = G$$

of G of length 3.

Next, let $N_0 := \{1_G\}$, $N_1 := \langle \sigma^2 \rangle \leq Z(G)$, $N_2 := \langle \sigma^2, \tau \rangle$ and $N_3 := G$. Then $N_0 \lneq N_1$, $N_1 \lneq N_2$, and $N_2 \lneq N_3$ (each N_i has index 2 in N_{i+1}). Moreover, N_1/N_0, N_2/N_1 and N_3/N_2 all have order 2, so each of these groups is isomorphic to C_2. Hence, each is simple, so the series

$$\{1_G\} = N_0 \lneq N_1 \lneq N_2 \lneq N_3 = G$$

is a composition series of G of length 3.

Finally, notice that $N_2 \neq G_2$, so these composition series are distinct.

Now, the composition series in the example above are different, but the lengths are the same (both 3), and the simple factors are the same up to isomorphism (each of the three factors are isomorphic to C_2, in each case). Composition series with these factors are thought of as "the same" via the following definition.

Definition 5.6. We shall say that the two composition series of a group G

$$\{1_G\} = A_0 \lneq A_1 \lneq \cdots \lneq A_r = G \text{ (I)} \quad \text{and} \quad \{1_G\} = B_0 \lneq B_1 \lneq \cdots \lneq B_s = G \text{ (II)}$$

are *equivalent* and write (I) $\sim$ (II) if $r = s$ and there is a bijection $\pi : \{1, \ldots, r\} \to \{1, \ldots, s\}$ such that $A_i/A_{i-1} \cong B_{\pi(i)}/B_{\pi(i)-1}$ for each $1 \leq i \leq r$.

Theorem 5.7 (The Jordan–Hölder theorem)**.** Let

$$\{1_G\} = A_0 \lneq A_1 \lneq \cdots \lneq A_r = G \text{ (I)} \quad \text{and} \quad \{1_G\} = B_0 \lneq B_1 \lneq \cdots \lneq B_s = G \text{ (II)}$$

be two composition series of a finite group G. Then (I) $\sim$ (II).

Proof. We may assume, without loss of generality, that $r \leq s$. We will now prove the result by induction on r. If $r = 0$, then $G = \{1_G\}$ and the result is clear. So assume that $r > 0$.

We now distinguish a number of cases.

Case 1. Suppose first that $A_{r-1} = B_{s-1}$. Then

$$\{1_G\} = A_0 \lneq A_1 \lneq \cdots \lneq A_{r-1} \quad \text{and} \quad \{1_G\} = B_0 \lneq B_1 \lneq \cdots \lneq B_{s-1}$$

are two composition series of $A_{r-1} = B_{s-1}$ of lengths $r-1$ and $s-1$, and so by induction they are equivalent, and hence so are (I) and (II).

Case 2. Suppose now that $A_{r-1} \neq B_{s-1}$. Consider $A_{r-1}B_{s-1}$. Since $A_{r-1} \lneq G$ and $B_{s-1} \lneq G$, we have $A_{r-1}B_{s-1} \trianglelefteq G$. Moreover, $A_{r-1} \not\leq B_{s-1}$: otherwise B_{s-1}/A_{r-1} would be a proper normal subgroup of the simple group G/A_{r-1}, forcing $B_{s-1} = A_{r-1}$, a contradiction. Hence

$$B_{s-1} \lneq A_{r-1}B_{s-1} \trianglelefteq G.$$

It follows from the Third Isomorphism Theorem that $A_{r-1}B_{s-1}/B_{s-1}$ is a normal subgroup of G/B_{s-1}. Since $B_{s-1} \lneq A_{r-1}B_{s-1}$, it is in fact a non-trivial normal subgroup, by the Correspondence Theorem. Thus since G/B_{s-1} is simple, we have $A_{r-1}B_{s-1}/B_{s-1} = G/B_{s-1}$, so $A_{r-1}B_{s-1} = G$ by the Correspondence Theorem.

Let $D = A_{r-1} \cap B_{s-1}$. By the Second Isomorphism Theorem,

$$G/A_{r-1} = A_{r-1}B_{s-1}/A_{r-1} \cong B_{s-1}/(A_{r-1} \cap B_{s-1}) = B_{s-1}/D$$

and as G/A_{r-1} is simple, so is B_{s-1}/D. Similarly, by the Second Isomorphism Theorem,

$$G/B_{s-1} = A_{r-1}B_{s-1}/B_{s-1} \cong A_{r-1}/(A_{r-1} \cap B_{s-1}) = A_{r-1}/D$$

and as G/B_{s-1} is simple, so is A_{r-1}/D.

Now, let

$$\{1_G\} = D_0 \lneq \cdots \lneq D_t := D$$

be a composition series for D. Then

$$\{1_G\} = D_0 \lneq \cdots \lneq D_t = D \lneq A_{r-1} \lneq G \text{ (III)}$$

and

$$\{1_G\} = D_0 \lneq \cdots \lneq D_t = D \lneq B_{s-1} \lneq G \text{ (IV)}$$

are also composition series for G.

Since we are assuming that $r \leq s$, we may apply Case 1 to the series (I) and (III), and deduce that (I) $\sim$ (III). In particular, we have $r = t + 2$, and hence series (IV) has length $t + 2 = r$. Thus we may also apply Case 1 to conclude that (II) $\sim$ (IV).

Finally, we saw above that $G/A_{r-1} \cong B_{s-1}/D$ and $A_{r-1}/D \cong G/B_{s-1}$. Thus (III) $\sim$ (IV), and hence (I) $\sim$ (II) as required. □

Thus, the length of a composition series of a finite group G, and the multiset of its factors (up to isomorphism), depend only on G. This justifies the following definition.

Definition 5.8. Let

$$\{1_G\} = G_0 \lneq G_1 \lneq \cdots \lneq G_r = G$$

be a composition series for a finite group G. The groups G_i/G_{i-1} for $1 \leq i \leq r$ are called the *composition factors* of G, while r is called the *composition length* of G.

When we write down the composition factors of a finite group, we usually list them up to isomorphism. For example, in the examples following Definition 5.1, we have the following:

(1) The group D_{2p} (for p an odd prime) has composition length 2, with composition factors C_p, C_2;

(2) The group S_n (for $n \geq 5$) has composition length 2, with composition factors A_n, C_2;

(3) The group D_8 has composition length 3, with composition factors C_2, C_2, C_2.

Definition 5.9. A finite group is called *soluble* if it is either trivial, or its composition factors are all cyclic groups of prime order.

Example 5.10.

(1) Let G be a finite abelian group. Then every subgroup of G, and every quotient of such a subgroup, is abelian. So the composition factors of G are abelian simple groups, i.e. cyclic groups of prime order. Thus, all finite abelian groups are soluble.

(2) Let $n \geq 5$, and let $G = A_n$. Then G is a nonabelian simple group, so it has precisely one composition factor: A_n itself (which is nonabelian). Thus, G is not soluble.

It is useful to be able to check whether a given finite group is soluble, especially in Galois theory. We give two techniques for doing this. The first is as follows (the second is the subject of the next section).

Lemma 5.11. Let G be a finite group, and let N be a normal subgroup of G. Then G is soluble if and only if both N and G/N are soluble.

Proof. Write $\mathrm{CF}(G)$ for the multiset of composition factors of G. By Corollary 5.4 and the Jordan–Hölder theorem, we have

$$\mathrm{CF}(G) = \mathrm{CF}(N) \cup \mathrm{CF}(G/N).$$

Thus, G is soluble if and only if both N and G/N are soluble. □

Example 5.12. Let $G = D_{2n} = \langle \sigma, \tau \rangle$ (see Definition 2.5), and let $N = \langle \sigma \rangle$. Then N is abelian, and $|G/N| = 2$, so G/N is abelian. Hence, N and G/N are both soluble by Example 5.10(1). So G is soluble by Lemma 5.11.

5.2 Commutators

In this section, we develop a second way to check whether a given finite group is soluble. To state it, we need the language of commutators (we saw a special case of these when we studied Lemma 4.11).

Definition 5.13. Let G be a group. For $g, h \in G$, the *commutator* $[g, h]$ of g and h is defined to be $ghg^{-1}h^{-1}$.

Example 5.14. Let $G = A_5$. Then

$$[(1,2,4),(1,3,5)] = (1,2,4)(1,3,5)(4,2,1)(5,3,1) = (1,2,3).$$

More generally, if $i, j, k, l, m \in \{1, \ldots, n\}$ are distinct, then

$$[(i,j,l),(i,k,m)] = (i,j,l)(i,k,m)(l,j,i)(m,k,i) = (i,j,k).$$

Note that $[g, h] = 1$ if and only if $gh = hg$ (i.e., g and h commute).

Definition 5.15. The *commutator subgroup* $[G, G]$ of G is the subgroup of G generated by all of its commutators:

$$[G, G] = \langle [g, h] \mid g, h \in G \rangle.$$

In the literature, $[G, G]$ is also called the *derived subgroup* of G.

More generally, if $H, K \leq G$, then we will write

$$[H, K] = \langle [h, k] \mid h \in H, k \in K \rangle.$$

We sometimes call $[H, K]$ the commutator subgroup of H and K.

In general, the set $\{[g, h] \mid g, h \in G\}$ is not a subgroup of G, so an element of $[G, G]$ may need to be written as a product of several commutators.

Example 5.16. The following are some examples of commutator subgroups.

(1) Let G be an abelian group. Then $gh = hg$ for all $g, h \in G$. Hence, $[g, h] = 1$ for all $g, h \in G$, and so $[G, G] = 1$.

(2) Let $G = A_5$. By the example above, every 3-cycle in A_5 is a commutator. But A_5 is generated by 3-cycles by Lemma 3.13(i), and so $[A_5, A_5] = A_5$.

Theorem 5.17. Let G be a group. Then the following conditions hold:

(i) $[G, G] \trianglelefteq G$;

(ii) $G/[G, G]$ is abelian;

(iii) If $N \trianglelefteq G$ and G/N is abelian, then $[G, G] \leq N$.

Proof. (i) For all $g, h, k \in G$:

$$\begin{aligned} g[h, k]g^{-1} &= ghkh^{-1}k^{-1}g^{-1} \\ &= (ghg^{-1})(gkg^{-1})(gh^{-1}g^{-1})(gk^{-1}g^{-1}) \\ &= [ghg^{-1}, gkg^{-1}] \in [G, G]. \end{aligned}$$

So, for $[h_1, k_1][h_2, k_2] \cdots [h_r, k_r] \in [G, G]$, we have

$$g[h_1, k_1] \cdots [h_r, k_r]g^{-1} = (g[h_1, k_1]g^{-1}) \cdots (g[h_r, k_r]g^{-1}) \in [G, G]$$

and hence, $[G, G] \trianglelefteq G$.

(ii) Let $g, h \in G$. Then

$$\begin{aligned} ghg^{-1}h^{-1} \in [G, G] &\implies [G, G]ghg^{-1}h^{-1} = [G, G] \\ &\implies [G, G]gh = [G, G]hg \\ &\implies [G, G]g[G, G]h = [G, G]h[G, G]g. \end{aligned}$$

So $G/[G, G]$ is abelian.

(iii) If G/N is abelian, then $NgNh = NhNg$ for all $g, h \in G$. Thus $N(ghg^{-1}h^{-1}) = N$, and so $[g, h] \in N$. Since N is a subgroup of G containing every commutator, $[G, G] \leq N$. □

Corollary 5.18. A group G is abelian if and only if $[G, G] = \{1_G\}$.

Example 5.19. Let $n \geq 3$, and let

$$G = D_{2n} = \langle \sigma, \tau \rangle$$

where σ and τ are as in Definition 2.5. Recall that ${}^{\tau}\sigma = \sigma^{-1}$. Then $[\tau, \sigma] = \tau\sigma\tau^{-1}\sigma^{-1} = \sigma^{-2}$. So $\sigma^2 = (\sigma^{-2})^{-1} \in [G, G]$. Consider $K = \langle \sigma^2 \rangle \leq G$. Then $K \leq [G, G]$. Note also that since $K \leq \langle \sigma \rangle \trianglelefteq G$, we have $K \trianglelefteq G$ (see the Problem Sheets).

Case 1: n odd. Since n is odd, $K = \langle \sigma^2 \rangle = \langle \sigma \rangle$. Thus $|G : K| = 2$, and so G/K is abelian. Hence, $[G, G] \leq K$ by Theorem 5.17(iii). Thus $[G, G] = K = \langle \sigma \rangle$ has order n.

Case 2: n even. Since n is even, $K = \langle \sigma^2 \rangle$ has order $n/2$. Thus $|G : K| = 4$, and so G/K is abelian by Corollary 2.25(i). Hence, $[G, G] \leq K$ by Theorem 5.17(iii). Thus, $K = [G, G]$ has order $n/2$.

Definition 5.20. For a group G, define $G^{(0)} = G$, $G^{(1)} = [G, G]$ and, inductively, $G^{(n+1)} = [G^{(n)}, G^{(n)}]$ for integers $n > 0$. Then the descending series

$$G^{(0)} \geq G^{(1)} \geq G^{(2)} \geq \cdots \geq G^{(n)} \geq G^{(n+1)} \geq \cdots$$

is called the *derived series* of G.

By definition of the derived series, for all non-negative integers n and m we have

(1) $(G^{(n)})^{(m)} = G^{(n+m)}$;

(2) if $H \leq G$, then $H^{(n)} \leq G^{(n)}$.

Both assertions follow by induction.

We can now state our second tool for deciding whether or not a given finite group is soluble.

Theorem 5.21. Let G be a finite group. Then G is soluble if and only if $G^{(n)} = \{1_G\}$ for some positive integer n.

Proof. Suppose first that G is soluble. We will prove, by induction on $|G|$, that $G^{(n)} = \{1_G\}$ for some positive integer n. If $G = \{1_G\}$, then this holds with $n = 1$, so assume that $|G| > 1$. Let $N := [G, G]$. Then $N \trianglelefteq G$ by Theorem 5.17(i), so N is soluble by Lemma 5.11. Also, G has a composition series

$$\{1_G\} =: G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_r := G$$

where all of the groups G_i/G_{i-1} are cyclic of prime order. This is by definition of solubility. It follows that G_{r-1} is a normal subgroup of G with $G_r/G_{r-1} = G/G_{r-1}$ cyclic of prime order. Thus, $N \leq G_{r-1}$ by Theorem 5.17(iii). In particular, $|N| < |G|$. So $N^{(m)} = \{1_G\}$ for some $m \in \mathbb{N}$ by the inductive hypothesis. Since $G^{(n)} = [G, G]^{(n-1)}$ by definition, it follows that $G^{(m+1)} = \{1_G\}$, as required.

Suppose next that $G^{(n)} = \{1_G\}$ for some $n \in \mathbb{N}$. We prove that G is soluble by induction on n. If $n = 1$, then $G^{(1)} = \{1_G\}$, so G is abelian, and hence soluble by Example 5.10(1).

So assume that $n > 1$, and that the result holds for smaller values of n. Let $N := [G, G] = G^{(1)}$. Then $N^{(n-1)} = (G^{(1)})^{(n-1)} = \{1_G\}$, so N is soluble by our inductive hypothesis. Also, $G/N = G/[G, G]$ is abelian by Theorem 5.17(ii). So both N and G/N are soluble. Hence, G is soluble by Lemma 5.11. This completes the induction, and the proof. □

5.3 Examples of soluble groups

In this section, we say more about which groups are soluble.

Theorem 5.22. Let G be a finite soluble group, and let $H \leq G$. Then H is soluble.

Proof. Since G is soluble, it follows from Theorem 5.21 that $G^{(n)} = \{1_G\}$ for some $n \in \mathbb{N}$. Since $H^{(n)} \leq G^{(n)}$ (see the properties listed after Definition 5.20), we have $H^{(n)} = \{1_G\}$. Thus, another application of Theorem 5.21 yields the result. □

Theorem 5.23. Let p be prime, let $n \in \mathbb{N}$, and let G be a group of order p^n. Then G is soluble, and all composition factors of G are isomorphic to C_p.

Proof. We prove the theorem by induction on $|G|$. If $|G| = p$, then G is cyclic of prime order, so G is soluble.

Assume now that $|G| > p$, and that the result holds for groups of order less than $|G|$. By Theorem 2.24, $Z := Z(G)$ is non-trivial. Since Z is abelian, Z is soluble. Also, G/Z is soluble: this is clear if $Z = G$, and follows from the inductive hypothesis otherwise. Thus, G is soluble by Lemma 5.11. Finally, the composition factors of G are cyclic of prime order, and their orders multiply to $|G| = p^n$, so each is isomorphic to C_p. □

Example 5.24. Let G be a finite group of order $4p^n$, with p prime. We claim that G is soluble. Indeed, let P be a Sylow p-subgroup of G, and let

$$K := \bigcap_{x \in G} {}^xP.$$

Since $K \leq P$ and P is soluble by Theorem 5.23, we have that K is soluble by Theorem 5.22.

Now, K is the kernel of the action of G on $G/P = \{gP \mid g \in G\}$. Thus, G/K is isomorphic to a subgroup of S_4 by Proposition 2.16(i). Now S_4 is soluble, since $\{1\} \lneq \langle (1,2)(3,4) \rangle \lneq V_4 \lneq A_4 \lneq S_4$ is a composition series with factors C_2, C_2, C_3 and C_2. It follows from Theorem 5.22 that G/K is soluble. Hence, G is soluble by Lemma 5.11.

Theorem 5.25. Let $G_1, G_2, \ldots, G_t$ be finite soluble groups. Then $G := G_1 \times \cdots \times G_t$ is soluble.

Proof. We prove the claim by induction on t. If $t = 1$, then $G = G_1$ and the result is trivial.

So assume that $t > 1$, and that the result holds for smaller values of t. Then

$$G = G_1 \times X$$

where $X = G_2 \times \cdots \times G_t$, and X is soluble by the inductive hypothesis. Consider the projection homomorphism $\pi : G \to X$. Its kernel is $N := G_1 \times \{1_X\} \cong G_1$, which is soluble, and its image is X, so $G/N \cong X$ is soluble by the First Isomorphism Theorem. Hence, G is soluble by Lemma 5.11. □

One can also prove the previous theorem by showing that if $G = G_1 \times \cdots \times G_t$, then $G^{(i)} = (G_1)^{(i)} \times \cdots \times (G_t)^{(i)}$ for all i.

In this chapter, we only defined finite soluble groups. More generally, you may see in the literature that an arbitrary group G is defined to be soluble if $G^{(n)} = \{1_G\}$ for some $n \in \mathbb{N}$. This agrees with Definition 5.9 on the set of finite groups, by Theorem 5.21.

5.4 Nilpotent groups

In the final section of the course, we study nilpotent groups, which form a special class of soluble groups.

Definition 5.26. For a group G, define $\gamma_1(G) := G$, and for $i \in \mathbb{N}$, define $\gamma_{i+1}(G) := [\gamma_i(G), G]$. The descending series

$$G = \gamma_1(G) \geq \gamma_2(G) \geq \cdots \geq \gamma_n(G) \geq \gamma_{n+1}(G) \geq \cdots$$

is called the *lower central series* of G.

Definition 5.27. A group G is said to be *nilpotent* if $\gamma_n(G) = \{1_G\}$ for some positive integer n. If G is nilpotent and c is maximal such that $\gamma_c(G) \neq 1$, then c is called the *nilpotency class* of G.

The following exercise will be useful for our next proposition.

Exercise. Let G be a group, $A \subseteq G$, $N \trianglelefteq G$. Show that $\langle A \rangle N/N = \langle B \rangle$, where $B := \{aN \mid a \in A\}$.

Proposition 5.28. Let G be a nilpotent group. Then

(i) Let $H \leq G$. Then H is nilpotent.

(ii) Let $N \trianglelefteq G$. Then $\gamma_n(G/N) = \gamma_n(G)N/N$ for all positive integers n. In particular, G/N is nilpotent.

Proof. (i) By induction on i, if $H \leq G$, then $\gamma_i(H) \leq \gamma_i(G)$ for all i. Thus, if G is nilpotent and $H \leq G$, then H is nilpotent.

(ii) Let $N \trianglelefteq G$, and let $g, h \in G$. Then $[gN, hN] = [g, h]N$. We will now prove, by induction on n, that $\gamma_n(G/N) = \gamma_n(G)N/N$. If $n = 1$, then $GN = G$, so $\gamma_1(G/N) = G/N = GN/N = \gamma_1(G)N/N$, as required.

Assume now that $n > 1$, and that the claim holds for indices less than n. Then, applying the inductive hypothesis to $\gamma_{n-1}(G/N)$, we have

$$\begin{aligned}\gamma_n(G/N) &= \langle [xN, gN] \mid xN \in \gamma_{n-1}(G/N) = \gamma_{n-1}(G)N/N, g \in G\rangle \\ &= \langle [xN, gN] \mid x \in \gamma_{n-1}(G), g \in G\rangle \\ &= \langle [x, g]N \mid x \in \gamma_{n-1}(G), g \in G\rangle = \gamma_n(G)N/N\end{aligned}$$

with the last equality following from the exercise above. This completes our induction.

Finally, assume G is nilpotent. Then $\gamma_n(G) = \{1_G\}$ for some positive integer n, so $\gamma_n(G/N) = \gamma_n(G)N/N = \{1_G\}N/N = \{1_{G/N}\}$. Thus, G/N is nilpotent. □

Corollary 5.29. Let G be a finite nilpotent group. Then G is soluble.

Proof. Consider the derived series

$$G = G^{(0)} \geq G^{(1)} \geq \dots$$

of G. By Theorem 5.21, it will suffice to prove that $G^{(m)} = \{1_G\}$ for some m. We first prove, by induction on n, that $G^{(n)} \leq \gamma_{n+1}(G)$ for all $n \geq 0$. If $n = 0$, this holds since $G^{(0)} = G = \gamma_1(G)$ by Definition 5.20 and Definition 5.26. So assume that $n \geq 1$, and that the claim holds for $n - 1$. Then $G^{(n)} = [G^{(n-1)}, G^{(n-1)}] \leq [\gamma_n(G), G] = \gamma_{n+1}(G)$, using the inductive hypothesis. Finally, since G is nilpotent, $\gamma_{m+1}(G) = \{1_G\}$ for some $m \geq 0$, so $G^{(m)} \leq \gamma_{m+1}(G) = \{1_G\}$, as required. □

With the more general definition of solubility given at the end of Section 5.3, our proof of Corollary 5.29 shows that every nilpotent group, finite or not, is soluble.

Proposition 5.30. Let G be a finite group of prime power order. Then G is nilpotent.

Proof. Write $|G| = p^n$ for p prime. We will prove that G is nilpotent by induction on n. If $n = 0$, then G is the trivial group, so G is nilpotent.

So assume that $n \geq 1$, and that the result holds for groups of order less than p^n. By Theorem 2.24, $Z := Z(G)$ is not trivial. Thus, G/Z is a group of p-power order and $|G/Z| < |G|$, so the inductive hypothesis applies: G/Z is nilpotent. By definition, this implies that there exists a positive integer m such that $\gamma_m(G/Z) = \{1_{G/Z}\}$. By Proposition 5.28, we deduce that $\gamma_m(G)Z/Z = \{1_{G/Z}\} = Z/Z$, so $\gamma_m(G)Z = Z$ by the Correspondence Theorem. That is, $\gamma_m(G) \leq Z$. It follows that $\gamma_{m+1}(G) = [\gamma_m(G), G] \leq [Z(G), G] = \{1_G\}$. Hence, $\gamma_{m+1}(G) = \{1_G\}$, as required. □

Example 5.31. Let $G = D_{2p}$, for p an odd prime. We claim that G is not nilpotent. Indeed, write $G = \langle \sigma, \tau \rangle \leq S_p$ as in Definition 2.5. Thus, $\sigma = (1, 2, \ldots, p) \in S_p$ and $\tau = (1, p)(2, p-1) \ldots (\frac{p-1}{2}, \frac{p+3}{2})$. In particular, $\tau\sigma\tau^{-1} = \sigma^{-1}$.

By Case 1 of Example 5.19, we have $\gamma_2(G) = [G, G] = N := \langle \sigma \rangle$. Thus, $\gamma_3(G) = [\gamma_2(G), G]$ contains $[\sigma, \tau] = \sigma^2$, so $\gamma_3(G)$ contains $\langle \sigma^2 \rangle = N = \gamma_2(G)$ (recall that $|\sigma|$ is odd). Hence, $\gamma_3(G) = \gamma_2(G) = N$, and it follows that $\gamma_i(G) = N$ for all $i \geq 2$. Hence, G is not nilpotent.

Exercise.

(1) Let $G_1, \ldots, G_t$ be finite groups. Prove that $\gamma_i(G_1 \times \cdots \times G_t) = \gamma_i(G_1) \times \cdots \times \gamma_i(G_t)$. Deduce that if G_i is nilpotent for all $1 \leq i \leq t$, then $G_1 \times \cdots \times G_t$ is nilpotent.

(2) Let $p_1, \ldots, p_t$ be primes, and let P_i be a group of p_i-power order. By the previous question and Proposition 5.30, $P_1 \times \cdots \times P_t$ is nilpotent. Prove that a finite group G is nilpotent if and only if $G \cong P_1 \times \cdots \times P_t$ for some groups P_i, where each P_i has prime power order. (Hint: Suppose that G is nilpotent. Using induction on the nilpotency class, prove that $|\operatorname{Syl}_p(G)| = 1$ for all primes p dividing $|G|$. The results on the Problem Sheets are useful here.)

Appendix A

Five tools for analysing element orders in finite groups

In Chapter 1 and Chapter 2, and on the Problem Sheets, we developed several tools for studying the structure of a finite group. The following five are particularly useful for finding the orders of elements in a finite group G.

Tool 1. If G is nonabelian, then $G/Z(G)$ is noncyclic (see the Problem Sheets).

Tool 2. If p is a prime divisor of $|G|$, then the number of elements of G of order p is congruent to -1 modulo p (see Theorem 2.26).

Tool 3. If $x \in G$, then $|^G x| = |G : C_G(x)|$ (by the Orbit–Stabiliser Theorem).

Tool 4. If $x, y \in G$ are such that

(i) $xy = yx$;

(ii) $\langle x \rangle \cap \langle y \rangle = \{1_G\}$,

then $|xy| = \text{lcm}(|x|, |y|)$. In particular, if $xy = yx$ and $\gcd(|x|, |y|) = 1$, then $|xy| = |x||y|$ (see Lemma 1.15(iv)).

Tool 5. If $m > 1$ is odd, then the number of elements of G of order m is even (see Question 2 below).

Sample questions (practice in applying the five tools above)

Question 1: Let p and q be distinct primes, and let G be a group of order pq. Prove that either

(i) G is cyclic; or

(ii) $Z(G) = \{1_G\}$.

Solution: Suppose first that G is abelian. By *Tool 2*, there exist $x \in G$ of order p and $y \in G$ of order q. Since G is abelian, we have $xy = yx$. Also, since $p \neq q$, Lagrange's theorem yields $\langle x \rangle \cap \langle y \rangle = \{1_G\}$. Thus, *Tool 4* yields $|xy| = \text{lcm}(|x|, |y|) = pq = |G|$. Thus, $G = \langle xy \rangle$.

Assume next that G is nonabelian. Then $Z(G) \neq G$, so Lagrange's theorem implies that $|Z(G)| \in \{1, p, q\}$. By *Tool 1*, $|G/Z(G)| = pq/|Z(G)|$ cannot be p or q (since all groups of prime order are cyclic). Thus, $|Z(G)|$ cannot be q or p. Hence, $|Z(G)| = 1$, i.e. $Z(G) = \{1_G\}$.

Question 2: Let G be a finite group, and let $m > 1$ be odd. Prove that the number of elements of G of order m is even.

Solution: Let X be the set of elements of G of order m. If X is empty, then we are done. Otherwise, define a relation on X by $x \sim y$ if and only if $y \in \{x, x^{-1}\}$. Since $|x^{-1}| = |x|$, this is an equivalence relation on X, and the equivalence class of x is $\{x, x^{-1}\}$. Since $m > 1$ is odd, $x^2 \neq 1_G$, so $x \neq x^{-1}$, and each equivalence class has size 2. Hence, $|X|$ is even.

Question 3: Let G be a group of order 12, and suppose that G contains no elements of order 6. Prove that

(i) G has precisely 8 elements of order 3;

(ii) G has precisely 3 elements of order 2.

Solution: For $p \in \{2, 3\}$, define

$$E_p(G) := \{x \in G \mid |x| = p\}.$$

So we need to prove that $|E_3(G)| = 8$ and $|E_2(G)| = 3$.

We start with $|E_3(G)|$. By *Tool 2* and Question 2, the possibilities for $|E_3(G)|$ are

$$|E_3(G)| \in \{2, 8\}.$$

(*Tool 2* implies that the possibilities are $|E_3(G)| \in \{2, 5, 8, 11\}$, as these are the only integers between 0 and $|G| = 12$ which are congruent to -1 modulo 3. Question 2 then rules out the possibilities $|E_3(G)| \in \{5, 11\}$.) Suppose, in pursuit of a contradiction, that $E_3(G)$ has size 2. Let $x \in E_3(G)$. By *Tool 3*, and the fact that $|{}^g x| = |x|$ for all $g \in G$ (Lemma 1.30), we have

$$|G : C_G(x)| = |{}^G x| \leq |E_3(G)| = 2, \text{ so } |G : C_G(x)| \in \{1, 2\}.$$

Since $|G| = 12$ and $|G : C_G(x)| = 12/|C_G(x)|$, we deduce that $|C_G(x)| \in \{6, 12\}$. In particular, 2 divides $|C_G(x)|$, so Cauchy's theorem implies that there exists $y \in C_G(x)$ of order 2. *Tool 4* then yields $|xy| = 6$, as in Question 1 above. This contradicts our assumption that G contains no elements of order 6. Thus, we have proved that $|E_3(G)| = 8$.

Finally, since $E_2(G)$ is contained in $G \setminus (E_3(G) \sqcup \{1_G\})$, we have $|E_2(G)| \leq 3$. It follows from *Tool 2* that $|E_2(G)| \in \{1, 3\}$. Suppose, in pursuit of a contradiction, that $|E_2(G)| = 1$, and let $z \in E_2(G)$. Then since $|{}^g z| = |z| = 2$ for all $g \in G$, and z is the unique element of order 2, we have that ${}^g z = z$ for all $g \in G$. That is, $z \in Z(G)$. Let $x \in E_3(G)$. Then $xz = zx$ since $z \in Z(G)$, so as in Question 1, we apply *Tool 4* to get $|xz| = 6$. Again, this is a contradiction. So $|E_2(G)| = 3$, as required.

More sample questions (practice in applying the five tools above)

Try using the methods in Question 3 above to come up with your own solutions to the following.

(1) Let G be a group of order 24 such that G has no elements of order 6. Prove that G has at least 8 elements of order 3 and at least 3 elements of order 2.

(2) Let G be a group of order 18 such that G has no elements of order 6. Prove that G has precisely 9 elements of order 2. Hence or otherwise, prove that G has either 2 or 8 elements of order 3.

There will likely be a question like one of these on the exam, so it is worth practising these methods.

Appendix B

The Sylow game

In Section 3.4, we studied two applications of Sylow's theorems (we called them Versions 1 and 2 of the Sylow game).

Application 1: Proving that a group of a particular order cannot be simple

The first application of Sylow's theorems has the following form:

Let G be a group of order m. Prove that G cannot be simple.

We start by assuming that G *is* simple, and look for a contradiction. In Section 3.4 we saw three strategies for doing this: Strategy 1 (the basic method), Strategy 2 (element counting) and Strategy 3 (the homomorphism method). We worked through groups of orders 20, 56 and 36 there, so let us do another example.

Example B.1. Prove that a group of order 80 cannot be simple.

Solution: Let G be a group of order $80 = 16 \cdot 5$, and assume that G is simple.

Trying Strategy 1: By Corollary 3.8(ii) and Sylow's theorem 4, $|\mathrm{Syl}_5(G)|$ divides 16 and is congruent to 1 modulo 5. Thus, $|\mathrm{Syl}_5(G)| \in \{1, 16\}$, and hence $|\mathrm{Syl}_5(G)| = 16$ since G is simple. Similarly, $|\mathrm{Syl}_2(G)|$ divides 5 and is odd, so $|\mathrm{Syl}_2(G)| \in \{1, 5\}$, and hence $|\mathrm{Syl}_2(G)| = 5$.

So Strategy 1 does not work, but it tells us that $|\mathrm{Syl}_5(G)| = 16$, which we will use below. Note that $|G|_5 = 5$, so Strategy 2 might be good to try.

Trying Strategy 2: We have $|\mathrm{Syl}_5(G)| = 16$. Then $|F_5(G)| = 16 \cdot 4 = 64$ by Corollary 3.9(iii). It follows that $|F_2(G)| \leq 80 - 64 - 1 = 15 = |G|_2 - 1$. Thus, G has a normal Sylow 2-subgroup, say Q, by Corollary 3.9(ii). Since $|Q| = 16$, we have $1 \neq Q \neq G$, so G is not simple.

Note that Strategy 3 would not have worked with $p = 5$ above, since 80 does divide $|S_{16}| = 16!$. However, Strategy 3 would have worked with $p = 2$. Indeed, we have $|\mathrm{Syl}_2(G)| = 5$, and 80 does not divide $5! = 120$.

We now look at the second version of the Sylow game.

Application 2: Proving that a particular group is simple

The second application of Sylow's theorems has the following form:

Let G be a particular group, or a group with a particular list of properties. Prove that G is simple.

The main tool that we use for this application is Corollary 3.10. The strategy is as follows. Suppose (in pursuit of a contradiction) that N is a normal subgroup of G with $1 \neq N \neq G$.

Step A. Choose a prime p with $|G|_p = p$, and suppose that p divides $|N|$. Then p does not divide $|G : N| = |G|/|N|$, since $|G|_p = p$. Thus,

(1) $|\operatorname{Syl}_p(G)| = |\operatorname{Syl}_p(N)|$ (Corollary 3.10(ii)(1));

(2) $|F_p(G)| = |F_p(N)|$ (Corollary 3.10(ii)(2)).

We now use Corollary 3.8(ii) and Sylow's theorem 4 to list the possibilities for $|\operatorname{Syl}_p(G)| = |\operatorname{Syl}_p(N)|$. By the above, this then gives us the possibilities for $|F_p(N)| = |F_p(G)|$. The idea is to prove that if $|\operatorname{Syl}_p(N)| = |\operatorname{Syl}_p(G)|$ is "small", then $|F_p(G)| = |F_p(N)|$ is "small", and this will usually contradict a property of the group G we are working with (see Proposition 3.11).

Thus, we conclude that $|F_p(G)| = |F_p(N)|$ is "large". Since $|N| \geq |F_p(N)| + 1$, we then have a lower bound on $|N|$.

Step B. Since $|N|$ divides $|G|$, we use our lower bound from Step A to give us a list of possibilities for $|N|$ (see Proposition 3.11). Note that if $q \neq p$ is prime, and we have shown that q must divide $|N|$, then N has an element x of order q by Cauchy's theorem. Moreover, ${}^G x \subseteq N$ by Corollary 3.10(i), so $|N| \geq |{}^G x| + 1 + |F_p(N)|$. Does this give us a contradiction?

If not, and $|G|_q = q$, then we can repeat Step A with q in place of p, and get a list of possibilities for $|F_q(N)|$. We then have $|N| \geq |F_q(N)| + 1 + |F_p(N)|$. Does this give us a contradiction?

Step C. Repeat Steps A and B for each prime p with $|G|_p = p$. Hopefully we get a contradiction in each case.

Step D. As we said, hopefully Steps A–C allow us to deduce that $|N|$ can only be divisible by primes p with $|G|_p \neq p$. The idea is then to use the properties of G to deduce a contradiction. For example, suppose we have shown that 2 is the only prime dividing $|N|$. Then we know that $|N|$ divides $|G|_2$. If we know that $|{}^G x| \geq |G|_2$ for all elements x of G of order 2, then we have a contradiction: N contains an element x of order 2 by Cauchy's theorem, and ${}^G x \sqcup \{1_G\} \subseteq N$ by Corollary 3.10(i), so $|N| > |G|_2$.

Beyond this, the methods vary a lot from group to group, so we cannot be more specific. There are also not many nonabelian finite simple groups of small order to practise on: by the Classification of Finite Simple Groups, the five smallest orders of nonabelian finite simple groups are 60, 168, 360, 504 and 660.

However, the approach above works very well if all but one of the prime divisors p of $|G|$ have the property that $|G|_p = p$ (for example, if $|G| \in \{60, 168, 660\}$). Some exercises applying these ideas are given below.

Exercises for Application 1

(1) Let G be a group of order 24. Prove that G cannot be simple.

(2) Let G be a group of order 28. Prove that G cannot be simple.

(3) Let G be a group of order 72. Prove that G cannot be simple.

(4) Let G be a group of order 160. Prove that G cannot be simple.

(5) Let G be a group of order 196. Prove that G cannot be simple.

(6) Let G be a group of order 112. Prove that G cannot be simple. (Hint: if G is a nonabelian simple group isomorphic to a subgroup H of S_n, consider $H \cap A_n$ to show that G is isomorphic to a subgroup of A_n.)

Exercises for Application 2

(1) Let G be a group of order 168, and assume that
   (i) all elements of G of order 2 are conjugate, and if x is such an element, then $|^G x| = 21$;
   (ii) G has at least 7 elements of order 7;
   (iii) G has 56 elements of order 3.

   Prove that G is simple.

(2) Let G be a group of order 660, and assume that
   (i) all elements of G of order 2 are conjugate, and if x is such an element, then $|^G x| = 55$;
   (ii) G has 120 elements of order 11;
   (iii) G has 264 elements of order 5;
   (iv) G has 110 elements of order 3.

   Prove that G is simple.

The simple groups $\mathrm{SL}(3,2)$ and $\mathrm{SL}(2,11)/Z(\mathrm{SL}(2,11))$ satisfy the conditions in the last two exercises, respectively.